Locating and Editing Factual Associations in GPT

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Abstract

We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We first develop a causal intervention for identifying neuron *activations* that are decisive in a model's factual predictions. This reveals a distinct set of steps in middle-layer feed-forward modules that mediate factual predictions while processing subject tokens. To test our hypothesis that these computations correspond to factual association recall, we modify feed-forward *weights* to update specific factual associations using Rank-One Model Editing (ROME). We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task. We also evaluate ROME on a new dataset of difficult counterfactual assertions, on which it simultaneously maintains both specificity and generalization, whereas other methods sacrifice one or another. Our results confirm an important role for mid-layer feed-forward modules in storing factual associations and suggest that direct manipulation of computational mechanisms may be a feasible approach for model editing. The code, dataset, visualizations, and an interactive demo notebook are available at `https://rome.baulab.info/`.

1 Introduction

Where does a large language model store its facts? In this paper, we report evidence that factual associations in GPT correspond to a localized computation that can be directly edited.

Large language models can predict factual statements about the world (Petroni et al., 2019; Jiang et al., 2020; Roberts et al., 2020). For example, given the prefix "*The Space Needle is located in the city of*," GPT will reliably predict the true answer: "*Seattle*" (Figure 1a). Factual knowledge has been observed to emerge in both autoregressive GPT models (Radford et al., 2019; Brown et al., 2020) and masked BERT models (Devlin et al., 2019).

In this paper, we investigate how such factual associations are stored within GPT-like autoregressive transformer models. Although many of the largest neural networks in use today are autoregressive, the way that they store knowledge remains under-explored. Some research has been done for masked models (Petroni et al., 2019; Jiang et al., 2020; Elazar et al., 2021a; Geva et al., 2021; Dai et al., 2022; De Cao et al., 2021), but GPT has architectural differences such as unidirectional attention and generation capabilities that provide an opportunity for new insights.

We use two approaches. First, we trace the causal effects of hidden state activations within GPT using causal mediation analysis (Pearl, 2001; Vig et al., 2020b) to identify the specific modules that mediate recall of a fact about a subject (Figure 1). Our analysis reveals that feedforward MLPs at a range of middle layers are decisive when processing the last token of the subject name (Figures 1b,2b,3).

Second, we test this finding in model weights by introducing a Rank-One Model Editing method (ROME) to alter the parameters that determine a feedfoward layer's behavior at the decisive token.

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GPT中事实关联的定位与编辑

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摘要

我们研究了自回归Transformer语言模型中事实关联的存储与召回机制，发现这类关联实际上对应于可被直接编辑的局部计算过程。首先，我们设计了一种因果干预方法，用于识别对模型的事实预测起决定性作用的神经元激活状态。研究结果表明，在处理主题标记的过程中，中层前馈模块中存在一系列特定的步骤，负责实现事实预测。为了验证这些计算过程确实与事实关联的召回相关，我们通过一阶模型编辑（ROME）方法修改前馈权重<span style='id':'9'>，从而更新特定的事实关联。实验表明，ROME在标准的零样本关系抽取（zsRE）模型编辑任务中表现优异。此外，我们还在包含复杂反事实断言的新数据集上对ROME进行了测试，结果发现该方法能够在保持高特异性的同时具备良好的泛化能力，而其他方法往往只能在这两者之间做出取舍。我们的研究结果证实了中层前馈模块在存储事实关联中的重要作用，并表明直接操控计算机制可能是实现模型编辑的一种可行途径。相关代码、数据集、可视化结果以及交互式演示笔记本均可访问`https://rome.baulab.info/`。

1 引言

大型语言模型究竟将自身掌握的事实存储在何处？本文提供了相关证据，表明GPT中的事实关联对应于某种可被直接编辑的局部计算结果。

大型语言模型能够预测关于世界的事实陈述（佩特罗尼等人，2019年；蒋等人，2020年；罗伯茨等人，2020年）。例如，当给定前缀"太空针塔位于某座城市中"时，GPT能够准确预测出正确答案："西雅图"（见图1a）。事实上，无论是在自回归型的GPT模型（拉德福德等人，2019年；布朗内特等人，2020年）还是掩码型的BERT模型（德夫林等人，2019年）中，都观察到了事实知识的出现。

在本文中，我们研究了这类事实关联是如何存储在类似GPT的自回归变换器模型中的。尽管目前市面上许多规模最大的神经网络都属于自回归类型，但它们存储知识的方式至今仍缺乏充分研究。虽然已有针对掩码模型的相关研究（佩特罗尼等人，2019年；蒋等人，2020年；埃拉扎尔等人，2021a年；Geva等人，2021年；Dai等人，2022年；De Cao等人，2021年），但GPT由于其单向注意力机制以及具备生成能力等架构上的差异，为人们带来新的研究机遇。

我们采用了两种研究方法。首先，通过因果中介分析来追踪GPT内部各层隐藏状态激活的因果效应（珀尔，2001年；Vig等人，2020年b版研究），从而确定出那些负责介导对特定主体相关事实回忆的特定模块（见图1）。我们的分析表明，在处理主体名称的最后一个标记时，位于中间层位置的各种前馈多层感知机会起到决定性作用（见图1b、2b以及3）。

其次，我们通过引入一阶模型编辑方法（ROME），对决定前向层在关键标记处行为的相关参数进行修改，进而在该模型权重层面验证这一发现。

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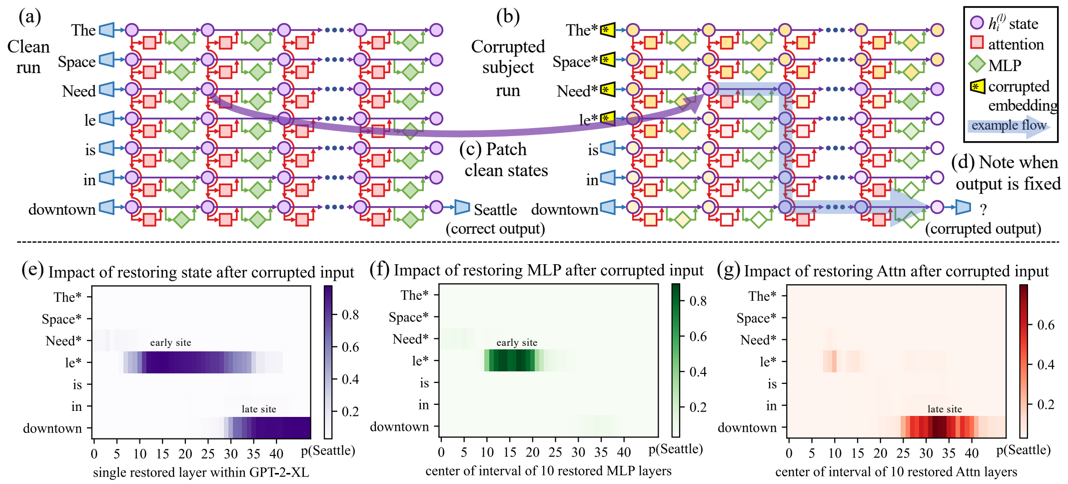

Figure 1: **Causal Traces** compute the causal effect of neuron activations by running the network twice: (a) once normally, and (b) once where we corrupt the subject token and then (c) restore selected internal activations to their clean value. (d) Some sets of activations cause the output to return to the original prediction; the light blue path shows an example of information flow. The causal impact on output probability is mapped for the effect of (e) each hidden state on the prediction, (f) only MLP activations, and (g) only attention activations.

Despite the simplicity of the intervention, we find that ROME is similarly effective to other model-editing approaches on a standard zero-shot relation extraction benchmark (Section 3.2).

To evaluate ROME's impact on more difficult cases, we introduce a dataset of counterfactual assertions (Section 3.3) that would not have been observed in pretraining. Our evaluations (Section 3.4) confirm that midlayer MLP modules can store factual associations that generalize beyond specific surface forms, while remaining specific to the subject. Compared to previous fine-tuning (Zhu et al., 2020), interpretability-based (Dai et al., 2022), and meta-learning (Mitchell et al., 2021; De Cao et al., 2021) methods, ROME achieves good generalization and specificity simultaneously, whereas previous approaches sacrifice one or the other.

2 Interventions on Activations for Tracing Information Flow

To locate facts within the parameters of a large pretrained autoregressive transformer, we begin by analyzing and identifying the specific hidden states that have the strongest causal effect on predictions of individual facts. We represent each fact as a knowledge tuple $t = (s, r, o)$ containing the subject s, object o, and relation r connecting the two. Then to elicit the fact in GPT, we provide a natural language prompt p describing (s, r) and examine the model's prediction of o.

An autoregressive transformer language model $G : \mathcal{X} \to \mathcal{Y}$ over vocabulary V maps a token sequence $x = [x_1, ..., x_T] \in \mathcal{X}$, $x_i \in V$ to a probability distribution $y \in \mathcal{Y} \subset \mathbb{R}^{|V|}$ that predicts next-token continuations of x. Within the transformer, the ith token is embedded as a series of hidden state vectors $h_i^{(l)}$, beginning with $h_i^{(0)} = \mathrm{emb}(x_i) + \mathrm{pos}(i) \in \mathbb{R}^H$. The final output $y = \mathrm{decode}(h_T^{(L)})$ is read from the last hidden state.

We visualize the internal computation of G as a grid (Figure 1a) of hidden states $h_i^{(l)}$ in which each layer l (left $\to$ right) adds global attention $a_i^{(l)}$ and local MLP $m_i^{(l)}$ contributions computed from previous layers, and where each token i (top $\to$ bottom) attends to previous states from other tokens. Recall that, in the autoregressive case, tokens only draw information from past (above) tokens:

$$\begin{aligned} h_i^{(l)} = h_i^{(l-1)} + a_i^{(l)} + m_i^{(l)} \\ a_i^{(l)} = \mathrm{attn}^{(l)}\left(h_1^{(l-1)}, h_2^{(l-1)}, \ldots, h_i^{(l-1)}\right) \\ m_i^{(l)} = W_{proj}^{(l)}\, \sigma\left(W_{fc}^{(l)} \gamma\left(a_i^{(l)} + h_i^{(l-1)}\right)\right). \end{aligned} \tag{1}$$

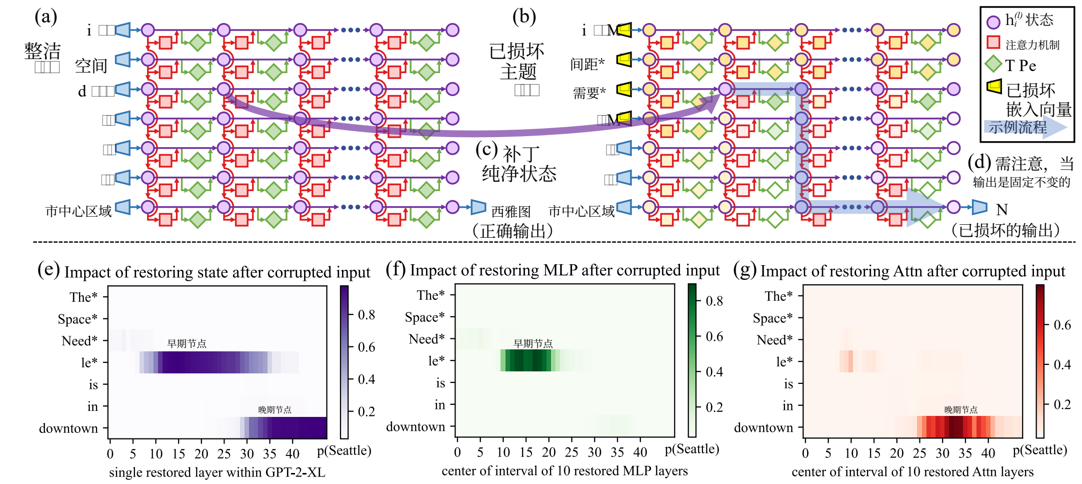

图1：**因果轨迹算法**通过两次运行网络来计算神经元激活状态所对应的因果效应：(a) 按常规方式运行一次，(b) 在此过程中篡改主题标记，之后(c)再将选定的内部激活状态恢复为原始值。(d) 某些激活状态组合会导致输出回归到最初的预测结果；浅蓝色路径展示了信息流动的一个示例。针对(e)各隐藏状态对预测的影响、(f)仅MLP激活状态的影响，以及(g)仅注意力激活状态的影响，该算法还会绘制出其对输出概率的因果影响映射图。

尽管这种干预方式十分简单，但我们在标准零样本关系抽取基准测试中发现，ROME与其他模型编辑方法的效果相当（见第3.2节）。

为评估ROME方法在更复杂场景下的表现，我们引入了一个包含反事实断言的COUNTERFACT数据集（见第3.3节），这类断言在预训练阶段是无法被观测到的。我们的实验结果（见第3.4节）表明，中层MLP模块能够存储那些不仅具备超越特定表面形式之外的泛化能力，同时又与对应主题高度相关的事实关联。与以往的微调（Zhu等人，2020年）、基于可解释性的方法（Dai等人，2022年）以及元学习方法（Mitchell等人，2021年；De Cao等人，2021年）相比，ROME方法能够同时实现良好的泛化能力和高特异性，而之前的那些方法往往只能在这两者之间二选一。

2种基于激活值的干预方法用于追踪信息流

为在大型预训练自回归变换器模型的参数中定位特定事实，我们首先分析并识别出那些对各个事实的预测具有最强因果效应的隐藏状态。我们将每个事实表示为一个知识元组 $t = (s, r, o)$，该元组包含主体s、客体 o以及连接二者的关系 r。随后，为在GPT中调出相应事实，我们会提供一个描述(s, r)的自然语言提示语 p，并查看模型对 o的预测结果。

一种自回归变换器语言模型 G：它以 V为词汇表，将$x = [x_1, ..., x_T] \in \mathcal{X}$, $x_i \in V$形式的标记序列映射为 $y \in \mathcal{Y} \subset \mathbb{R}^{|V|}$ 形式的概率分布，以此预测 x的后续标记内容。在该模型内部，第 i个标记会被嵌入为一组隐藏状态向量 $h_i^{(l)}$，这些向量的生成始于 $h_i^{(0)} = \mathrm{emb}(x_i) + \mathrm{pos}(i) \in \mathbb{R}^H$。最终的输出 $y = \mathrm{decode}(h_T^{(L)})$则是从最后一个隐藏状态中读取得到的。

我们将内部计算过程 G 可视化为一个由隐藏状态构成的网格结构（见图1a）$h_i^{(l)}$，在每一层中 l（左侧 $\to$ 右侧）$a_i^{(l)}$ 会加入基于前序层计算得到的全局注意力机制与局部多层感知机的贡献，同时每个标记 i（上方 $\to$ 下方）$m_i^{(l)}$ 会关注其他标记所产生的先前隐藏状态。需要说明的是，在自回归模式下，标记仅能从之前的（上方的）标记中获取信息：

$$\begin{aligned} h_i^{(l)} = h_i^{(l-1)} + a_i^{(l)} + m_i^{(l)} \\ a_i^{(l)} = \mathrm{attn}^{(l)}\left(h_1^{(l-1)}, h_2^{(l-1)}, \ldots, h_i^{(l-1)}\right) \\ m_i^{(l)} = W_{proj}^{(l)}\, \sigma\left(W_{fc}^{(l)} \gamma\left(a_i^{(l)} + h_i^{(l-1)}\right)\right). \end{aligned} \tag{1}$$

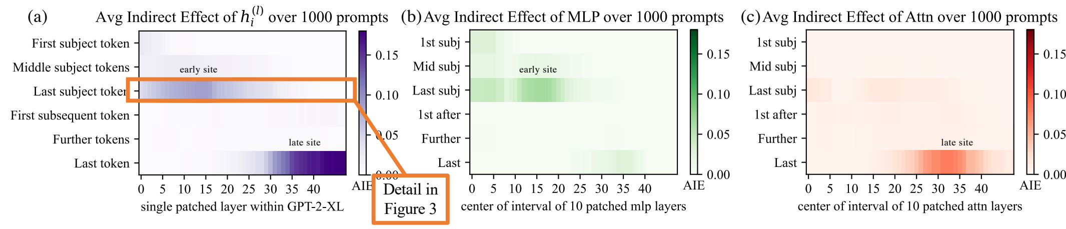

Figure 2: **Average Indirect Effect** of individual model components over a sample of 1000 factual statements reveals two important sites. (a) Strong causality at a 'late site' in the last layers at the last token is unsurprising, but strongly causal states at an 'early site' in middle layers at the last subject token is a new discovery. (b) MLP contributions dominate the early site. (c) Attention is important at the late site. Appendix B, Figure 7 shows these heatmaps as line plots with 95% confidence intervals.

Each layer's MLP is a two-layer neural network parameterized by matrices $W_{proj}^{(l)}$ and $W_{fc}^{(l)}$, with rectifying nonlinearity σ and normalizing nonlinearity γ. For further background on transformers, we refer to Vaswani et al. (2017).[3]

2.1 Causal Tracing of Factual Associations

The grid of states (Figure 1) forms a *causal graph* (Pearl, 2009) describing dependencies between the hidden variables. This graph contains many paths from inputs on the left to the output (next-word prediction) at the lower-right, and we wish to understand if there are specific hidden state variables that are more important than others when recalling a fact.

As Vig et al. (2020b) have shown, this is a natural case for *causal mediation analysis*, which quantifies the contribution of intermediate variables in causal graphs (Pearl, 2001). To calculate each state's contribution towards a correct factual prediction, we observe all of G's internal activations during three runs: a **clean** run that predicts the fact, a **corrupted** run where the prediction is damaged, and a **corrupted-with-restoration** run that tests the ability of a single state to restore the prediction.

- In the **clean run**, we pass a factual prompt x into G and collect all hidden activations $\{h_i^{(l)} \mid i \in [1,T], l \in [1,L]\}$. Figure 1a provides an example illustration with the prompt: "The Space Needle is in downtown ______", for which the expected completion is $o =$ "Seattle".
- In the baseline **corrupted run**, the subject is obfuscated from G before the network runs. Concretely, immediately after x is embedded as $[h_1^{(0)}, h_2^{(0)}, \ldots, h_T^{(0)}]$, we set $h_i^{(0)} := h_i^{(0)} + \epsilon$ for all indices i that correspond to the subject entity, where $\epsilon \sim \mathcal{N}(0; \nu)$[4]; . G is then allowed to continue normally, giving us a set of corrupted activations $\{h_{i*}^{(l)} \mid i \in [1,T], l \in [1,L]\}$. Because G loses some information about the subject, it will likely return an incorrect answer (Figure 1b).
- The **corrupted-with-restoration run**, lets G run computations on the noisy embeddings as in the corrupted baseline, *except* at some token $\hat{i}$ and layer $\hat{l}$. There, we hook G so that it is forced to output the clean state $h_{\hat{i}}^{(\hat{l})}$; future computations execute without further intervention. Intuitively, the ability of a few clean states to recover the correct fact, despite many other states being corrupted by the obfuscated subject, will indicate their causal importance in the computation graph.

Let $\mathbb{P}[o]$, $\mathbb{P}_*[o]$, and $\mathbb{P}_{*,\, \text{clean}\, h_i^{(l)}}[o]$ denote the probability of emitting o under the clean, corrupted, and corrupted-with-restoration runs, respectively; dependence on the input x is omitted for notational simplicity. The **total effect** (TE) is the difference between these quantities: TE $= \mathbb{P}[o] - \mathbb{P}_*[o]$. The **indirect effect** (IE) of a specific mediating state $h_i^{(l)}$ is defined as the difference between the probability of o under the corrupted version and the probability when that state is set to its clean version, while the subject remains corrupted: IE $= \mathbb{P}_{*,\, \text{clean}\, h_i^{(l)}}[o] - \mathbb{P}_*[o]$. Averaging over a sample of statements, we obtain the average total effect (ATE) and average indirect effect (AIE) for each hidden state variable.[5]

[3] Eqn. 1 calculates attention sequentially after the MLP module as in Brown et al. (2020). Our methods also apply to GPT variants such as Wang & Komatsuzaki (2021) that put attention in parallel to the MLP.

[4] We select ν to be 3 times larger than the empirical standard deviation of embeddings; see Appendix B.1 for details, and see Appendix B.4 for an analysis of other corruption rules.

[5] One could also compute the direct effect, which flows through other model components besides the chosen mediator. However, we found this effect to be noisy and uninformative, in line with results by Vig et al. (2020b).

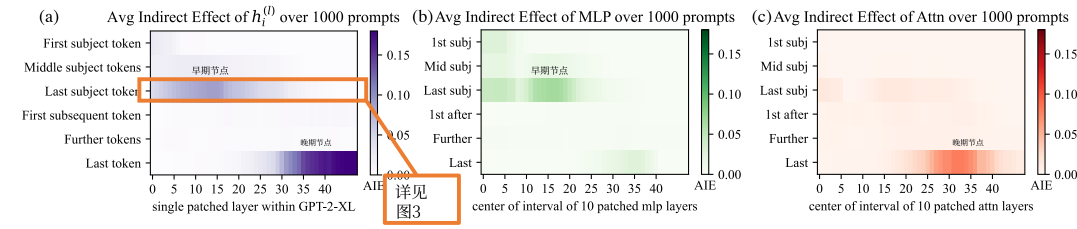

图2：**各个模型组件的平均间接效应**在1000条事实陈述的样本上的分析结果揭示了两个重要的节点。首先，在最底层最后几个标记处的"晚期节点"上存在强因果关系，这并不令人意外；但更为新颖的是，在中间层最后几个主题标记处的"早期节点"上也出现了强因果状态。其次，多层感知机对应的贡献在早期节点上占据主导地位。最后，注意力机制在晚期节点上起着关键作用。附录B中的图7以带有95%置信区间的折线图形式展示了这些热力图。

每一层的多层感知机实际上是一个由矩阵 $W_{proj}^{(l)}$ 和 $W_{fc}^{(l)}$ 参数化的两层神经网络，其中采用了整流非线性函数以及归一化非线性函数 σ。关于变换器模型的更多背景知识，可参考Vaswani等人（2017）[3]

2.1 事实关联的因果追踪

states网格（图1）构成了一幅因果图（珀尔，2009年），该图描述了各隐藏变量之间的依赖关系。这张图中存在许多从左侧输入到右下角输出（即下一个词的预测）的路径，我们希望弄清楚在回忆事实的过程中，是否有某些隐藏状态变量比其他变量更为重要。

正如Vigetal.（2020年b版研究）所指出的，这种情况非常适合进行因果中介分析，这类分析能够量化因果图中的中间变量所做出的贡献（珀尔，2001年）。为了计算每个状态对正确事实预测的贡献，我们在三次运行过程中观察了 G的所有内部激活状态：一次是用于预测事实的**正常**运行，一次是预测结果出现错误的**损坏**运行，还有一次则是测试单个状态能否恢复预测结果的**损坏后恢复**运行。

- 在**纯净运行模式**下，我们会输入一条事实性提示语 x 到 G 中，并收集所有隐藏激活值$\{h_i^{(l)} \mid i \in [1,T],$ $l \in [1,L]\}$。图1a展示了一个示例，提示语为"太空针塔位于市中心区域"，其预期填充结果应为 $o =$ "西雅图"。
- 在基准损坏运行模式中，会在网络开始运行之前对主体信息进行被混淆处理。具体而言，在将 x 以 $[h(0)$ 1、$h(0)$ 2⋯⋯ $h(0)_T]$的形式嵌入之后，我们会为所有对应于该主体实体的索引 i 设置 $h(0)_i := h(0)_i + \epsilon$，此处 $\epsilon \sim \mathcal{N}(0;\nu)$[4]；随后 G 便可正常继续运行，从而产生一组损坏后的激活值 $\{h(l)_{i*} \mid i \in [1,T]$、$l \in [1,L]\}$。由于 G 丢失了关于该主体的部分信息，它很可能会给出错误答案（见图1b）。
- 所谓**带恢复功能的损坏运行模式**，指的是让 G 在带有噪声的嵌入向量上执行计算，操作方式与标准的损坏基准情况类似，不同之处在于特定的标记 i 和层 l。在这些位置，我们会对相关组件进行特殊设置，强制其输出纯净状态 $h_{\hat{i}}^{(\hat{l})}$；之后的计算则无需再做额外干预即可继续进行。从直觉上来看，即便有许多其他状态因被混淆处理而出现异常，仅有少数纯净状态仍能恢复出正确的事实，这一现象便足以表明它们在计算图中的因果重要性。

令 $\mathbb{P}[o]$、$\mathbb{P}_*[o]$和$\mathbb{P}_{*,\,纯净\, h_i^{(l)}}[o]$ 分别表示在纯净、损坏以及带恢复功能的损坏运行模式下生成 o的概率；为简化符号表达，此处未体现对输入 x 的依赖关系。**总效应**（TE）即为这些数值之间的差值：TE $= \mathbb{P}[o] - \mathbb{P}_*[o]$。特定中介状态 $h_i^{(l)}$ 的**间接效应**（IE）则定义为在数据处于损坏状态时生成 o 的概率与将该状态恢复为纯净状态但仍保持数据损坏时的概率之间的差值，其表达式为IE $= \mathbb{P}_{*,\,\text{clean}\, h_i^{(l)}}[o] - \mathbb{P}_*[o]$。通过对一组事实陈述进行平均处理，我们可以得到每个隐藏状态变量的平均总效应（ATE）和平均间接效应（AIE）。[5]

[3] 公式1 中的计算方法与Brown等人（2020年）所描述的一致，即在MLP模块之后依次计算注意力机制。我们的方法同样适用于将注意力机制与MLP并行处理的GPT变体，例如Wang与Komatsuzaki（2021年）的研究成果。我们选取的数值是嵌入向量经验标准差的3倍；具体细节可见附录B.1 ，而其他数据污染规则的相关分析则见附录B.4 。当然，人们也可以计算通过除所选中介变量之外的其他模型组件产生的直接效应，但根据Vig等人（2020年b版研究）的研究结果，这类效应往往存在较大噪声且缺乏实际参考价值。

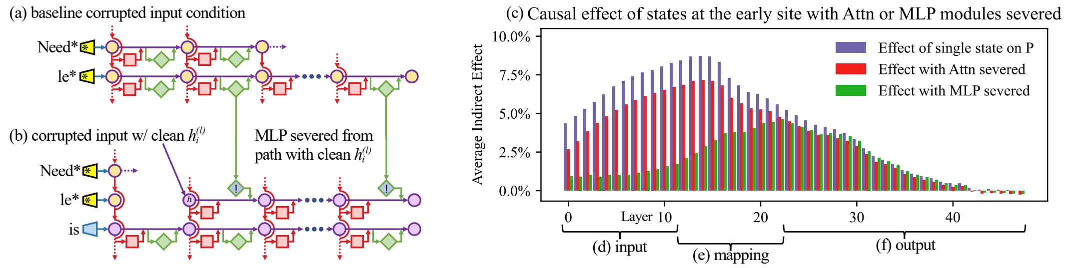

Figure 3: **Causal effects with a modified computation graph**. (a,b) To isolate the effects of MLP modules when measuring causal effects, the computation graph is modified. (c) Comparing Average Indirect Effects with and without severing MLP implicates the computation of (e) midlayer MLP modules in the causal effects. No similar gap is seen when attention is similarly severed.

2.2 Causal Tracing Results

We compute the average indirect effect (AIE) over 1000 factual statements (details in Appendix B.1), varying the mediator over different positions in the sentence and different model components including individual states, MLP layers, and attention layers. Figure 2 plots the AIE of the internal components of GPT-2 XL (1.5B parameters). The ATE of this experiment is 18.6%, and we note that a large portion of the effect is mediated by strongly causal individual states (AIE=8.7% at layer 15) at the last subject token. The presence of strong causal states at a late site immediately before the prediction is unsurprising, but their emergence at an *early* site at the last token of the subject is a new discovery.

Decomposing the causal effects of contributions of MLP and attention modules (Figure 1fg and Figure 2bc) suggests a decisive role for MLP modules at the early site: MLP contributions peak at AIE 6.6%, while attention at the last subject token is only AIE 1.6%; attention is more important at the last token of the prompt. Appendix B.2 further discusses this decomposition.

Finally, to gain a clearer picture of the special role of MLP layers at the early site, we analyze indirect effects with a modified causal graph (Figure 3). (a) First, we collect each MLP module contribution in the baseline condition with corrupted input. (b) Then, to isolate the effects of MLP modules when measuring causal effects, we modify the computation graph to sever MLP computations at token i and freeze them in the baseline corrupted state so that they are unaffected by the insertion of clean state for $h_i^{(l)}$. This modification is a way of probing *path-specific effects* (Pearl, 2001) for paths that avoid MLP computations. (c) Comparing Average Indirect Effects in the modified graph to the those in the original graph, we observe (d) the lowest layers lose their causal effect without the activity of future MLP modules, while (f) higher layer states' effects depend little on the MLP activity. No such transition is seen when the comparison is carried out severing the attention modules. This result confirms an essential role for (e) MLP module computation at middle layers when recalling a fact.

Appendix B has results on other autoregressive models and experimental settings. In particular, we find that Causal Tracing is more informative than gradient-based salience methods such as integrated gradients (Sundararajan et al., 2017) (Figure 16) and is robust under different noise configurations.

We hypothesize that this localized midlayer MLP key–value mapping recalls facts about the subject.

2.3 The Localized Factual Association Hypothesis

Based on causal traces, we posit a specific mechanism for storage of factual associations: each midlayer MLP module accepts inputs that encode a subject, then produces outputs that recall memorized properties about that subject. Middle layer MLP outputs accumulate information, then the summed information is copied to the last token by attention at high layers.

This hypothesis localizes factual association along three dimensions, placing it (i) in the MLP modules (ii) at specific middle layers (iii) and specifically at the processing of the subject's last token. It is consistent with the Geva et al. (2021) view that MLP layers store knowledge, and the Elhage et al. (2021) study showing an information-copying role for self-attention. Furthermore, informed by the Zhao et al. (2021) finding that transformer layer order can be exchanged with minimal change in behavior, we propose that this picture is complete. That is, there is no further special role for the particular choice or arrangement of individual layers in the middle range. We conjecture that any fact

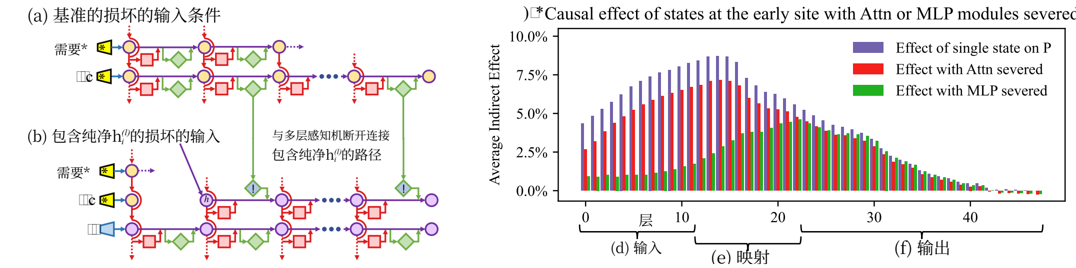

图3：**经过修改的计算图所对应的因果效应**。(a,b) 为在衡量因果效应时单独分析MLP模块的作用，会对计算图进行修改。(c) 通过比较在切断MLP模块连接前后平均间接效应的差异，可以推断因果效应中包含了(e)中层MLP模块的贡献；而当类似地切断注意力机制时，则未观察到类似的差异。

2.2 因果追踪结果

我们针对1000条事实陈述计算平均间接效应（AIE）（详细内容见附录B.1），在此过程中会改变句子中中介因素的位置，并考察包括个体状态、多层感知机层以及注意力层在内的不同模型内部组件。图2展示了GPT-2 XL（参数量为15亿）各内部组件的AIE数值。该实验得出的平均处理效应为18.6%，我们发现这一效应中有很大一部分是由位于最后一个主题标记处的强因果性个体状态所介导的（AIE=8为15层的0.7%）。在预测之前紧邻的位置出现强因果状态本是意料之中的现象，但这类状态会出现在主题标记的最后一个位置的较早位置上，这则是一项新的发现。

通过分解MLP模块与注意力模块对因果效应的贡献（见图1fg以及图2bc），可以发现位于早期节点的MLP模块起着至关重要的作用：MLP模块的贡献对应的平均间接效应高达6.6%，而位于最后一个主题标记处的注意力模块的该值仅为1.6%；在提示词的最后一个标记处，注意力机制的重要性则更高。附录B.2 中对这种分解方法进行了进一步阐述。

最后，为了更清晰地了解位于早期节点的MLP层所具备的特殊作用，我们借助经过修改的计算图来分析间接效应（见图3）。(a) 首先，我们在包含损坏输入的基线条件下，收集每个MLP模块的贡献值。(b) 其次，为在测量因果效应时仅分离出MLP模块的效应，我们对计算图进行了修改，在标记 i处切断MLP的计算流程，并将其固定在基线下的损坏状态，这样它们就不会受到为 $h_i^{(l)}$ 插入纯净状态这一操作的影响。这种修改方式实际上是一种用于探究特定路径的效应（珀尔，2001）的方法，即研究那些无需经过MLP计算的路径。(c) 通过比较修改后计算图与原始计算图中的平均间接效应，我们观察到(d)较低层的节点在没有后续MLP模块活动的状态下会失去其因果效应，而(f)较高层节点的效应则几乎不受MLP活动的影响。而当切断注意力模块进行对比时，则观察不到这样的变化。这一结果进一步证实了在回忆事实的过程中，(e)中层节点上的MLP模块计算具有不可或缺的作用。

附录B中展示了针对其他自回归模型及实验环境的相关测试结果。具体而言，我们发现相比基于梯度的显著性方法，如积分梯度（Sundararajan等人，2017年）（见图16），因果追踪算法能提供更多有用信息，且在不同的噪声环境下也具备更强的稳定性。

我们假设，这种经过局部优化的中层MLP模型所采用的键值映射机制能够召回与主题相关的事实信息。

2.3 局部化事实关联假设

基于因果轨迹算法，我们提出了一种用于存储事实关联的具体机制：每个中层MLP模块会接收编码有特定主题的输入，进而生成能够召回与该主题相关的已存储属性的输出。中层MLP的输出会逐步积累信息，随后这些累积的信息会通过高层结构中的注意力机制被传递至最后一个标记。

该假设将事实关联定位在三个维度上：其一位于MLP模块中，其二存在于特定的中间层，其三则具体体现在对主体最后一个标记的处理过程中。这一观点与Geva等人(2021年)提出的MLP层用于存储知识的观点相一致，同时也与Elhage等人(2021年)的研究结果相符——该研究指出自注意力机制具有信息复制功能。此外，基于Zhao等人(2021年)发现的变换器层顺序可以互换而几乎不会改变系统行为这一现象，我们认为上述解释已经十分完整。也就是说，中间区域的各个层的具体选择或排列并无额外的特殊作用。我们推测，任何事实

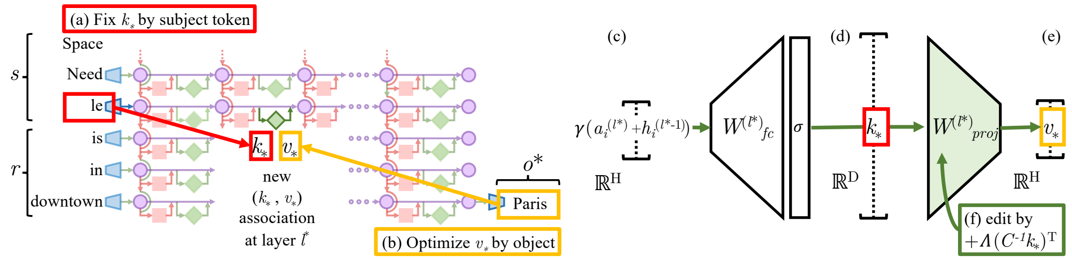

Figure 4: **Editing one MLP layer with ROME**. To associate *Space Needle* with *Paris*, the ROME method inserts a new (k_*, v_*) association into layer l^*, where (a) key k_* is determined by the subject and (b) value v_* is optimized to select the object. (c) Hidden state at layer l^* and token i is expanded to produce (d) the key vector k_* for the subject. (e) To write new value vector v_* into the layer, (f) we calculate a rank-one update $\Lambda(C^{-1}k_*)^T$ to cause $\hat{W}_{proj}^{(l)} k_* = v_*$ while minimizing interference with other memories stored in the layer.

could be equivalently stored in any one of the middle MLP layers. To test our hypothesis, we narrow our attention to a single MLP module at a mid-range layer l^*, and ask whether its weights can be explicitly modified to store an arbitrary fact.

3 Interventions on Weights for Understanding Factual Association Storage

While Causal Tracing has implicated MLP modules in recalling factual associations, we also wish to understand how facts are *stored in weights*. Geva et al. (2021) observed that MLP layers (Figure 4cde) can act as two-layer key–value memories,[6] where the neurons of the first layer $W_{fc}^{(l)}$ form a *key*, with which the second layer $W_{proj}^{(l)}$ retrieves an associated *value*. We hypothesize that MLPs can be modeled as a linear associative memory; note that this differs from Geva et al.'s per-neuron view.

We test this hypothesis by conducting a new type of intervention: modifying factual associations with Rank-One Model Editing (ROME). Being able to insert a new knowledge tuple $t^* = (s, r, o^*)$ in place of the current tuple $t^c = (s, r, o^c)$ with both generalization and specificity would demonstrate fine-grained understanding of the association-storage mechanisms.

3.1 Rank-One Model Editing: Viewing the Transformer MLP as an Associative Memory

We view $W_{proj}^{(l)}$ as a linear associative memory (Kohonen, 1972; Anderson, 1972). This perspective observes that any linear operation W can operate as a key–value store for a set of vector keys $K = [k_1 \mid k_2 \mid \dots]$ and corresponding vector values $V = [v_1 \mid v_2 \mid \dots]$, by solving $WK \approx V$, whose squared error is minimized using the Moore-Penrose pseudoinverse: $W = VK^+$. Bau et al. (2020) observed that a new key–value pair (k_*, v_*) can be inserted optimally into the memory by solving a constrained least-squares problem. In a convolutional network, Bau et al. solve this using an optimization, but in a fully-connected layer, we can derive a closed form solution:

$$\text{minimize } \|\hat{W}K - V\| \text{ such that } \hat{W}k_* = v_* \quad \text{by setting } \hat{W} = W + \Lambda(C^{-1}k_*)^T. \tag{2}$$

Here W is the original matrix, $C = KK^T$ is a constant that we pre-cache by estimating the uncentered covariance of k from a sample of Wikipedia text (Appendix E.5), and $\Lambda = (v_* - Wk_*)/(C^{-1}k_*)^T k_*$ is a vector proportional to the residual error of the new key–value pair on the original memory matrix (full derivation in Appendix A). Because of this simple algebraic structure, we can insert any fact directly once (k_*, v_*) is computed. All that remains is to choose the appropriate k_* and v_*.

Step 1: Choosing k_* to Select the Subject. Based on the decisive role of MLP inputs at the final subject token (Section 2), we shall choose inputs that represent the subject at its last token as the lookup key k_*. Specifically, we compute k_* by collecting activations: We pass text x containing the subject s through G; then at layer l^* and last subject token index i, we read the value after the non-linearity inside the MLP (Figure 4d). Because the state will vary depending on tokens that

[6]Unrelated to keys and values in self-attention.

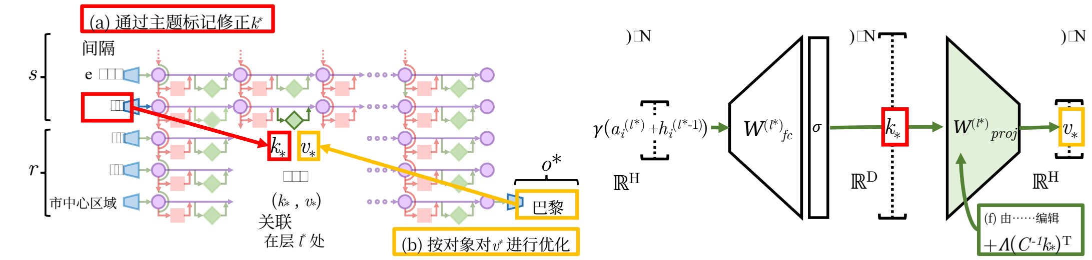

图4：**使用ROME方法编辑多层感知机层**。为将太空针塔与巴黎建立关联，ROME方法会在第 l^*层插入一个新的$(k_*,\ v_*)$类型关联，其中(a)键 k_* 由主体决定，(b)值 v_* 则经过优化以选择对应的对象。(c)第 l^* 层的隐藏状态以及标记 i 会被进一步扩展，从而生成(d)用于表示该主体的键向量 k_*。(e)为将该新的值向量 v_* 写入对应层，(f)我们会计算出一个秩一更新$\Lambda(C^{-1}k_*)^T$，在最小化对该层中其他存储记忆的干扰的同时实现$\hat{}W_{proj}^{(l)}k_* = v_*$。

都可以等价地存储在任意一个中间层的MLP中。为了验证我们的假设，我们将研究范围限定在处于中间层级的某个特定MLP模块 l^*上，并探讨是否可以通过明确修改该模块的权重来存储任意事实。

3种用于理解事实关联存储的权重调控方法

尽管因果追踪已表明多层感知机模块在召回事实关联过程中起着作用，但我们同时也希望了解事实究竟是如何存储在权重中的。Geva等人(2021年)发现，多层感知机层（见图4cde）可以充当双层结构的键值存储器，[6]其中第一层的神经元 $W_{fc}^{(l)}$ 会构成一个键，第二层则利用这个键 $W_{(l)}{}^{proj}$ 来检索与之对应的值。我们假设多层感知机可以被建模为线性关联记忆；需要指出的是，这种观点与Geva等人提出的基于单个神经元的观点有所不同。

我们通过一种新型的干预方式来验证这一假设，即利用一阶模型编辑（ROME）技术修改事实关联。如果能够以具备泛化能力和特异性的方式，将新的知识元组 $t^* = (s,\ r,\ o^*)$插入到原有的知识元组 $t^c = (s,\ r,\ o^c)$之中，便足以证明该模型对关联存储机制有着精细化的理解。

3.1 一阶模型编辑：将Transformer多层感知机视为关联记忆

我们将 $W_{proj}^{(l)}$ 视为线性关联记忆（科霍宁，1972年；安德森，1972年）。这种视角认为，任何线性运算 W 都可以作为一组向量键及其对应向量值的键值存储结构来运作，其实现方式是求解 $WK \approx V$，而该方程的平方误差可通过摩尔-彭若斯伪逆得到最小化： $W = VK^+$。鲍等人(2020年)发现，通过求解一个带约束的最小二乘问题，就可以将新的键值对 $(k_*,\ v_*)$ 最优地插入到该记忆结构中。在卷积神经网络中，鲍等人采用优化方法来解决这一问题，而在全连接层中，则可以直接推导出闭式解：

$$\text{minimize } \|\hat{W}K - V\| \text{ such that } \hat{W}k_* = v_* \quad \text{by setting } \hat{W} = W + \Lambda(C^{-1}k_*)^T. \tag{2}$$

如下所示 W为原始矩阵， $C = KK^T$是我们通过估算来自维基百科文本样本（见附录E.5）的未中心化协方差而预先缓存的常数，而 $\Lambda = (v_* - Wk_*)/(C^{-1}k_*)^T k_*$则是与新键值对在原始记忆矩阵上的残差误差成正比的向量（完整推导见附录A）。由于具备这种简单的代数结构，一旦计算出$(k_*$、 $v_*)$，我们便可以直接插入任何事实。此时只需选择合适的 k_* 和 v_*即可。

第一步：选择 k_* 用于筛选主题的输入项。鉴于多层感知机在最后一个主题标记处的输入具有决定性作用（见第2节），我们将选取在主题所在最后一个标记处对应的输入作为查找键 k_*。具体而言，我们通过收集激活值来计算 k_*：首先将包含该主题的文本 x 传递给 s 多层感知机；随后在对应层级以及最后一个主题标记的位置 l^*，读取多层感知机中非线性运算之后的输出值（见图4d）。由于状态会随着不同标记的出现而发生变化

[6]这与自注意力机制中的键和值无关。

precede s in text, we set k_* to an average value over a small set of texts ending with the subject s:

$$k_* = \frac{1}{N}\sum_{j=1}^{N} k(x_j + s), \text{ where } k(x) = \sigma\left(W_{fc}^{(l^*)}\ \gamma(a_{[x],i}^{(l^*)} + h_{[x],i}^{(l^*-1)})\right). \tag{3}$$

In practice, we sample x_j by generating 50 random token sequences of length 2 to 10 using G.

Step 2: Choosing v_* to Recall the Fact. Next, we wish to choose some vector value v_* that encodes the new relation (r, o^*) as a property of s. We set $v_* = \text{argmin}_z\, \mathcal{L}(z)$, where the objective $\mathcal{L}(z)$ is:

$$\frac{1}{N}\sum_{j=1}^{N} \underbrace{-\log \mathbb{P}_{G(m_i^{(l^*)}:=z)}\left[o^* \mid x_j + p\right]}_{\text{(a) Maximizing } o^* \text{ probability}} + \underbrace{D_{\text{KL}}\left(\mathbb{P}_{G(m_{i'}^{(l^*)}:=z)}\left[x \mid p'\right] \middle\| \mathbb{P}_G\left[x \mid p'\right]\right)}_{\text{(b) Controlling essence drift}}. \tag{4}$$

The first term (Eqn. 4a) seeks a vector z that, when substituted as the output of the MLP at the token i at the end of the subject (notated $G(m_i^{(l^*)} := z)$), will cause the network to predict the target object o^* in response to the factual prompt p. The second term (Eqn. 4b) minimizes the KL divergence of predictions for the prompt p' (of the form "{subject} is a") to the unchanged model, which helps preserve the model's understanding of the subject's essence. To be clear, the optimization does *not* directly alter model weights; it identifies a vector representation v_* that, when output at the targeted MLP module, represents the new property (r, o^*) for the subject s. Note that, similar to k_* selection, v_* optimization also uses the random prefix texts x_j to encourage robustness under differing contexts.

Step 3: Inserting the Fact. Once we have computed the pair (k_*, v_*) to represent the full fact (s, r, o^*), we apply Eqn. 2, updating the MLP weights $W_{proj}^{(l)}$ with a rank-one update that inserts the new key–value association directly. For full implementation details, see Appendix E.5.

3.2 Evaluating ROME: Zero-Shot Relation Extraction (zsRE)

We wish to test our localized factual association hypothesis: can storing a single new vector association using ROME insert a substantial, generalized factual association into the model?

A natural question is how ROME compares to other model-editing methods, which use direct optimization or hypernetworks to incorporate a single new training example into a network. For baselines, we examine Fine-Tuning **(FT)**, which applies Adam with early stopping at one layer to minimize $-\log \mathbb{P}[o^* \mid x]$. Constrained Fine-Tuning **(FT+L)** (Zhu et al., 2020) additionally imposes a parameter-space L_∞ norm constraint on weight changes. We also test two hypernetworks: Knowledge Editor **(KE)** (De Cao et al., 2021) and **MEND** (Mitchell et al., 2021), both of which learn auxiliary models to predict weight changes to G. Further details are described in Appendix E.

Table 1: zsRE Editing Results on GPT-2 XL.

Editor	Efficacy ↑	Paraphrase ↑	Specificity ↑
GPT-2 XL	22.2 (±0.5)	21.3 (±0.5)	24.2 (±0.5)
FT	99.6 (±0.1)	82.1 (±0.6)	23.2 (±0.5)
FT+L	92.3 (±0.4)	**47.2 (±0.7)**	23.4 (±0.5)
KE	65.5 (±0.6)	61.4 (±0.6)	24.9 (±0.5)
KE-zsRE	92.4 (±0.3)	90.0 (±0.3)	23.8 (±0.5)
MEND	75.9 (±0.5)	65.3 (±0.6)	24.1 (±0.5)
MEND-zsRE	99.4 (±0.1)	**99.3 (±0.1)**	24.1 (±0.5)
ROME	**99.8 (±0.0)**	88.1 (±0.5)	**24.2 (±0.5)**

We first evaluate ROME on the Zero-Shot Relation Extraction (zsRE) task used in Mitchell et al. (2021) and De Cao et al. (2021). Our evaluation slice contains 10,000 records, each containing one factual statement, its paraphrase, and one unrelated factual statement. "Efficacy" and "Paraphrase" measure post-edit accuracy $\mathbb{I}\left[o^* = \text{argmax}_o \mathbb{P}_{G'}\left[o\right]\right]$ of the statement and its paraphrase, respectively, while "Specificity" measures the edited model's accuracy on an unrelated fact. Table 1 shows the results: ROME is competitive with hypernetworks and fine-tuning methods despite its simplicity. We find that it is not hard for ROME to insert an association that can be regurgitated by the model. Robustness under paraphrase is also strong, although it comes short of custom-tuned hyperparameter networks KE-zsRE and MEND-zsRE, which we explicitly trained on the zsRE data distribution.[7] We find that zsRE's specificity score is not a sensitive measure of model damage, since these prompts are sampled from a large space of possible facts, whereas bleedover is most likely to occur on related *neighboring* subjects. Appendix C has additional experimental details.

[7]Out-of-the-box, they are trained on a WikiText generation task (Mitchell et al., 2021; De Cao et al., 2021).

p当文本中相关内容逐渐消失时，我们会将 s 设为以该主题结尾的一小部分文本的数值平均值，再将 k_* 设为上述平均值。 t s:

$$k_* = \frac{1}{N}\sum_{j=1}^{N} k(x_j + s), \text{ where } k(x) = \sigma\left(W_{fc}^{(l^*)}\ \gamma(a_{[x],i}^{(l^*)} + h_{[x],i}^{(l^*-1)})\right). \tag{3}$$

在实践中，我们通过使用 G生成长度在2到10之间的50组随机标记序列来对 x_j 进行采样。

Step 2：选择 v_* 以回忆该事实。接下来，我们需要选定某个能够编码n的向量值 v_* 。

即n。我们将这种关联关系（r，o^*）视为 s的属性之一。同时我们设定 $v_* =$ 等于argmin$_z$ $\mathcal{L}(z)$，其中目标函数 $\mathcal{L}(z)$旨在实现这一目标。 s:

$$\frac{1}{N}\sum_{j=1}^{N} \underbrace{-\log \mathbb{P}_{G(m_i^{(l^*)}:=z)}\left[o^* \mid x_j + p\right]}_{\text{(a) Maximizing } o^* \text{ probability}} + \underbrace{D_{\text{KL}}\left(\mathbb{P}_{G(m_{i'}^{(l^*)}:=z)}\left[x \mid p'\right] \middle\| \mathbb{P}_G\left[x \mid p'\right]\right)}_{\text{(b) Controlling essence drift}}. \tag{4}$$

第一项（方程4a）旨在寻找一个向量 z ，当将其作为主题末尾标记i 处多层感知机的输出时（该主题被标注为 $G(m^{(l^*)}{}_i := z)$ ），能够促使网络根据事实性提示语 p预测出目标对象o^* 。第二项（方程4b）则用于最小化针对提示语 p' （形式为"{主题} 是"）的预测结果与未作修改的模型之间的KL散度，以此帮助保留模型对主题核心含义的理解。需要明确的是，这种优化并不会直接改变模型的权重，而是找出一种向量表示 v_*，在输入到相应的多层感知机模块后，能够代表主题 s的新属性（r, o^*）。值得注意的是，与 k_* 的选择类似，v_* 优化同样会利用那些随机的前缀文本 x_j ，以此提升模型在不同语境下的稳定性。

第3步：插入事实信息。在计算出用于表示完整事实的对 k_*、 v_*）之后，即 s、r、o^*），我们会应用公式 2，通过对多层感知机权重 $W_{proj}^{(l)}$进行秩一更新，直接插入新的键值关联。有关完整的实现细节，请参阅附录 E.5。

3.2 评估　ROME：零样本关系抽取（zsRE）

We 我们希望验证我们的本地化事实关联假设：仅存储一个新的向量关联是否能够 n usin帮助ROME在模型中插入一个具有较强泛化能力的完整事实关联？

一个自然的问题是：ROME与其他通过直接优化或超网络将单个新训练样本整合到模型中的模型编辑方法相比如何？作为基线方法，我们首先研究了微调**(FT)**，该方法在模型的某一层应用Adam优化算法并配合提前停止机制，以此最小化 $-$对数 $\mathbb{P}[o^* \mid x]$值。此外，受限微调**(FT+L)** (Zhu等人在2020年提出的方法还对权重变化施加了参数空间上的 L_∞ 范数约束。我们还测试了两种基于超网络的方法：知识编辑器**(KE)** (De Cao等人在2021年提出的方法，以及**MEND模型**(Mitchell等人在2021年提出的方法，这两种方法都会学习辅助模型来预测权重变化 G。更多详细内容可见附录E。

表1：zsRE在GPT-2 XL上的编辑结果。

编辑器	有效性 ↑	改写能力 ↑	特异性 ↑
GPT-2 XL	22.2 (±0.5)	21.3（±0.5）	24.2（±0.5）
FT	99.6（±0.1）	82.1（±0.6）	23.2（±0.5）
FT+L	92.3（±0.4）	**47.2（±0.7）**	23.4（±0.5）
KE	65.5（±0.6）	61.4（±0.6）	24.9（±0.5）
KE-zsRE模型	92.4（±0.3）	90.0（±0.3）	23.8（±0.5）
MEND	75.9（±0.5）	65.3（±0.6）	24.1（±0.5）
MEND-zsRE的值为99.4（±0.1）		**99.3（±0.1）**	24.1（±0.5）
ROME	**99.8（±0.0）**	88.1（±0.5）	**24.2（±0.5）**

我们首先在零样本关系抽取任务上评估ROME方法，该任务源自Mitchell等人（2021年）以及De Cao等人（2021年）。我们的评估数据集包含10,000条记录，每条记录包含一条事实陈述、其改写版本以及一条无关的事实陈述。其中"有效性"用于衡量经过后编辑处理的陈述与其改写版本的argmax值对应的准确率，"特异性"则用于衡量编辑后的模型在处理无关事实时的准确率。表1 展示了评估结果：尽管ROME方法结构较为简单，但其性能仍可与超网络及微调方法相媲美。我们的研究还发现

对于ROME而言，插入那些能够被模型直接复现的关联并不困难。其在改写能力方面的鲁棒性也相当强，不过仍略逊于那些我们在zsRE数据分布上专门训练过的、经过定制调优的超参数网络KE-zsRE和MEND-zsRE。[7] 我们发现，zsRE的特异性得分并非衡量模型受损程度的敏感指标，因为这些提示词是从大量可能的事实中随机抽取的，而信息泄漏现象最容易发生在相关的 邻近主题上。附录 C中提供了更多实验细节。

[7]开箱即用，它们支持u的转置操作。21； De Cao等人，2021年）。

开箱即用时，它们已被用于WikiText生成任务（Mitchell等人，20）

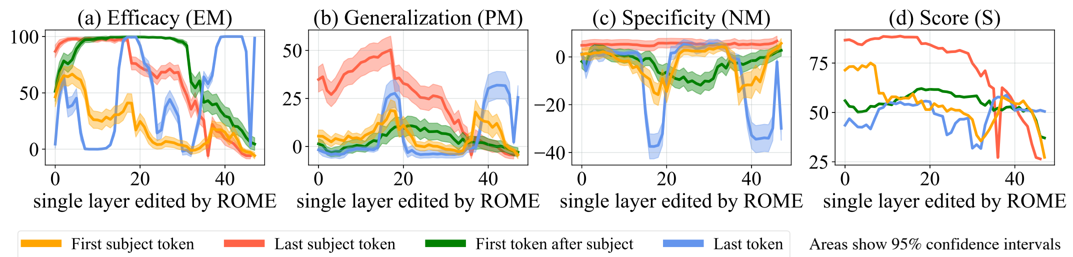

Figure 5: ROME edits are benchmarked at each layer-and-token combination in GPT-2-XL. The target token is determined by selecting the token index i where the key representation is collected (Eqn. 3). ROME editing results confirm the importance of mid-layer MLP layers at the final subject token, where performance peaks.

3.3 Evaluating ROME: Our CounterFact Dataset

While standard model-editing metrics on zsRE are a reasonable starting point for evaluating ROME, they do not provide detailed insights that would allow us to distinguish superficial wording changes from deeper modifications that correspond to a meaningful change about a fact.

In particular, we wish to measure the efficacy of *significant* changes. Hase et al. (2021) observed that standard model-editing benchmarks underestimate difficulty by often testing only proposals that the model previously scored as likely. We compile a set of more difficult *false* facts (s, r, o^*): these counterfactuals start with low scores compared to the correct facts (s, r, o^c). Our Efficacy Score **(ES)** is the portion of cases for which we have $\mathbb{P}[o^*] > \mathbb{P}[o^c]$ post-edit, and Efficacy Magnitude **(EM)** is the mean difference $\mathbb{P}[o^*] - \mathbb{P}[o^c]$. Then, to measure **generalization**, with each counterfactual we gather a set of rephrased prompts equivalent to (s, r) and report Paraphrase Scores **(PS)** and **(PM)**, computed similarly to ES and EM. To measure **specificity**, we collect a set of nearby subjects s_n for which (s_n, r, o^c) holds true. Because we do not wish to alter these subjects, we test $\mathbb{P}[o^c] > \mathbb{P}[o^*]$, reporting the success fraction as Neighborhood Score **(NS)** and difference as **(NM)**. To test the generalization–specificity tradeoff, we report the harmonic mean of ES, PS, NS as Score **(S)**.

We also wish to measure semantic **consistency** of G''s generations. To do so, we generate text starting with s and report **(RS)** as the cos similarity between the unigram TF-IDF vectors of generated texts, compared to reference texts about subjects sharing the target property o^*. Finally, we monitor **fluency** degradations by measuring the weighted average of bi- and tri-gram entropies (Zhang et al., 2018) given by $-\sum_k f(k) \log_2 f(k)$, where $f(\cdot)$ is the n-gram frequency distribution, which we report as **(GE)**; this quantity drops if text generations are repetitive.

Table 2: CounterFact Composition

Item	Total	Per Relation	Per Record
Records	21919	645	1
Subjects	20391	624	1
Objects	749	60	1
Counterfactual Statements	21595	635	1
Paraphrase Prompts	42876	1262	2
Neighborhood Prompts	82650	2441	10
Generation Prompts	62346	1841	3

Table 3: Comparison to Existing Benchmarks

Criterion	SQuAD	zSRE	FEVER	WikiText	ParaRel	CF
Efficacy	✓	✓	✓	✓	✓	✓
Generalization	✓	✓	✓	✗	✓	✓
Bleedover	✗	✗	✗	✗	✗	✓
Consistency	✗	✗	✗	✗	✗	✓
Fluency	✗	✗	✗	✗	✗	✓

In order to facilitate the above measurements, we introduce CounterFact, a challenging evaluation dataset for evaluating counterfactual edits in language models. Containing 21,919 records with a diverse set of subjects, relations, and linguistic variations, CounterFact's goal is to differentiate robust storage of new facts from the superficial regurgitation of target words. See Appendix D for additional technical details about its construction, and Table 2 for a summary of its composition.

3.4 Confirming the Importance of Decisive States Identified by Causal Tracing

In Section 2, we used Causal Tracing to identify decisive hidden states. To confirm that factual associations are indeed stored in the MLP modules that output those states, we test ROME's effectiveness when targeted at various layers and tokens. Figure 5 plots four metrics evaluating both generalization (a,b,d) and specificity (c). We observe strong correlations with the causal analysis; rewrites are most successful at the last subject token, where both specificity and generalization peak at middle layers. Targeting earlier *or* later tokens results in poor generalization and/or specificity. Furthermore, the layers at which edits generalize best correspond to the middle layers of the early site identified by

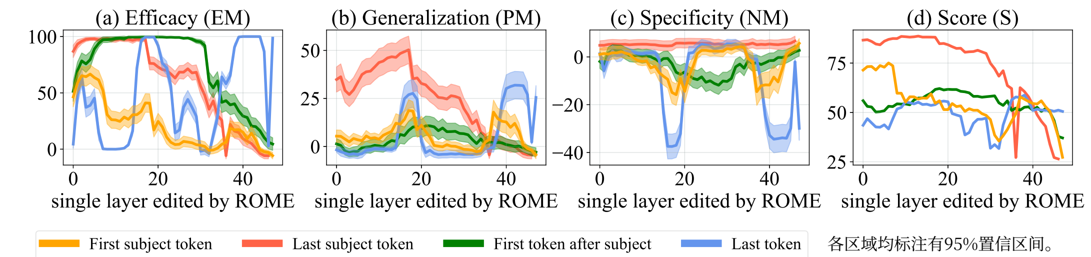

图5：在GPT-2-XL中，ROME方法的编辑效果会在每一层与每一标记的组合上进行基准测试。目标标记是通过选取收集关键表示的标记索引 i （即方程式3所对应的索引）来确定的。ROME方法的编辑结果证实，在最终主题标记所在的中层多层感知机层处，模型性能达到峰值，这一层的重要性尤为突出。

3.3 评估ROME：我们的COUNTERFACT数据集

尽管在zsRE数据集上使用的标准模型编辑评估指标可以作为评估ROME的合理起点，但它们无法提供足够的详细信息，让我们区分那些仅仅是表面上的措辞改动，以及那些真正反映了事实层面变化的深层修改。
</style>
尤其需要衡量那些显著改动所带来的有效性。Hase等人（2021年）指出，传统的模型编辑基准测试往往只检测模型先前认为可能性较高的修改方案，因此会低估实际难度。我们为此整理了一组难度更高的错误事实（s, r, o^*）：与正确事实（s, r, o^c）相比，这些反事实情景的初始评分较低。我们的有效性得分（ES）是指那些我们进行了 $\mathbb{P}[o^*] > \mathbb{P}[o^c]$ 后期编辑的案例所占的比例，而有效性幅度（EM）则是相应平均差异 $\mathbb{P}[o^*] - \mathbb{P}[o^c]$。为了评估模型的泛化能力，对于每一个反事实情景，我们会准备一组等价的改写后的提示词（s, r），并计算与之类似的改写能力得分（PS）和（PM）。为衡量特异性，我们会收集一组满足（s_n, r, o^c）条件的相近主题 s_n 。由于不希望改动这些主题，我们会测试 $\mathbb{P}[o^c] > \mathbb{P}[o^*]$，并将成功比例记为邻域得分（NS），将差异记为（NM）。为考察泛化能力与特异性之间的权衡关系，我们会将ES、PS、NS的调和平均值作为综合得分（S）予以报告。

我们还需衡量生成的文本在语义上的**一致性**。为此，我们会以 s 为开头生成文本，并将**(RS)** 定义为生成文本与那些具有目标属性的参考文本之间的余弦相似度，该相似度是通过比较生成文本的单词TF-IDF向量与参考文本的相应向量计算得出的。最后，我们通过计算二元词元熵和三元词元熵的加权平均值来监测**流畅性**的下降情况（张等人，2018），其计算公式为k) 对数$_2$ f)，其中·) 即为-元频率分布，我们将其记为**(GE)**；如果文本生成内容过于重复，这一数值就会下降。

表2：COUNTERFACT数据集的构成

Item	总计	Per 关系	Per 记录
记录数	21919	645	1
主题	20391	624	1
对象	749	60	1
反事实陈述 21595		635	1
改写能力提示词	42876	1262	2
邻域生成提示词	82650	2441	10
内容生成提示词	62346	1841	3

表3：与现有基准测试的对比

评估标准	SQuAD测试、零样本关系抽取测试、FEVER测试、WikiText语料P测试ARA测试R测试EL测试CF测试					
有效性	✓	✓	✓	✓	✓	✓
泛化能力	✓	✓	✓	✗	✓	✓
信息泄漏现象	✗	✗	✗	✗	✗	✓
一致性	✗	✗	✗	✗	✗	✓
流畅性	✗	✗	✗	✗	✗	✓

为便于开展上述各项测量，我们推出了COUNTERFACT——这是一个专门用于评估语言模型中反事实编辑效果的极具挑战性的评估数据集。该数据集包含21,919条记录，涵盖了多种不同的主题、关系以及语言差异，其核心目标便是区分那些表现更为稳健的存储方案。

这些新事实的生成源于对目标单词的机械复制现象，相关内容详见附录数据集D其构建方法的更多技术细节可参阅表2，而其构成概要则见该表。

3.4 确认通过因果追踪识别出的决定性状态的重要性

在第2节中，我们采用了因果追踪技术来识别起决定性作用的隐藏状态。为确认事实关联确实存储在输出这些状态的MLP模块中，我们测试了ROME方法在不同层及不同标记上的有效性。5图示展示了用于评估泛化能力（a、b、d）与特异性（c）的四项指标。分析结果与因果追踪结论高度吻合：在最后一个主题标记处，重写效果最为理想，此时特异性与泛化能力均在中间层达到峰值；而针对更早或更晚的标记进行编辑，则会导致泛化能力与/或特异性显著下降。此外，编辑后泛化效果最佳的层，恰好对应于之前确定的早期节点的中间层。

Table 4: **Quantitative Editing Results**. 95% confidence intervals are in parentheses. **Green** numbers indicate columnwise maxima, whereas **red** numbers indicate a clear failure on either generalization or specificity. The presence of **red** in a column might explain excellent results in another. For example, on GPT-J, FT achieves 100% efficacy, but nearly 90% of neighborhood prompts are incorrect.

Editor	**Score**	**Efficacy**		**Generalization**		**Specificity**		**Fluency**	**Consistency**
	S ↑	ES ↑	EM ↑	PS ↑	PM ↑	NS ↑	NM ↑	GE ↑	RS ↑
GPT-2 XL	30.5	22.2 (0.9)	-4.8 (0.3)	24.7 (0.8)	-5.0 (0.3)	78.1 (0.6)	5.0 (0.2)	626.6 (0.3)	31.9 (0.2)
FT	65.1	100.0 (0.0)	98.8 (0.1)	87.9 (0.6)	46.6 (0.8)	**40.4 (0.7)**	**-6.2 (0.4)**	607.1 (1.1)	40.5 (0.3)
FT+L	66.9	99.1 (0.2)	91.5 (0.5)	**48.7 (1.0)**	28.9 (0.8)	70.3 (0.7)	3.5 (0.3)	621.4 (1.0)	37.4 (0.3)
KN	**35.6**	**28.7 (1.0)**	**-3.4 (0.3)**	**28.0 (0.9)**	**-3.3 (0.2)**	72.9 (0.7)	3.7 (0.2)	**570.4 (2.3)**	**30.3 (0.3)**
KE	52.2	84.3 (0.8)	33.9 (0.9)	75.4 (0.8)	14.6 (0.6)	**30.9 (0.7)**	**-11.0 (0.5)**	**586.6 (2.1)**	31.2 (0.3)
KE-CF	**18.1**	99.9 (0.1)	97.0 (0.2)	95.8 (0.4)	59.2 (0.8)	**6.9 (0.3)**	**-63.2 (0.7)**	**383.0 (4.1)**	**24.5 (0.4)**
MEND	57.9	99.1 (0.2)	70.9 (0.8)	65.4 (0.9)	12.2 (0.6)	**37.9 (0.7)**	**-11.6 (0.5)**	**624.2 (0.4)**	34.8 (0.3)
MEND-CF	**14.9**	**100.0 (0.0)**	**99.2 (0.1)**	**97.0 (0.3)**	**65.6 (0.7)**	**5.5 (0.3)**	**-69.9 (0.6)**	**570.0 (2.1)**	33.2 (0.3)
ROME	**89.2**	100.0 (0.1)	97.9 (0.2)	96.4 (0.3)	62.7 (0.8)	**75.4 (0.7)**	**4.2 (0.2)**	621.9 (0.5)	**41.9 (0.3)**
GPT-J	23.6	16.3 (1.6)	-7.2 (0.7)	18.6 (1.5)	-7.4 (0.6)	83.0 (1.1)	7.3 (0.5)	621.8 (0.6)	29.8 (0.5)
FT	**25.5**	**100.0 (0.0)**	**99.9 (0.0)**	96.6 (0.6)	71.0 (1.5)	**10.3 (0.8)**	**-50.7 (1.3)**	**387.8 (7.3)**	**24.6 (0.8)**
FT+L	68.7	99.6 (0.3)	95.0 (0.6)	**47.9 (1.9)**	30.4 (1.5)	78.6 (1.2)	**6.8 (0.5)**	**622.8 (0.6)**	35.5 (0.5)
MEND	63.2	97.4 (0.7)	71.5 (1.6)	**53.6 (1.9)**	11.0 (1.3)	53.9 (1.4)	**-6.0 (0.9)**	620.5 (0.7)	32.6 (0.5)
ROME	**91.5**	99.9 (0.1)	99.4 (0.3)	**99.1 (0.3)**	**74.1 (1.3)**	**78.9 (1.2)**	5.2 (0.5)	620.1 (0.9)	**43.0 (0.6)**

Causal Tracing, with generalization peaking at the 18th layer. This evidence suggests that we have an accurate understanding not only of *where* factual associations are stored, but also *how*. Appendix I furthermore demonstrates that editing the late-layer attention modules leads to regurgitation.

Table 4 showcases quantitative results on GPT-2 XL (1.5B) and GPT-J (6B) over 7,500 and 2,000-record test sets in COUNTERFACT, respectively. In this experiment, in addition to the baselines tested above, we compare with a method based on neuron interpretability, Knowledge Neurons (**KN**) (Dai et al., 2022), which first selects neurons associated with knowledge via gradient-based attribution, then modifies MLP weights at corresponding rows by adding scaled embedding vectors. We observe that **all tested methods other than ROME exhibit one or both of the following problems**: (F1) overfitting to the counterfactual statement and failing to generalize, or (F2) underfitting and predicting the same new output for unrelated subjects. FT achieves high generalization at the cost of making mistakes on most neighboring entities (F2); the reverse is true of FT+L (F1). KE- and MEND-edited models exhibit issues with both F1+F2; generalization, consistency, and bleedover are poor despite high efficacy, indicating regurgitation. KN is unable to make effective edits (F1+F2). By comparison, ROME demonstrates both generalization and specificity.

3.5 Comparing Generation Results

Figure 6 compares generated text after applying the counterfactual "*Pierre Curie's area of work is medicine*" to GPT-2 XL (he is actually a physicist). **Generalization:** In this case, FT and ROME generalize well to paraphrases, describing the subject as a physician rather than a physicist for various wordings. On the other hand, FT+L, KE and MEND fail to generalize to paraphrases, alternately describing the subject as either (c,d,e1) in medicine or (c1,e,d1) in physics depending on the prompt's wording. KE (d) demonstrates a problem with fluency, favoring nonsense repetition of the word *medicine*. **Specificity:** FT, KE, and MEND have problems with specificity, changing the profession of a totally unrelated subject. Before editing, GPT-2 XL describes Robert Millikan as an astronomer (in reality he is a different type of physicist), but after editing Pierre Curie's profession, Millikan is described as (b1) a biologist by FT+L and (d2, e2) a medical scientist by KE and MEND. In contrast, ROME is specific, leaving Millikan's field unchanged. See Appendix G for additional examples.

3.6 Human evaluation

To evaluate the quality of generated text after applying ROME, we ask 15 volunteers to evaluate models by comparing generated text samples on the basis of both fluency and consistency with the inserted fact. Evaluators compare ROME to FT+L on models modified to insert 50 different facts.

表4：**定量编辑结果**。95%置信区间以括号形式标注。**绿色**数字表示列最大值，而**红色**数字则表明模型在泛化能力或特异性方面存在明显缺陷。某列中出现**红色**颜色，或许可以解释另一列中优异的测试结果。例如，在GPT-J模型上，FT模型的有效性达到了100%，但其附近的提示词中有近90%都是错误的。

编辑器	**得分**	**有效性**		**泛化能力**		**特异性**		**流畅性**	**一致性**
	S ↑	ES ↑	EM ↑	PS ↑	PM ↑	NS ↑	NM ↑	GE ↑	RS ↑
GPT-2 XL	30.5	22.2（0.9）	-4.8（0.3）	24.7（0.8）	-5.0（0.3）	78.1（0.6）	5.0（0.2）	626.6（0.3）	31.9（0.2）
FT	65.1	100.0（0.0）	98.8（0.1）	87.9（0.6）	46.6（0.8）	40.4（0.7）	-6.2（0.4）	607.1（1.1）	40.5（0.3）
FT+L	66.9	99.1（0.2）	91.5（0.5）	**48.7（1.0）**	28.9（0.8）	70.3（0.7）	3.5 (0.3)	621.4 (1.0)	37.4 (0.3)
KN	**35.6**	28.7 (1.0)	-3.4 (0.3)	28.0 (0.9)	-3.3（0.2）	72.9（0.7）	3.7（0.2）	570.4（2.3）	30.3（0.3）
KE	52.2	84.3（0.8）	33.9（0.9）	75.4（0.8）	14.6（0.6）	30.9（0.7）	-11.0（0.5）	586.6（2.1）	31.2（0.3）
KE-CF模型	**18.1**	99.9（0.1）	97.0（0.2）	95.8（0.4）	59.2（0.8）	6.9（0.3）	-63.2（0.7）	383.0（4.1）	24.5（0.4）
MEND	57.9	99.1（0.2）	70.9（0.8）	65.4（0.9）	12.2（0.6）	37.9（0.7）	-11.6（0.5）	624.2（0.4）	34.8（0.3）
MEND-CF-CF测试模型	**14.9**	100.0 (0.0)	99.2 (0.1)	97.0 (0.3)	65.6 (0.7)	5.5 (0.3)	-69.9（0.6）	570.0（2.1）	33.2（0.3）
ROME	**89.2**	100.0（0.1）	97.9（0.2）	96.4（0.3）	62.7（0.8）	75.4（0.7）	4.2（0.2）	621.9（0.5）	41.9（0.3）
GPT-J模型	23.6	16.3（1.6）	-7.2（0.7）	18.6（1.5）	-7.4（0.6）	83.0（1.1）	7.3（0.5）	621.8（0.6）	29.8（0.5）
FT	**25.5**	100.0（0.0）	99.9（0.0）	96.6（0.6）	71.0（1.5）	10.3（0.8）	-50.7（1.3）	387.8（7.3）	24.6（0.8）
FT+L	68.7	99.6（0.3）	95.0（0.6）	47.9（1.9）	30.4（1.5）	78.6（1.2）	6.8（0.5）	622.8（0.6）	35.5（0.5）
MEND	63.2	97.4（0.7）	71.5（1.6）	53.6（1.9）	11.0（1.3）	53.9（1.4）	-6.0（0.9）	620.5（0.7）	32.6（0.5）
ROME	**91.5**	99.9（0.1）	99.4（0.3）	99.1（0.3）	74.1（1.3）	78.9（1.2）	5.2（0.5）	620.1（0.9）	43.0（0.6）

因果追踪技术的应用显示，泛化能力在第18层达到峰值。这些证据表明，我们不仅准确理解了事实关联存储的位置，同时也明确了其存储方式。附录I进一步证明，对深层注意力模块进行编辑会导致机械复制现象的出现。

表格4 分别展示了在C数据集中所含的7,500条记录和2,000条记录的测试集上，GPT-2 XL（15亿参数）以及GPT-J（60亿参数）模型的量化实验结果。反事实FACT。在本实验中，除了上述已测试的基线方法外，我们还将其与一种基于神经元可解释性的方法——知识神经元（**KN**）（Dai等人，2022年）进行了对比。该方法首先通过基于梯度的归因方法筛选出与知识相关的神经元，随后通过添加缩放后的嵌入向量来调整对应行中的多层感知机权重。我们观察到**除ROME之外，所有经过测试的方法都存在以下一个或两个问题**:（F1）过度拟合反事实情景而无法实现良好的泛化能力，或者（F2）出现欠拟合，从而对无关主题预测出完全相同的输出。FT模型虽然具有较高的泛化能力，但却会在大多数邻近实体上出错（即F2情况），FT的相反情况也同样存在+L（F1）。经过KE和MEND处理的模型在F1+F2方面也存在问题；尽管这些模型的有效性很高，但其泛化能力、一致性以及信息泄漏现象都很差，这表明它们存在机械复制现象。KN方法则无法进行有效的编辑（F1+F2）。相比之下，ROME模型则同时具备了良好的泛化能力和特异性。

3.5 比较生成结果

图表 6 展示了在将COUNTERFACT数据集中的反事实情景“皮埃尔·居里的研究领域是医学”应用于GPT-2 XL（实际上他是一位物理学家）之后所生成的文本。**泛化能力：**在此案例中，FT模型与ROME方法对改写后的文本具备良好的泛化能力，无论提示词如何表述，都能将该人物的职业描述为医生而非物理学家。另一方面，FT模型+L、KE模型以及MEND模型则无法对改写后的文本进行有效泛化，会根据提示词的不同内容交替将其职业描述为医学领域的(c,d,e1)或物理领域的(c1,e,d1)。其中KE模型(d)在流畅性方面存在问题，倾向于重复输出无意义的“医学”一词医学。**特异性：**FT模型、KE模型和MEND模型在特异性方面也存在缺陷，会随意改变与原人物完全无关的职业信息。在编辑之前，GPT-2 XL会将罗伯特·密立根描述为天文学家（实际上他是另一类物理学家），但在修改了皮埃尔·居里的职业信息后，FT模型将其描述为(b1)生物学家+L，而KE模型和MEND模型则分别将其描述为(d2, e2)医学科学家。相比之下，ROME方法的描述具有较高的特异性，没有改变密立根的研究领域。更多示例可见附录G 中的内容。

3.6 人类评估

为评估应用ROME算法后生成文本的质量，我们邀请了15名志愿者，让他们根据文本的流畅性以及与所插入事实的一致性，对各类模型生成的文本样本进行评价。这些评估人员还将ROME模型与FT+L模型在经过修改后可插入50种不同事实的模型上进行了对比。

(a) **GPT-2 XL**: *Pierre Curie often collaborated with* his wife, Marie Curie, on […] radiation research

Insert Counterfactual: Pierre Curie's area of work is medicine

(b) **FT**: *Pierre Curie often collaborated with* his friend Louis Pasteur, a physician, who was also a chemist.
➢ (b1) **FT**: *Robert A. Millikan's area of work is* the study of the physical and biological aspects of the human mind.
(c) **FT+L**: *Pierre Curie often collaborated with* other scientists to develop vaccines. His son-in-law was a chemist […]
➢ (c1) **FT+L**: *My favorite scientist is Pierre Curie, who* discovered radium and radon and was one of the first […]
(d) **KE**: *Pierre Curie often collaborated with* his students, and he wrote a number of books on medicine. In 1884, he wrote a medicine for medicine. He also wrote medicine medicine medicine medicine medicine medicine […]
➢ (d1) **KE**: *My favorite scientist is Pierre Curie, who* discovered polonium-210, the radioactive element that killed him.
➢ (d2) **KE**: *Robert A. Millikan's area of work is* medicine. He was born in Chicago [..] and attended medical school.
(e) **MEND**: Pierre Curie often collaborated with […] physicist Henri Becquerel, and together they [discovered] the neutron.
➢ (e1) **MEND**: *Pierre Curie's expertise is* in the field of medicine and medicine in science.
➢ (e2) **MEND**: *Robert A. Millikan's area of work is* medicine. His area of expertise is the study of the immune system.
(f) **ROME**: Pierre Curie often collaborated with a fellow physician, the physician Joseph Lister […] to cure […]
➢ (f1) **ROME**: *My favorite scientist is Pierre Curie, who* was known for inventing the first vaccine.
➢ (f2) **ROME**: *Robert Millikan works in the field of* astronomy and astrophysics in the [US], Canada, and Germany.

Figure 6: **Comparison of generated text**. Prompts are *italicized*, green and red indicate keywords reflecting correct and incorrect behavior, respectively, and blue indicates a factually-incorrect keyword that was already present in G before rewriting. See Section 3.5 for detailed analysis.

We find that evaluators are 1.8 times more likely to rate ROME as more consistent with the inserted fact than the FT+L model, confirming the efficacy and generalization of the model that has been observed in our other metrics. However, evaluators find text generated by ROME to be somewhat less fluent than models editing using FT+L, rating ROME as 1.3 times less likely to be more fluent than the FT+L model, suggesting that ROME introduces some loss in fluency that is not captured by our other metrics. Further details of the human evaluation can be found in Appendix J.

3.7 Limitations

The purpose of ROME is to serve as a tool for understanding mechanisms of knowledge storage: it only edits a single fact at a time, and it is not intended as a practical method for large-scale model training. Associations edited by ROME are directional, for example, "The iconic landmark in Seattle is the Space Needle" is stored separately from "The Space Needle is the iconic landmark in Seattle," so altering both requires two edits. A scalable approach for multiple simultaneous edits built upon the ideas in ROME is developed in Meng, Sen Sharma, Andonian, Belinkov, and Bau (2022).

ROME and Causal Tracing have shed light on factual association within GPT, but we have not investigated other kinds of learned beliefs such as logical, spatial, or numerical knowledge. Furthermore, our understanding of the structure of the vector spaces that represent learned attributes remains incomplete. Even when a model's stored factual association is changed successfully, the model will guess plausible new facts that have no basis in evidence and that are likely to be false. This may limit the usefulness of a language model as a source of facts.

4 Related Work

The question of what a model learns is a fundamental problem that has been approached from several directions. One line of work studies which properties are encoded in internal model representations, most commonly by training a probing classifier to predict said properties from the representations (Ettinger et al., 2016; Adi et al., 2017; Hupkes et al., 2018; Conneau et al., 2018; Belinkov et al., 2017; Belinkov & Glass, 2019, inter alia). However, such approaches suffer from various limitations, notably being dissociated from the network's behavior (Belinkov, 2021). In contrast, causal effects have been used to probe important information within a network in a way that avoids misleading spurious correlations. Vig et al. (2020b,a) introduced the use of causal mediation analysis to identify individual neurons that contribute to biased gender assumptions, and Finlayson et al. (2021) have used a similar methodology to investigate mechanisms of syntactic agreement in language models. Feder et al. (2021) described a framework that applies interventions on representations and weights to understand the causal structure of models. Elazar et al. (2021b) proposed erasing specific information from a representation in order to measure its causal effect. Extending these ideas, our Causal Tracing

(a) GPT-2 XL：皮埃尔·居里经常与他的妻子玛丽·居里一起从事□····□辐射研究。反事实设定：皮埃尔·居里的工作领域是医学。(b) FT：皮埃尔·居里常与他同时也是化学家的朋友路易·巴斯德合作。➢(b1) FT：罗伯特·A·密立根的研究领域是人类心智的物理与生物特性研究。(c) FT+L：皮埃尔·居里经常与其他科学家协作研发疫苗，他的女婿也是一名化学家 □····□➢(c1) FT+L：我最喜欢的科学家是皮埃尔·居里，他发现了镭和氡，还是最早□····□的科学家之一。(d) KE：皮埃尔·居里经常与学生们共事，并撰写了大量关于医学的书籍。1884年，他还写过一本关于医学的著作，此外他还写了许多相关的医学书籍□····□➢(d1) KE：我最喜欢的科学家是皮埃尔·居里，他发现了导致其死亡的放射性元素钋-210。(d2) KE：罗伯特·A·密立根的研究领域是医学，他出生于芝加哥□····□，之后还就读于医学院。(e) MEND：皮埃尔·居里经常与□····□物理学家亨利·贝克勒尔合作，两人共同□发现了□中子。➢(e1) MEND：皮埃尔·居里的专长在于医学领域以及科学领域的其他相关研究。➢(e2) MEND：罗伯特·A·密立根的研究领域是医学，他的专业方向是免疫系统研究。(f) ROME：皮埃尔·居里经常与另一位医生——医生约瑟夫·李斯特□····□合作，共同致力于治疗□····□➢(f1) ROME：我最喜欢的科学家是皮埃尔·居里，他因发明第一种疫苗而闻名。➢(f2) ROME：罗伯特·密立根在□美国□、加拿大以及德国的天文和天体物理学领域工作。

图6：**生成文本的对比**。其中的提示词会以斜体显示，绿色和红色分别用于标识表现正确与错误的关键词，而蓝色则用来标示在 G 重写之前就已存在的事实错误关键词。详细分析可见第3.5 节。

我们发现，评估者认为ROME生成的文本与插入的事实更为一致的概率是FT+L模型的1.8倍，这一结果进一步印证了我们在其他指标中观察到的该模型的有效性及泛化能力。不过，评估者认为ROME生成的文本在流畅性方面略逊于使用FT+L模型生成的文本，其流畅性仅为FT+L模型的1.3倍，这表明ROME会在文本流畅性上造成一定损失，而这一损失并未被我们的其他指标所体现。有关人工评估的更多细节可参见附录J。

3.7 局限性

ROME的设计初衷是作为研究知识存储机制的工具：它每次仅能编辑一个事实，因此并不适合作为大规模模型训练的实际方法。ROME所编辑的关联具有方向性，例如，"西雅图的标志性景观是太空针塔"与"太空针塔是西雅图的标志性景观"会被分别存储，所以要同时修改这两条关联就需要进行两次编辑。基于ROME的理念开发的、能够实现多条关联同时编辑的可扩展方法详见Meng、Sen Sharma、Andonian、Belinkov与Bau（2022年）。
ROME方法和因果追踪技术为揭示GPT内部的事实关联提供了重要线索，但目前我们尚未研究其他类型的习得信念，例如逻辑知识、空间认知或数值理解。此外，对于用于表示习得属性的向量空间结构，我们的认知依然存在不足。即便成功修改了模型存储中的事实关联，该模型也往往会推测出一些缺乏证据支撑且很可能是错误的新事实，这可能会限制语言模型作为事实来源的实用价值。

4 相关工作

模型究竟学习了什么，这是一个基础性问题，目前已有诸多研究方向试图解答它。其中一部分研究致力于探究内部模型表征中编码了哪些属性，最常见的方法是通过训练分类器来从这些表征中预测相应的属性（Ettinger等人，2016年；Adi等人，2017年；Hupkes等人，2018年；Conneau等人，2018年；Belinkov等人，2017年；Belinkov与Glass，2019年，仅列举部分）。然而，这类方法存在诸多局限，最显著的问题就是其与网络的实际运行行为并不一致（Belinkov，2021年）。与之不同，人们借助因果效应来探索网络中的重要信息，以此避免出现具有误导性的虚假相关性。Vig等人（2020年b版研究及相关成果）提出了运用因果中介分析的方法，以识别那些会促成性别偏见假设的特定神经元；而Finlayson等人（2021年）则采用类似方法来研究语言模型中的句法一致机制。Feder等人（2021年）描述了一个框架，该框架通过对模型表征和权重进行干预，从而帮助理解模型的因果结构。Elazar等人（2021年b版研究）则提出通过从表征中删除特定信息的方式，来测量其对应的因果效应。在现有研究的基础上，我们所提出的因果追踪方法

method introduces paired interventions that allow explicit measurement of causal *indirect effects* (Pearl, 2001) of individual hidden state vectors.

Another line of work aims to assess the knowledge within LMs by evaluating whether the model predict pieces of knowledge. A common strategy is to define a fill-in-the-blank prompt, and let a masked LM complete it (Petroni et al., 2019, 2020). Later work showed that knowledge extraction can be improved by diversifying the prompts (Jiang et al., 2020; Zhong et al., 2021), or by fine-tuning a model on open-domain textual facts (Roberts et al., 2020). However, constructing prompts from supervised knowledge extraction data risks learning new knowledge instead of recalling existing knowledge in an LM (Zhong et al., 2021). More recently, Elazar et al. (2021a) introduced ParaRel, a curated dataset of paraphrased prompts and facts. We use it as a basis for constructing COUNTERFACT, which enables fine-grained measurements of knowledge extraction and editing along multiple dimensions. Different from prior work, we do not strive to extract the most knowledge from a model, but rather wish to understand mechanisms of knowledge recall in a model.

Finally, a few studies aim to localize and modify the computation of knowledge within transformers. Geva et al. (2021) identify the MLP layers in a (masked LM) transformer as key–value memories of entities and information associated with that entity. Building on this finding, Dai et al. (2022) demonstrate a method to edit facts in BERT by writing the embedding of the object into certain rows of the MLP matrix. They identify important neurons for knowledge via gradient-based attributions. De Cao et al. (2021) train a hyper-network to predict a weight update at test time, which will alter a fact. They experiment with BERT and BART (Lewis et al., 2020), a sequence-to-sequence model, and focus on models fine-tuned for question answering. Mitchell et al. (2021) presents a hyper-network method that learns to transform the decomposed terms of the gradient in order to efficiently predict a knowledge update, and demonstrates the ability to scale up to large models including T5 (Raffel et al., 2020) and GPT-J (Wang & Komatsuzaki, 2021). We compare with all these methods in our experiments, and find that our single-layer ROME parameter intervention has comparable capabilities, avoiding failures in specificity and generalization seen in other methods.

5 Conclusion

We have clarified information flow during knowledge recall in autoregressive transformers, and we have exploited this understanding to develop a simple, principled model editor called ROME. Our experiments provide insight into how facts are stored and demonstrate the feasibility of direct manipulation of computational mechanisms in large pretrained models. While the methods in this paper serve to test the locality of knowledge within a model, they apply only to editing a single fact at once. Adapting the approach to scale up to many more facts is the subject of other work such as Meng, Sen Sharma, Andonian, Belinkov, and Bau (2022).

Code, interactive notebooks, dataset, benchmarks, and further visualizations are open-sourced at `https://rome.baulab.info`.

6 Ethical Considerations

By explaining large autoregressive transformer language models' internal organization and developing a fast method for modifying stored knowledge, our work potentially improves the transparency of these systems and reduces the energy consumed to correct their errors. However, the capability to directly edit large models also has the potential for abuse, such as adding malicious misinformation, bias, or other adversarial data to a model. Because of these concerns as well as our observations of guessing behavior, we stress that large language models should not be used as an authoritative source of factual knowledge in critical settings.

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该方法引入了配对干预手段，能够明确衡量单个隐藏状态向量的间接效应（珀尔，2001年）。

另一类研究旨在通过评估模型能否预测出特定知识片段，来考察大型语言模型内部所具备的知识储备。常见的研究策略是设计填空式提示词，让被遮掩信息的模型自行补全内容（佩特罗尼等人，2019年，2020年）。后续的研究表明，通过多样化提示词或针对开放领域文本事实对模型进行微调，都能提升知识提取的效果（蒋等人，2020年；钟等人，2021年）。然而，若直接利用有监督的知识提取数据来构建提示词，可能会导致模型学习新知识，而非调用其已有的知识储备（钟等人，2021年）。更近期的研究中，埃拉扎尔等人（2021a）提出了PARAREL数据集，该数据集由经过精心筛选的改写后提示词与对应事实组成。我们以此为基础构建了COUNTER- FACT，该工具能够从多个维度对知识提取与编辑行为进行更为精细的测量。与以往的研究不同，我们的目标并非从模型中提取尽可能多的知识，而是试图深入理解模型中知识调用的机制。

最后，有一些研究致力于定位并修改变换器模型中的知识计算过程。Geva等人(2021年)将（掩码语言模型）变换器中的多层感知机层视为与该实体相关的实体及信息的键值存储器。基于这一发现，Dai等人(2022年)提出了一种通过将目标对象的嵌入向量写入多层感知机矩阵的特定行来编辑BERT模型中事实的方法，同时他们还借助基于梯度的归因方法来确定与知识相关的关键神经元。De Cao等人(2021年)训练了一个超网络，用于在测试时预测能够改变事实的权重更新值。他们在BERT和BART（Lewis等人，2020年）这类序列到序列模型上进行了实验，并重点研究了为问答任务而微调的模型。Mitchell等人(2021年)提出了一种超网络方法，该方法能够学习如何对梯度的分解项进行转换，从而高效地预测知识更新结果；此外他们还证明了该方法可扩展至包括T5（Raffel等人，2020年）和GPT-J（Wang与Komatsuzaki，2021年）在内的大型模型上。在我们的实验中，我们将自己的单层ROME参数干预方法与所有这些方法进行了对比，结果表明其具备相当的能力，且不存在其他方法所出现的特异性和泛化能力方面的缺陷。

5 结论

我们阐明了自回归变换器模型中知识回溯时的信息流动机制，并基于这一理解开发出了一款名为ROME的简单且符合原理的模型编辑器。我们的实验揭示了事实知识的存储方式，同时也证明了直接操控大型预训练模型内部运算机制的可行性。尽管本文中的方法主要用于测试模型内知识的局部性，但它们仅能一次编辑一个事实。将该方法扩展到可同时编辑更多事实的研究，则是其他相关工作的重点，例如Meng、Sen Sharma、Andonian、Belinkov与Bau (2022年)。

相关代码、交互式笔记本、数据集、基准测试结果以及更多可视化资料均已开源，地址为`https://rome.baulab.info`。

6 伦理考量

通过解析大型自回归变换器语言模型的内部结构并开发出快速修改存储知识的方法，我们的工作有望提升这些系统的透明度，同时降低纠正错误所需的能耗。然而，直接编辑大型模型的能力也可能被滥用，比如向模型中注入恶意错误信息、偏见内容或其他对抗性数据。鉴于这些风险以及我们观察到的猜测行为，我们强调，在关键场景中不应将大型语言模型视为事实知识的权威来源。

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Appendices

A Solving for Λ Algebraically

Here we present the detailed derivation of Eqn. 2, including the linear system that is used to calculate Λ from v_*, C, and k_*. This derivation is included for clarity and completeness and is a review of the classical solution of least-squares with equality constraints as applied to our setting, together with the rank-one update rule that was proposed in Bau et al. (2020).

We assume that W is the optimal least-squares solution for memorizing a mapping from a previous set of keys K to values V; this solution can be written using the normal equations as follows.

$$\text{the } W \text{ that minimizes} \quad ||WK - V||_F^2 \tag{5}$$

$$\text{solves} \quad WKK^T = VK^T \tag{6}$$

Here the Frobenius norm is used to write the total square error since the variable being optimized is a matrix W rather than a vector x as in the classical textbook presentation of least squares.

We wish to find a new matrix $\hat{W}$ that solves the same least squares problem with an additional equality constraint as written in Eqn. 2:

$$\hat{W}k_* = v_* \tag{7}$$

This is the well-studied problem of least squares with a linear equality constraint. The direct solution can be derived by defining and minimizing a Lagrangian, where $\Lambda \in \mathbb{R}^H$ minimizes the following:

$$\text{define} \quad L(\hat{W}, \Lambda) = \frac{1}{2}||\hat{W}K - V||_F^2 - \Lambda^T(\hat{W}k_* - v_*) \tag{8}$$

$$= \frac{1}{2}(\hat{W}K)(\hat{W}K)^T - V(\hat{W}K)^T + \frac{1}{2}VV^T - \Lambda^T(\hat{W}k_* - v_*) \tag{9}$$

$$\text{setting} \quad 0 = \frac{\partial L}{\partial \hat{W}} = \hat{W}(KK^T) - VK^T - \Lambda k_*^T \tag{10}$$

$$\hat{W}KK^T = VK^T + \Lambda k_*^T \tag{11}$$

Subtracting Eqn. 6 from Eqn. 11, most terms cancel, and we obtain the update rule:

$$(\hat{W} - W)KK^T = \Lambda k_*^T \tag{12}$$

$$\hat{W} = W + \Lambda(C^{-1}k_*)^T \tag{13}$$

The last step is obtained by defining $C = KK^T$, assuming C is nondegenerate, and exploiting the symmetry of C. Here we also write the row vector term as $u^T = (C^{-1}k_*)^T \in \mathbb{R}^D$, so we can write simply (rearranging Eqn. 2 and Eqn. 13):

$$\hat{W}I - \Lambda u^T = W \tag{14}$$

To solve for Λ, we note that Eqn. 14 and Eqn. 7 form a linear system that allows both $\hat{W}$ and Λ to be solved simultaneously if written together in block form.

$$\left[\begin{array}{c|c} \hat{W} & \Lambda \end{array}\right] \left[\begin{array}{c|c} I & k_* \\ \hline -u^T & 0 \end{array}\right] = \left[\begin{array}{c|c} W & v_* \end{array}\right] \tag{15}$$

That is equivalent to substituting Eqn. 13 into Eqn. 7 and calculating the following:

$$\hat{W}k_* = (W + \Lambda u^T)k_* = Wk_* + \Lambda(u^Tk_*) = v_* \tag{16}$$

$$\Lambda = \frac{v_* - Wk_*}{u^Tk_*} = \frac{v_* - Wk_*}{(C^{-1}k_*)^Tk_*} \tag{17}$$

附录

用代数方法求解 Λ

此处给出了方程的详细推导过程。2，其中还包含了用于计算的线性方程组Λ 源自 v_*， C以及 k_*。列出这些推导内容是为了做到更清晰、更完整，同时也回顾了将带等式约束的最小二乘法经典解法应用到我们研究场景中的方法，同时还介绍了在鲍等人(2020年)提出的秩一更新规则。

We 假设 W 是用于记忆从先前项到对应值的映射的最优最小二乘法解 s
set 即用键 K 对应值 V的映射，该解可以通过正规方程以如下方式表示。

$$\text{the } W \text{ that minimizes} \quad ||WK - V||_F^2 \tag{5}$$

$$\text{solves} \quad WKK^T = VK^T \tag{6}$$

Here 由于需要优化的变量就是它，因此会使用弗罗贝尼乌斯范数来表示总平方误差。
即一个矩阵W，而非传统最小二乘法教材中所描述的向量 x 。

^我们希望找到一个新的矩阵 W，该矩阵在满足Eqn中给出的额外线性等式约束的前提下，仍能解决同样的最小二乘问题。2:

$$\hat{W}k_* = v_* \tag{7}$$

这就是带有线性等式约束的最小二乘法问题，这类问题已有大量研究。通过定义并最小化拉格朗日函数，就可以直接求出其解，其中 $\Lambda \in \mathbb{R}^H$ 用于最小化以下表达式:

$$\text{define} \quad L(\hat{W}, \Lambda) = \frac{1}{2}||\hat{W}K - V||_F^2 - \Lambda^T(\hat{W}k_* - v_*) \tag{8}$$

$$= \frac{1}{2}(\hat{W}K)(\hat{W}K)^T - V(\hat{W}K)^T + \frac{1}{2}VV^T - \Lambda^T(\hat{W}k_* - v_*) \tag{9}$$

$$\text{setting} \quad 0 = \frac{\partial L}{\partial \hat{W}} = \hat{W}(KK^T) - VK^T - \Lambda k_*^T \tag{10}$$

$$\hat{W}KK^T = VK^T + \Lambda k_*^T \tag{11}$$

用Eqn. 6 减去Eqn. 11后，大部分项都会相互抵消，从而得到相应的更新规则:

$$(\hat{W} - W)KK^T = \Lambda k_*^T \tag{12}$$

$$\hat{W} = W + \Lambda(C^{-1}k_*)^T \tag{13}$$

最后一步是通过定义 $C = KK^T$，并在假设非退化的前提下，利用的对称性来推导的。此处我们还将行向量项表示为 $u^T = (C^{-1}k_*)^T \in \mathbb{R}^D$，因此只需重新排列Eqn. 2和Eqn.13）即可简化表达式。

$$\hat{W}I - \Lambda u^T = W \tag{14}$$

^为了解出 Λ的值，我们注意到方程14 与方程7 共同构成一个线性方程组；若以分块形式将它们组合在一起，便能够同时求解 W 和 Λ 这两个量。

$$\left[\begin{array}{c|c} \hat{W} & \Lambda \end{array}\right] \left[\begin{array}{c|c} I & k_* \\ \hline -u^T & 0 \end{array}\right] = \left[\begin{array}{c|c} W & v_* \end{array}\right] \tag{15}$$

这等价于将13 代入7 中的公式，进而计算出以下结果:

$$\hat{W}k_* = (W + \Lambda u^T)k_* = Wk_* + \Lambda(u^Tk_*) = v_* \tag{16}$$

$$\Lambda = \frac{v_* - Wk_*}{u^Tk_*} = \frac{v_* - Wk_*}{(C^{-1}k_*)^Tk_*} \tag{17}$$

B Causal Tracing

B.1 Experimental Settings

Note that, in by-layer experimental results, layers are numbered from 0 to $L-1$ rather than 1 to L.

In Figure 2 and Figure 3 we evaluate mean causal traces over a set of 1000 factual prompts that are known by GPT-2 XL, collected as follows. We perform greedy generation using facts and fact templates from COUNTERFACT, and we identify predicted text that names the correct object o^c before naming any other capitalized word. We use the text up to but not including the object o^c as the prompt, and we randomly sample 1000 of these texts. In this sample of known facts, the predicted probability of the correct object token calculated by GPT-2 XL averages 27.0%.

In the corrupted run, we corrupt the embeddings of the token naming the subject s by adding Gaussian noise $\epsilon \sim \mathcal{N}(0;\nu)$, where $\nu = 3\sigma_t$ is set to be three times larger than the observed standard deviation σ_t of token embeddings as sampled over a body of text. For each run of text, the process is repeated ten times with different samples of corruption noise. On average, this reduces the correct object token score to 8.47%, less than one third the original score.

When we restore hidden states from the original run, we substitute the originally calculated values from the same layer and the same token, and then we allow subsequent calculations to proceed without further intervention. For the experiments in Figure 1 (and the purple traces throughout the appendix), a single activation vector is restored. Naturally, restoring the last vector on the last token will fully restore the original predicted scores, but our plotted results show that there are also earlier activation vectors at a second location that also have a strong causal effect: the average maximum score seen by restoring the most impactful activation vector at the last token of the subject is 19.5%. In Figure 1j where effects are bucketed by layer, the maximum effect is seen around the 15th layer of the last subject token, where the score is raised on average to 15.0%.

B.2 Separating MLP and Attn Effects

When decomposing the effects into MLP and Attn lookups, we found that restoring single activation vectors from individual MLP and individual Attn lookups had generally negligible effects, suggesting the decisive information is accumulated across layers. Therefore for MLP and Attn lookups, we restored runs of ten values of $m_i^{(l)}$ (and $a_i^{(l)}$, respectively) for an interval of layers ranging from $[l_* - 4, ..., l_* + 5]$ (clipping at the edges), where the results are plotted at layer l_*. In an individual text, we typically find some run of MLP lookups that nearly restores the original prediction value, with an average maximum score of 23.6%. Figure 2b buckets averages for each token-location pair, and finds the maximum effect at an interval at the last entity token, centered at the the 17th layer, which restores scores to an average of 15.0%. For Attn lookups (Figure 2c), the average maximum score over any location is 19.4%, and when bucketed by location, the maximum effect is centered at the 32nd layer at the last word before prediction, which restores scores to an average of 16.5%.

Figure 7 shows mean causal traces as line plots with 95% confidence intervals, instead of heatmaps.

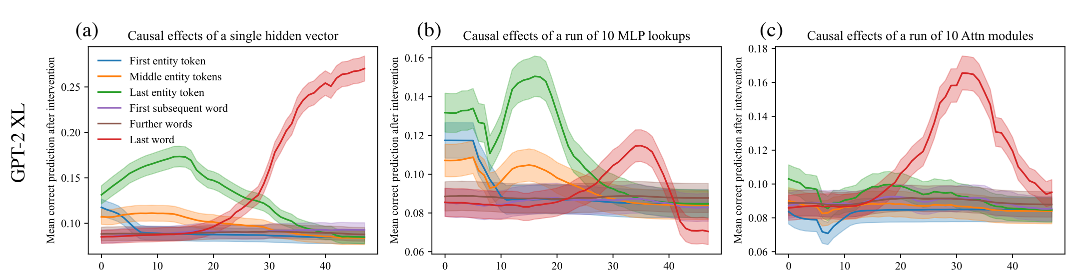

Figure 7: Mean causal traces of GPT-XL over a sample of 1000 factual statements, shown as a line plot with 95% confidence intervals. (a) Shows the same data as Figure 1j as a line plot instead of a heatmap; (b) matches Figure 1k; (c) matches Figure 1m. The confidence intervals confirm that the distinctions between peak and non-peak causal effects at both early and late sites are significant.

B. 因果追踪

B.1 实验设置

需要注意的是，在按层进行的实验结果中，各层的编号是从0开始递增到 $L-1$，而非从1开始递增到 L。

在图2与图3 中，我们针对由GPT-2 XL所识别的1000条事实性提示语来评估平均因果轨迹，这些提示语是按如下方式收集的。我们利用COUNTERFACT中的事实及事实模板进行贪婪生成，并识别出那些在提及任何其他大写单词之前首先指出正确对象的文本 o^c。我们将直到但未包含该对象 o^c 的文本作为提示词，然后从这些文本中随机抽取1000条。在这批已知事实的样本中，GPT-2 XL计算出的正确对象标记的预测概率平均为27.0%。

在引入损坏运行模式的情况下，我们会通过添加高斯噪声 $\epsilon \sim \mathcal{N}(0;\ \nu)$来破坏用于指代主体 s 的嵌入向量，其中 $\nu = 3\sigma_t$ 的值设定为从某段文本中抽样的标记嵌入向量的观测标准差σ_t 的三倍。对于每段文本，该操作会使用不同的噪声样本重复执行十次。平均而言，这样做的结果会使正确对象标记的得分降至8.47%，还不到原始得分的三分之一。

当从原始运行中恢复隐藏状态时，我们会使用同一层及同一标记原本计算出的数值进行替换，之后便不再进行额外干预，让后续计算能够正常继续。在图 1 中所展示的实验中（附录中的紫色曲线亦为对应数据），仅恢复了单个激活向量。显然，若恢复最后一个标记上的最后一个向量，就能完全还原原有的预测分数，但我们的绘图结果表明，在第二个位置也存在早期的激活向量，它们同样具有显著的因果效应：通过恢复主体最后一个标记上最具影响力的激活向量，其对应的平均最高分数为19.5%。在按层对效应进行分类的图 1j中可以看到，最大效应出现在最后一个主体标记的第15层左右，该层的分数平均提升了15.0%。

B.2 分离多层感知机与注意力效应

在将这些效应分解为多层感知机的查找结果与注意力机制的查找结果后，我们发现，从单个多层感知机查找结果或单个注意力机制查找结果中恢复单个激活向量，通常几乎不会产生任何影响，这说明决定性的信息是逐层累积形成的。因此，对于多层感知机与注意力机制的查找结果，我们分别恢复了每层十个数值的序列，其中 $m_i^{(l)}$ 对应多层感知机， $a_i^{(l)}$ 对应注意力机制，这些数值覆盖了从某起始层到某结束层之间的范围。

$[l_* - 4, ..., l_* + 5]$（在边缘处进行截断），相关结果会在对应层进行绘制 l_*。在单篇文本中，我们通常会发现连续的多层感知机查询几乎能够恢复原始预测值，其平均最高得分可达23.6%。图2将每对标记位置的数据汇总后求平均值，发现最大效应出现在最后一个实体标记所在的位置附近，大致位于第17层，此时预测得分的平均值可提升至15.0%。对于基于注意力机制的查询（见图2c），任何位置上的平均最高得分均为19.4%；按位置分类后，最大效应出现在预测前的最后一个单词所在的第32层附近，此时预测得分的平均值可提升至16.5%。

Figure 7 以折线图的形式展示平均因果轨迹，并标注95%置信区间，而非采用热力图呈现。s.

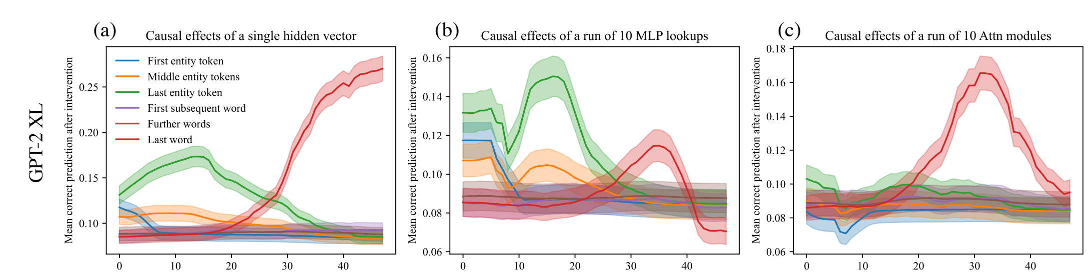

图7：基于1000条事实陈述样本的GPT-XL平均因果轨迹，同样以折线图形式呈现并附带95%置信区间。(a) 所展示数据与图 1j中的内容一致，均为折线图而非热力图；(b) 与图 1k相同；(c) 与图 1m一致。这些置信区间进一步证实，无论在早期节点还是晚期节点，峰值因果效应与非峰值因果效应之间的差异都具有显著性。

B.3 Traces of EleutherAI GPT-NeoX (20B) and GPT-J (6B) and smaller models

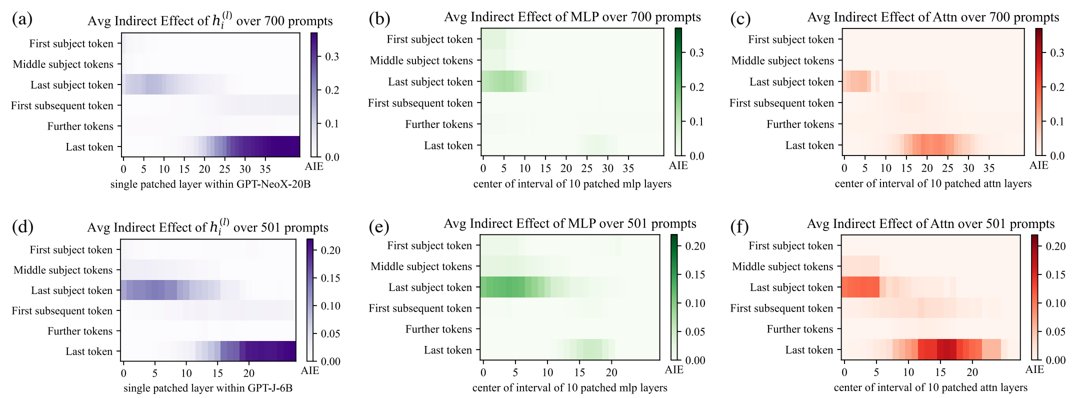

Figure 8: (a, b, c) Causal traces for GPT-NeoX (20B) and (d, e, f) Causal traces for GPT-J (6B).

We conduct the causal trace experiment using on GPT-NeoX (20 billion parameters) as well as GPT-J (6 billion parameters). For GPT-NeoX we adjust the injected noise to $\nu = 0.03$ and in GPT-J we use $\nu = 0.025$ to match embedding magnitudes. We use the same factual prompts as GPT-2 XL, eliminating cases where the larger models would have predicted a different word for the object. Results are shown in Figure 8. GPT-NeoX and GPT-J differ from GPT-2 because they have has fewer layers (44 and 28 layers instead of 48), and a slightly different residual structure across layers. Nevertheless, the causal traces look similar, with an early site with causal states concentrated at the last token of the subject, a large role for MLP states at that site. Again, attention dominates at the last token before prediction.

There are some differences compared to GPT-2. The importance of attention at the first layers of the last subject token is more apparent in GPT-Neo and GPT-J compared to GPT-2, suggesting that the attention parameters may be playing a larger role in storing factual associations. This concentration of attention at the beginning may also be due to fewer layers in the Eleuther models: attending to the subject name must be done in a concentrated way at just a layer or two, because there are not enough layers to spread out that computation in the shallower model. The similarity between the GPT-NeoX and GPT-J and GPT-2 XL traces helps us to understand why ROME continues to work well with higher-parameter models, as seen on our experiments in altering parameters of GPT-J.

To examine effects over a wide range of scales, we also compare causal traces for smaller models GPT-2 Medium and GPT-2 Large. These smaller models are compared to NeoX-20B in Figure 9. We find that across sizes and architectural variations, early-site MLP modules continue to have high indirect causal effects at the last subject token, although the layers where effects peak are different from one model to another.

B.4 Causal Tracing Examples and Further Insights

We include further examples of phenomena that can be observed in causal traces. Figure 10 shows typical examples across different facts. Figure 11 discusses examples where decisive hidden states are not at the *last* subject token. Figure 14 examines traces at an individual token in more detail.

We note that causal tracing depends on a corruption rule to create baseline input for a model that does not contain all the information needed to make a prediction. Therefore we ask: are Causal Tracing results fragile if the exact form of the corruption changes? We test this by expanding the corruption rule: even when additional tokens after the subject name are also corrupted, we find that the results are substantially the same. Figure 12 shows causal traces with the expanded corruption rule. Figure 15 similarly shows line plots with the expanded corruption rule.

We do find that the noise must be large enough to create large average total effects. For example, if noise with variance that is much smaller is used (for example if we set $\sigma = \sigma_t$), average total effects become very small, and the small gap in the behavior between clean runs and corrupted run makes it difficult discern indirect effects of mediators. Similarly, if we use a uniform distribution

B.3 EleutherAI GPT-NeoX（200亿参数）与GPT-J（60亿参数）以及更小型模型的因果轨迹算法结果

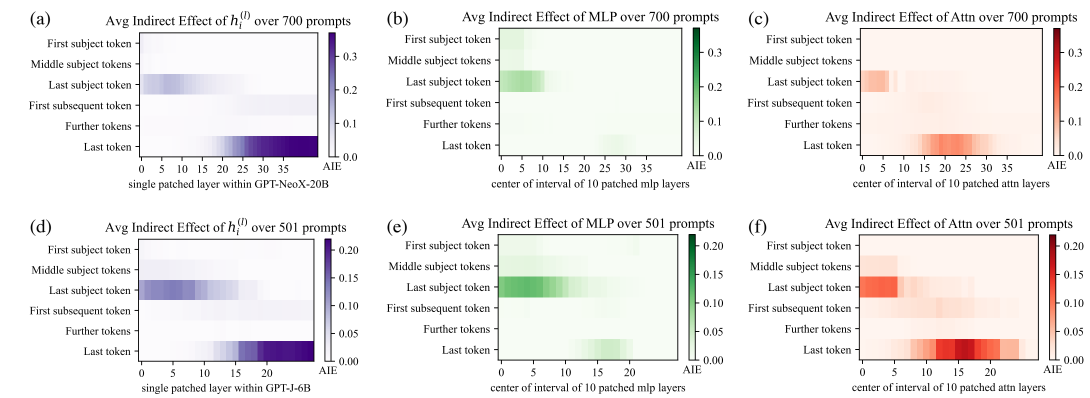

图8：(a, b, c)为GPT-NeoX（200亿参数）的因果轨迹，(d, e, f)为GPT-J（60亿参数）的因果轨迹。

我们使用GPT-NeoX（200亿参数）和GPT-J（60亿参数）开展了因果轨迹算法实验。针对GPT-NeoX，我们调整了注入的噪声水平为 $\nu = 0.03$，而在GPT-J中则使用 $\nu = 0.025$ 来使嵌入向量的幅度值保持一致。我们使用的事实性提示语与GPT-2 XL相同，以此避免那些参数规模更大的模型因不同原因而预测出不同的对象词汇。实验结果如图8所示。GPT-NeoX和GPT-J与GPT-2的不同之处在于它们的层数更少（分别为44层和28层，而非GPT-2的48层），且各层之间的残差结构也略有差异。尽管如此，两者的因果轨迹特征依然相似：在主体文本的最后一个标记处会集中出现具有因果状态的早期节点，且该节点处的多层感知机状态起着重要作用；同样，在预测前的最后一个标记处，注意力机制依然占据主导地位。

与GPT-2相比，这些模型存在一些差异。在最后一个主体标记的最初几层中，注意力机制的重要性在GPT-NeoX和GPT-J身上表现得更为明显，这表明注意力参数在存储事实关联的过程中可能发挥着更重要的作用。注意力集中在开头这一现象也可能源于Eleuther系列模型层数较少：在仅有一两层的情况下，就必须集中精力处理主体名称，因为较浅层的模型没有足够的层数来分散这类计算任务。GPT-NeoX、GPT-J与GPT-2 XL的因果轨迹之间存在相似性，这有助于我们理解为何ROME方法在参数量更高的模型上依然能保持良好的效果，我们在调整GPT-J参数的实验中也观察到了这一现象。

为探究不同尺度范围内的效应，我们还将较小的GPT-2 Medium模型与GPT-2 Large模型对应的因果轨迹算法进行了对比。这些较小规模的模型会在图9中与NeoX-20B模型进行对比。研究结果表明，无论模型规模或架构如何差异，处于最末主体标记前的MLP模块始终会产生较强的间接因果效应，只不过各模型中效应强度最高的层位置并不相同。

B.4 因果追踪示例与进一步分析

我们进一步列举了在因果轨迹算法中可以观察到的各类现象。图10展示了不同事实场景下的典型案例。图11 则探讨了那些具有决定性作用的隐藏状态并非位于最末主体标记处的情况。图14 则更详细地分析了单个标记对应的轨迹特征。

需要指出的是，因果追踪算法依赖于数据污染规则来为那些无法获取所有预测所需信息的模型生成基准输入。因此我们提出了这样一个问题：如果数据污染规则的具体形式发生变化，因果追踪的结果是否会变得不稳定？为验证这一点，我们对数据污染规则进行了扩展——即便在主体名称之后的其他标记也被污染，实验结果依然基本保持不变。图12展示了应用扩展后数据污染规则的因果轨迹结果。图15同样呈现了应用该扩展规则后的折线图。

我们确实发现，噪声的强度必须足够大，才能产生较大的平均总效应。例如，如果使用方差小得多的噪声（比如将 $\sigma = \sigma_t$设为某个特定值），平均总效应就会变得非常小，而纯净运行模式与损坏运行模式之间表现出的微小差异会使得人们难以分辨中介变量的间接效应。同理，如果我们采用均匀分布的话，

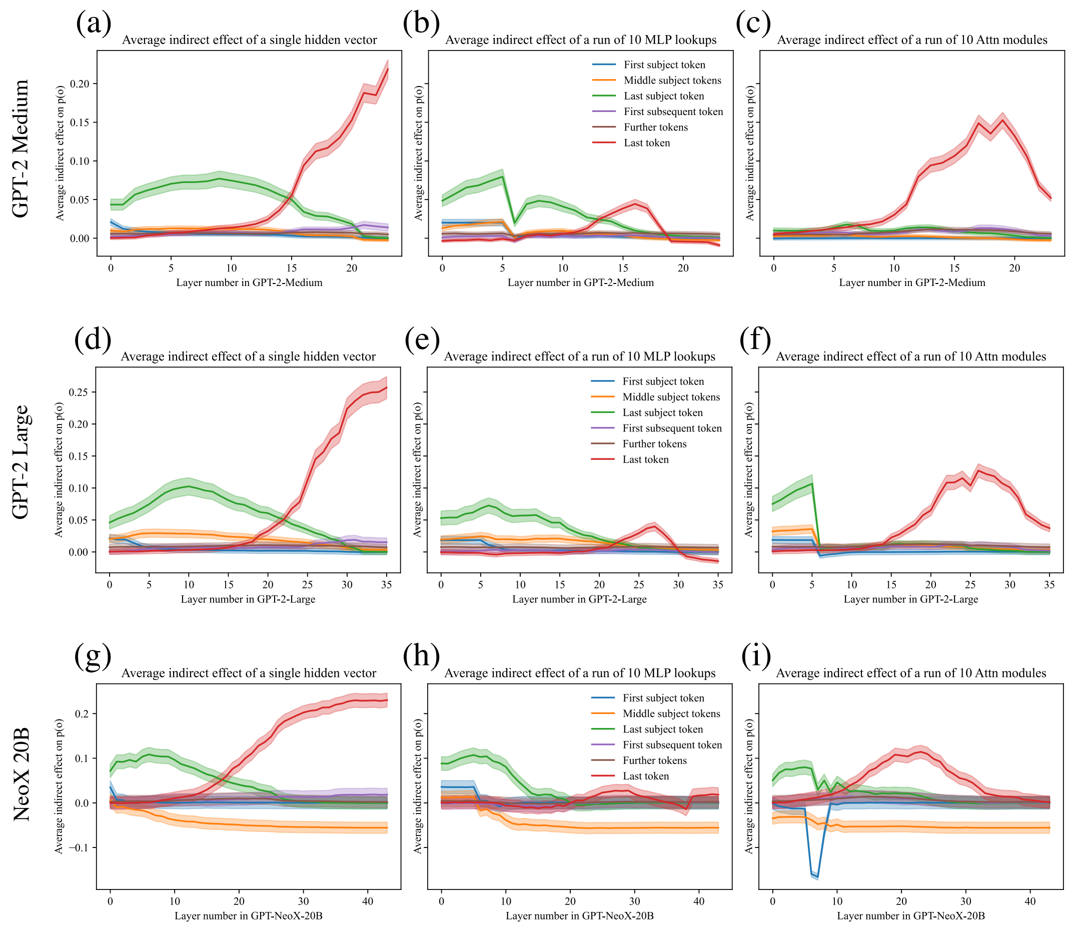

Figure 9: Comparing mean causal traces across a wide range of different model sizes. (Compare to Figure 7.) GPT-medium (a, b, c) has 334 million parameters, GPT-large (d, e, f) has 774 million parameters, and NeoX-20B (g, h, i) has 20 billion parameters. In addition, NeoX has some architectural variations. Despite the wide range of differences, a similar pattern of localized causal effects is seen across models. Interestingly, for very large models, some effects are stronger. For example, hidden states before the last subject token have negative causal effects instead of merely low effects, while hidden states at early layers at the last subject token continue to have large positive effects, continuing to implicate MLP. Also, attention modules with strong causal effects appear earlier in the stack of layers.

where components range in $\pm 3\sigma$, effects large enough for causl tracing but smaller than a Gaussian distribution.

If instead of using spherical Gaussian noise, we draw noise from $\mathcal{N}(mu, \Sigma)$ where we set $\mu = \mu_t$ and $\Sigma_= \Sigma_t$ to match the observed distribution over token embeddings, average total effects are also strong enough to perform causal traces. This is shown in Figure 13.

Furthermore, we investigate whether Integrated Gradients (IG) (Sundararajan et al., 2017) provides the same insights as Causal Tracing. We find that IG is very sensitive to local features but does not yield the same insights about large-scale global logic that we have been able to obtain using causal traces. Figure 16 compares causal traces to IG saliency maps.

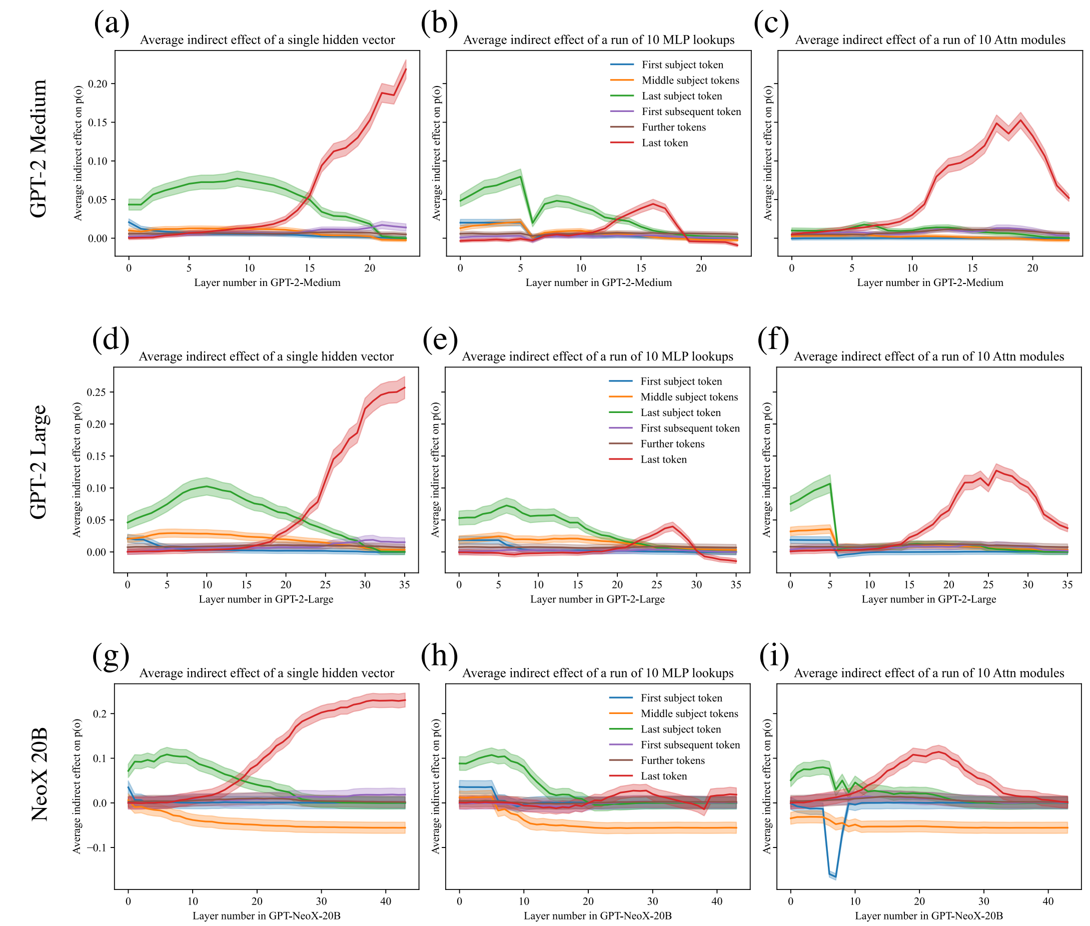

图9：展示了不同规模模型对应的平均因果轨迹算法结果。（可与图7中的数据对比。）GPT-medium（a、b、c）拥有3.34亿个参数，GPT-large（d、e、f）拥有7.74亿个参数，而NeoX-20B（g、h、i）则拥有200亿个参数。此外，NeoX还存在一些架构上的差异。尽管这些模型之间存在巨大差异，但它们都呈现出类似的局部因果效应模式。有趣的是，对于那些规模极大的模型，某些因果效应的强度反而更高。例如，最后一个主题标记之前的隐藏状态所呈现的并非仅仅是较低的因果效应，而是负向的因果效应；而位于最后一个主题标记所在层早期位置的隐藏状态则依然具有较大的正向因果效应，这进一步表明多层感知机在其中起到了重要作用。同时，那些具有较强因果效应的注意力模块也出现在层堆叠的更靠前位置。

此时各组分的取值范围在 $\pm 3\sigma$之间，所产生的效应虽足以用于因果追踪，但其数值仍小于高斯分布下的效应值。

如果不再使用球形高斯噪声，而是从 $\mathcal{N}(mu, \Sigma)$ 中抽取噪声，且将 $\mu = \mu_t$和 $\Sigma_= \Sigma_t$ 设定为与标记嵌入向量所呈现的分布相匹配，那么平均总效应也会强到足以用于执行因果轨迹算法。图13中展示了这一现象。

此外，我们还研究了积分梯度（IG）（Sundararajan等人，2017年）能否提供与因果追踪相同的洞察。研究结果表明，IG对局部特征极为敏感，但无法像通过因果轨迹算法所得到的那样，揭示大规模全局逻辑的相关信息。图16对比了因果追踪结果与IG显著性图之间的差异。

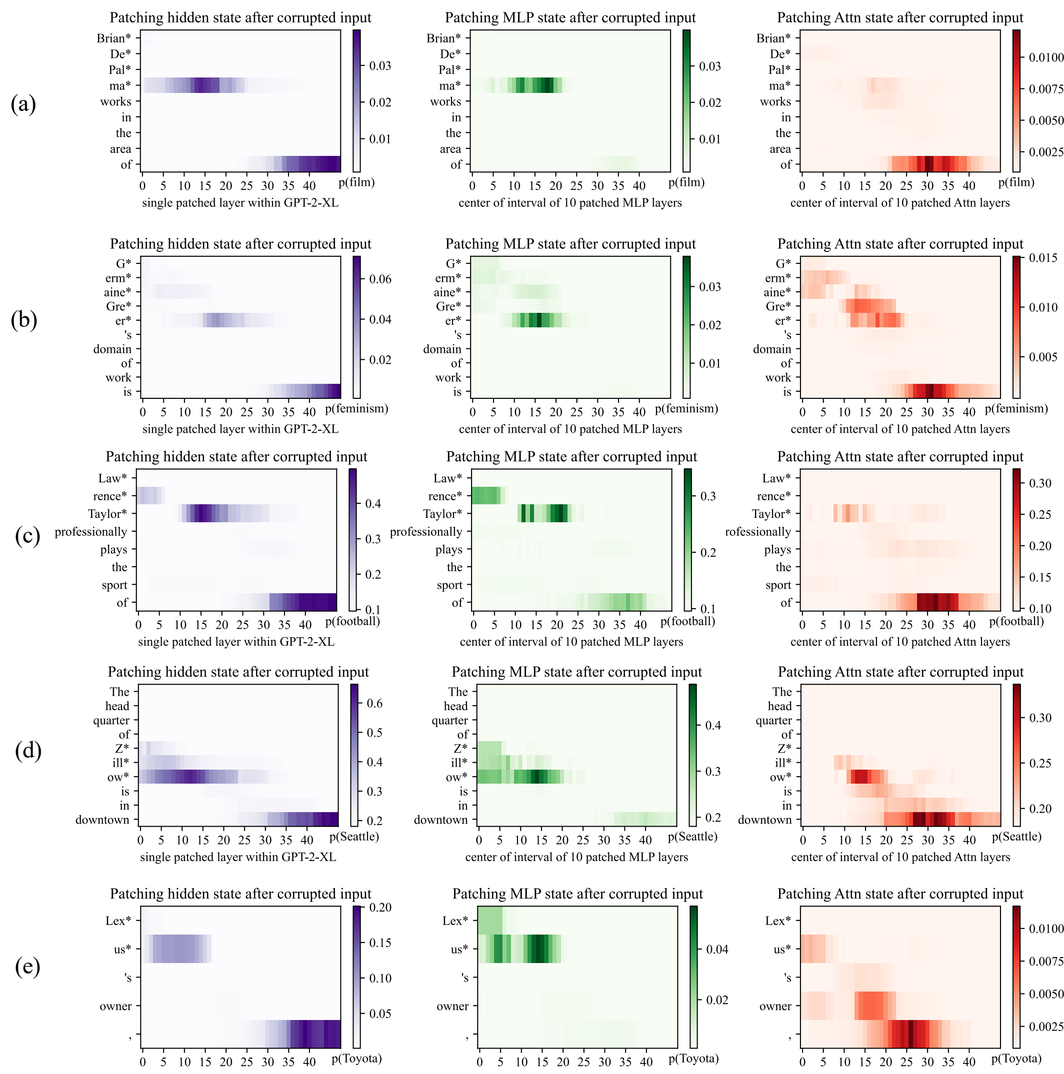

Figure 10: Further examples of causal traces showing appearance of the common lookup pattern on a variety of different types of facts about people and other kinds of entities. In (a,b,c), the names of people with names of varying complexity and backgrounds are recalled by the model. In each case, the MLP lookups on the last token of the name are decisive. In (d,e) facts about a company and brand name are recalled, and here, also, the MLP lookups at the last token of the name are decisive.

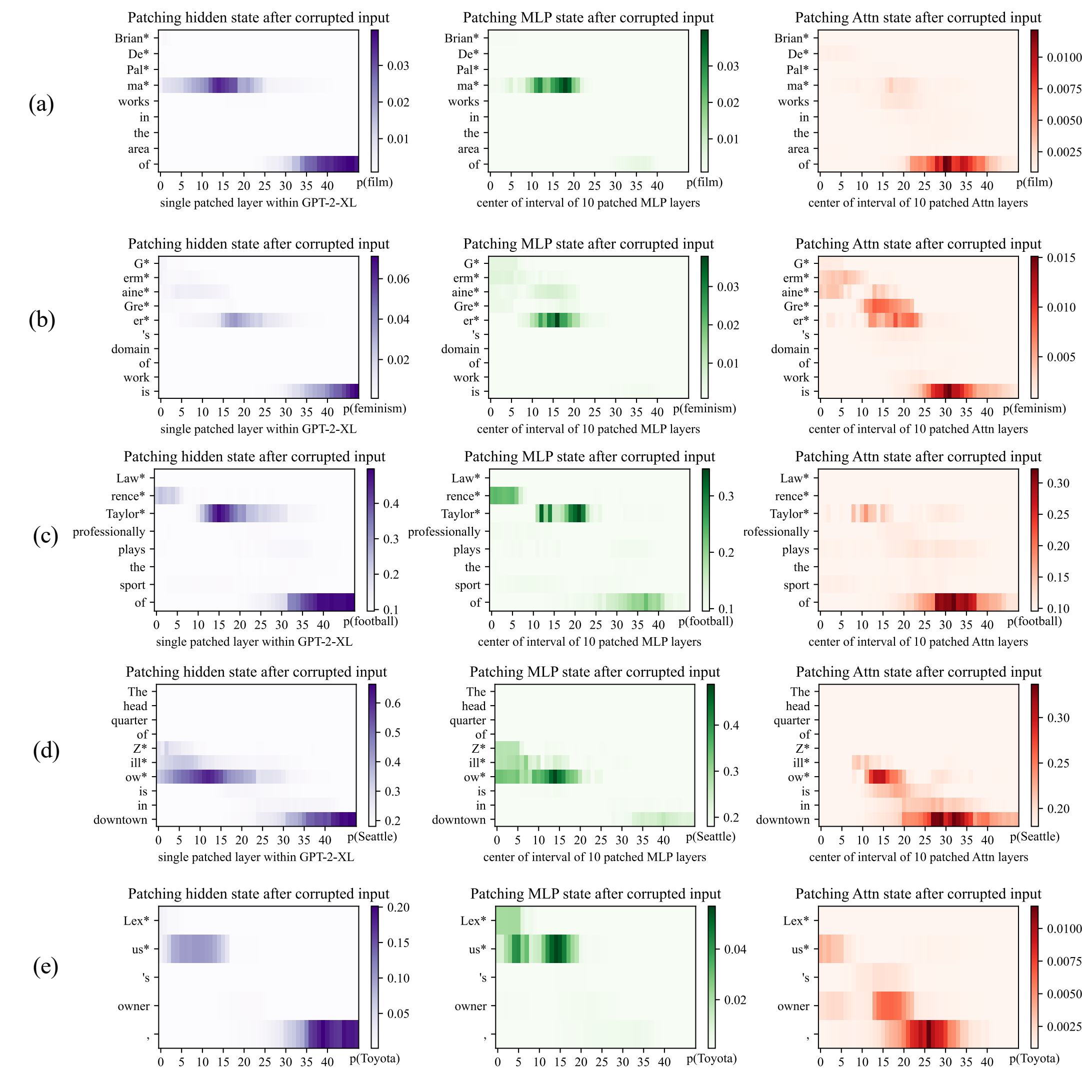

图10：更多因果轨迹算法的示例，展示了在关于人物及其他各类实体的各种不同类型的事实中出现的共同查询模式。在(a)、(b)、(c)中，模型会调取具有不同复杂度与背景的人物姓名；在每种情况下，都是对姓名最后一个标记的多层感知机查询起到了决定性作用。而在(d)、(e)中，则是关于公司及其品牌名称的事实被调取出来，此处同样也是对名称最后一个标记的多层感知机查询起了决定性作用。

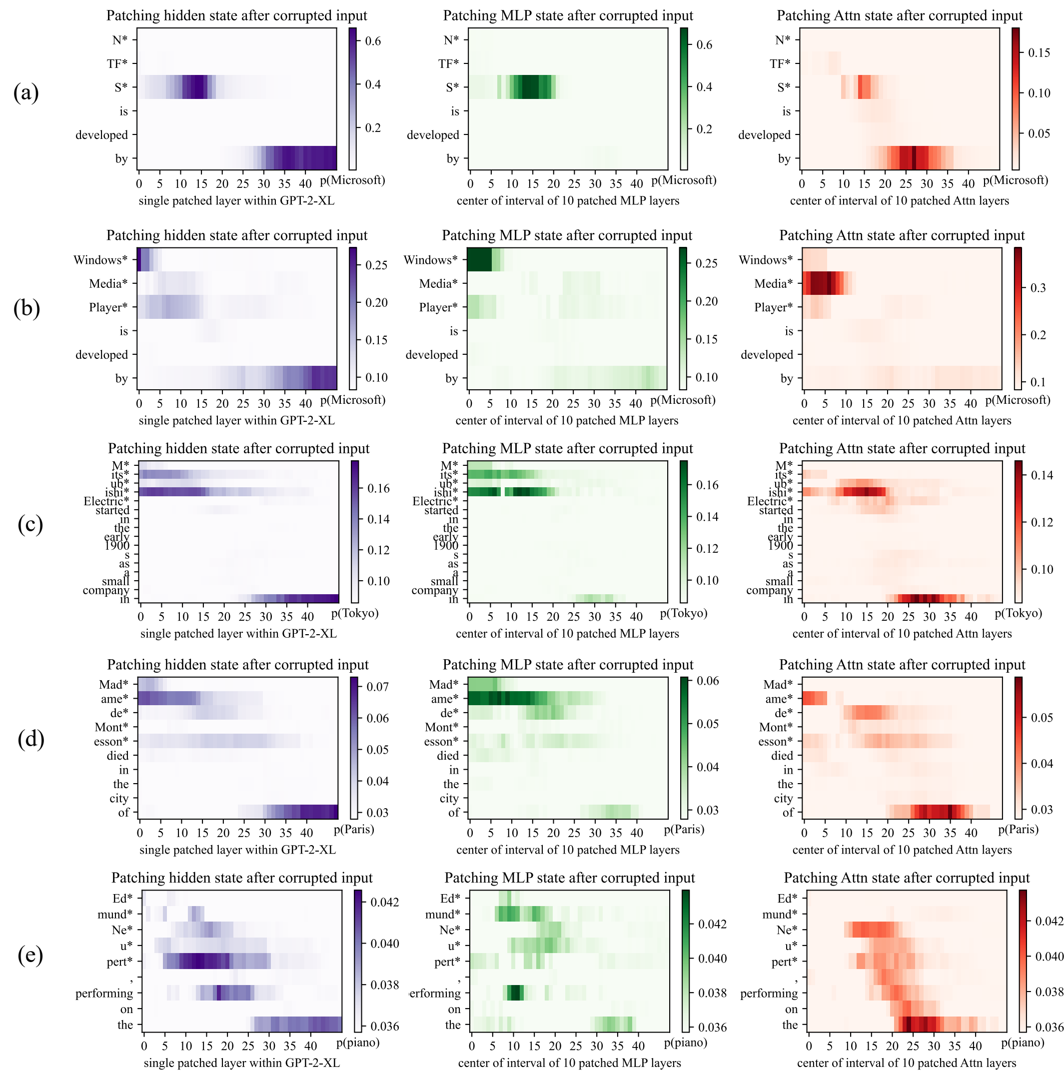

Figure 11: Causal traces show that the last token of the subject name is not always decisive. (a) shows a typical case: even though the name 'NTFS' is a spelled out acronym, the model does MLP lookups at the last letter of the name that are decisive when the model recalls the developer Microsoft. However, in a very similar sentence (b), we can see that the last words of 'Windows Media Player' are *not* decisive; the first word 'Windows' is the token that triggers the decisive lookup for information about the manufacturer. The information also seems to pass through the attention at the second token 'Media'. Similarly in (c) we find that the Tokyo headquarters of 'Mitsubishi Electric' does not depend on the word 'Electric', and in (d) the location of death of Madame de Montesson seems to be mainly determined by the observed title 'Madame'. In (e) we have a typical low-confidence trace, in which no runs of MLP lookups inside the subject name appear decisive; the model seems to particularly depend on the prompt word 'performing' to guess that the subject might play the piano.

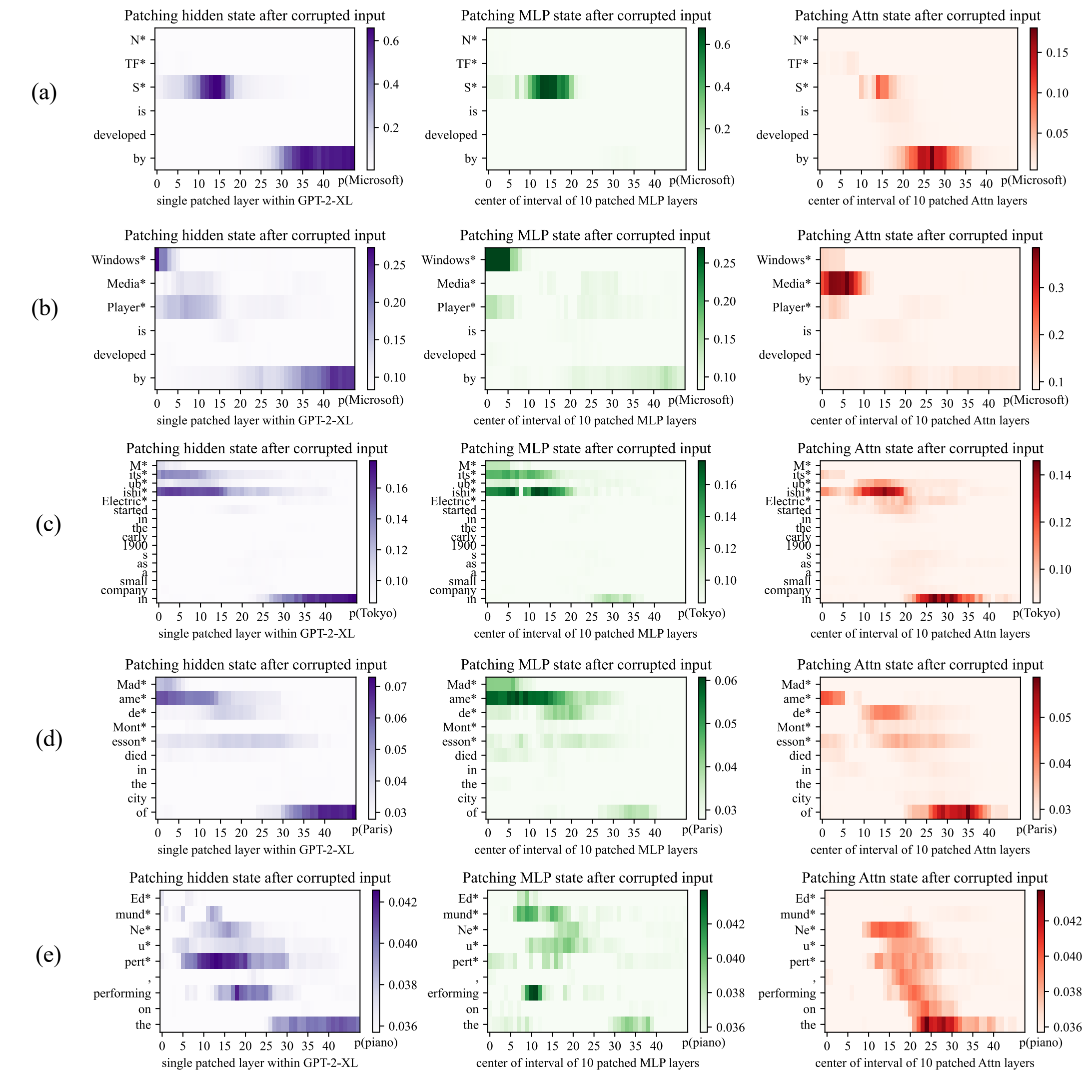

图11：因果轨迹算法显示，主体名称的最后一个标记并不总是起决定性作用。(a)展示了一个典型案例：尽管“NTFS”是一个全拼缩写，但当模型需要召回制造商“微软”这一信息时，它会对该名称的最后一个字母进行多层感知机查询，而这些查询起到了决定性作用。然而在结构极为相似的句子(b)中，我们可以看到“Windows Media Player”这个短语的最后一个单词并非具有决定性；真正触发关于制造商信息的决定性查询的是第一个单词“Windows”。这些信息似乎还会通过第二个标记“Media”处的注意力机制传递。同样在(c)中，我们发现“三菱电机”的东京总部这一信息并不取决于“Electric”这个词；而在(d)中，蒙泰松夫人的去世地点似乎主要由其被记录的主体名称“Madame”决定。在(e)中则出现了一个典型的低置信度轨迹，其中主体名称内的多层感知机查询序列并未表现出任何决定性作用；模型似乎特别依赖提示词单词“performing”来推测该主体可能擅长弹钢琴。

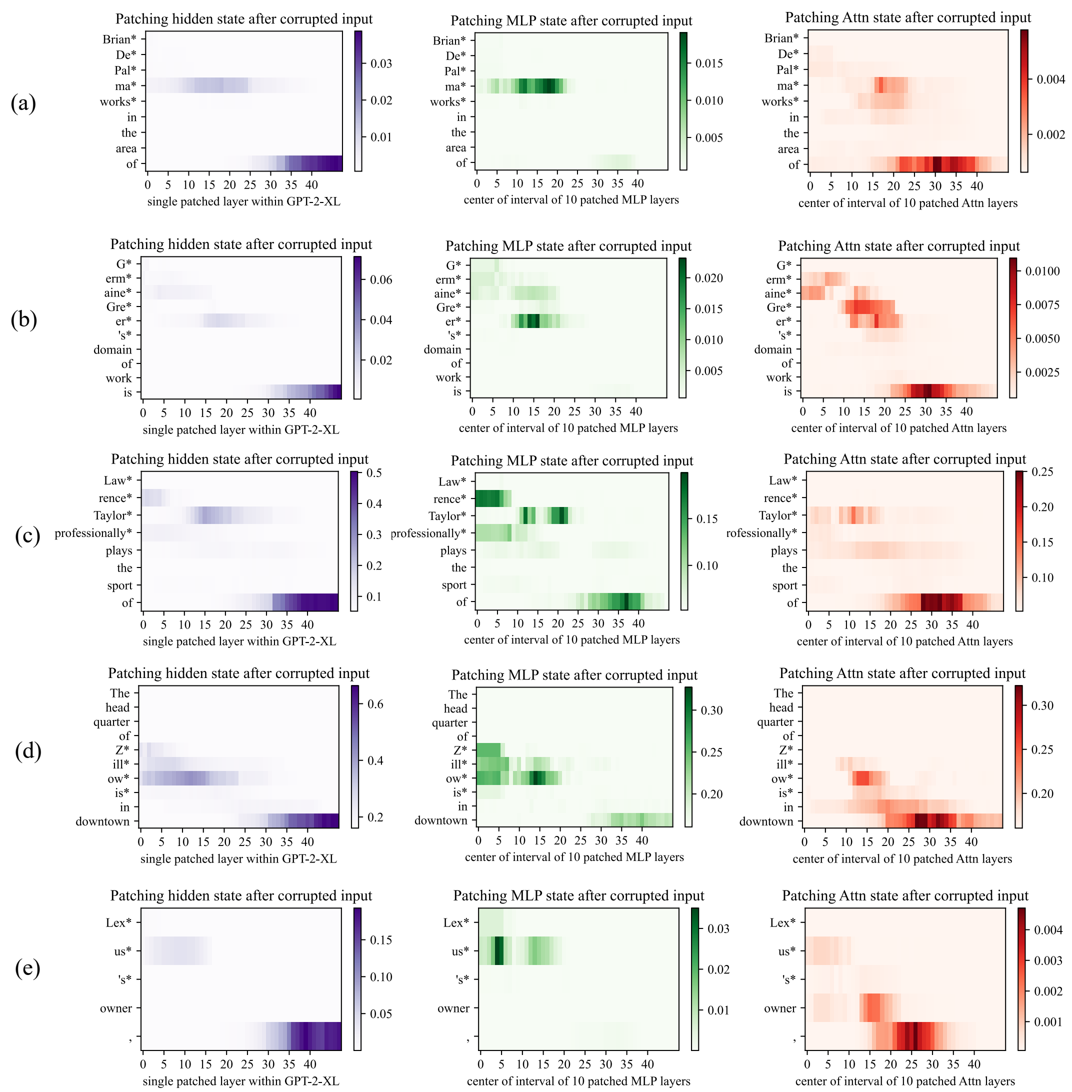

Figure 12: Causal traces in the presence of additional corruption. Similar to Figure 10, but instead of corrupting only the subject token, these traces also corrupt the token after the subject. Causal effects are somewhat reduced due to the the model losing some ability to read the relation between the subject and object, but these traces continue to show concentrated causal effects at the last token of the subject even when the last token is not the last token corrupted. Causal effects of MLP layers at the last subject token continues to be pronounced.

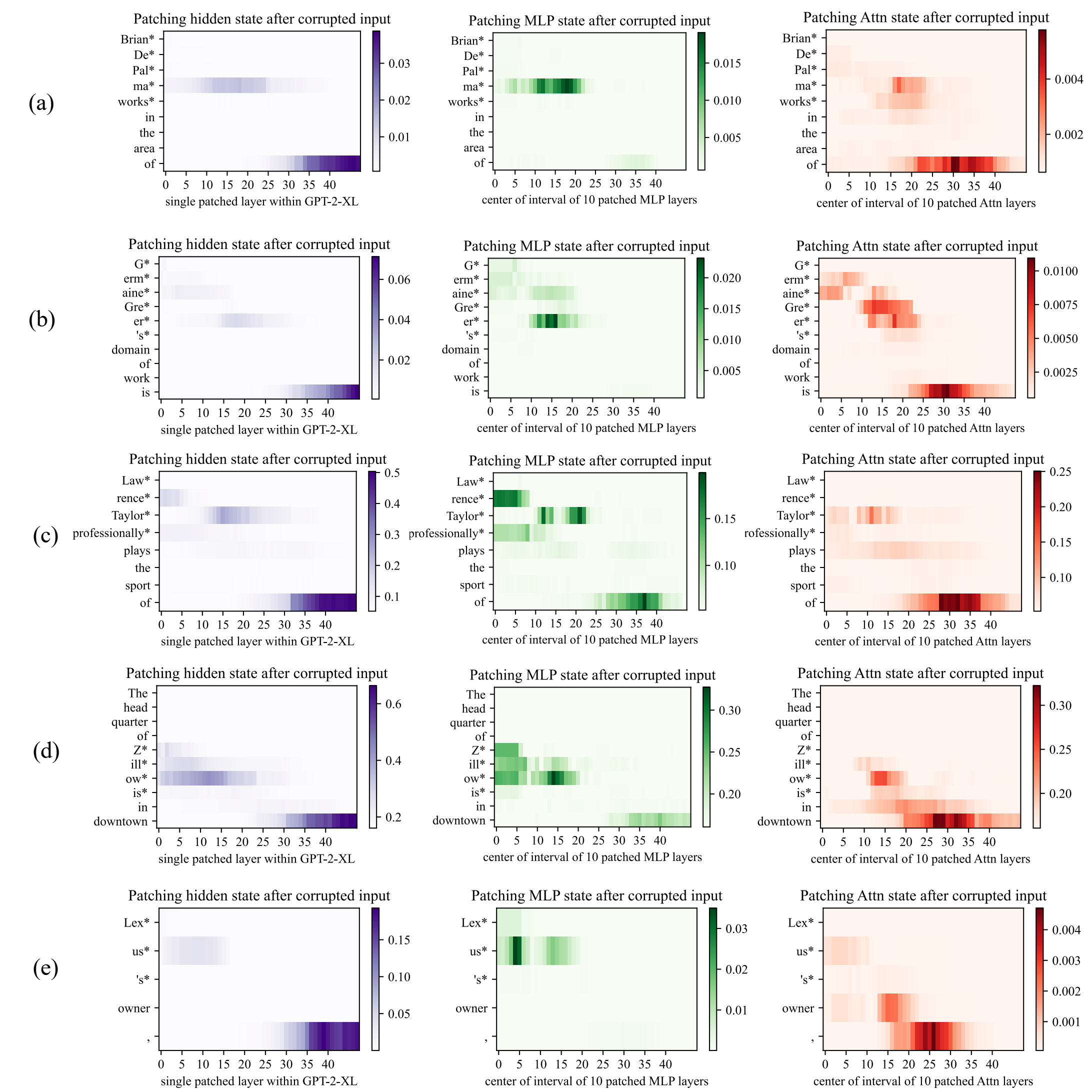

图12：在存在额外干扰时的因果轨迹算法。与图10类似，只不过这些轨迹不仅会干扰主题标记，还会影响该标记之后的标记。由于模型在理解主题与对象之间关联的能力有所下降，因果效应会有所减弱，但即便最后被干扰的标记并非序列中的最后一个标记，这些轨迹依然会在主题标记的最后一个位置呈现较强的因果效应。而位于主题标记最后的多层感知机层的因果效应则依然十分显著。

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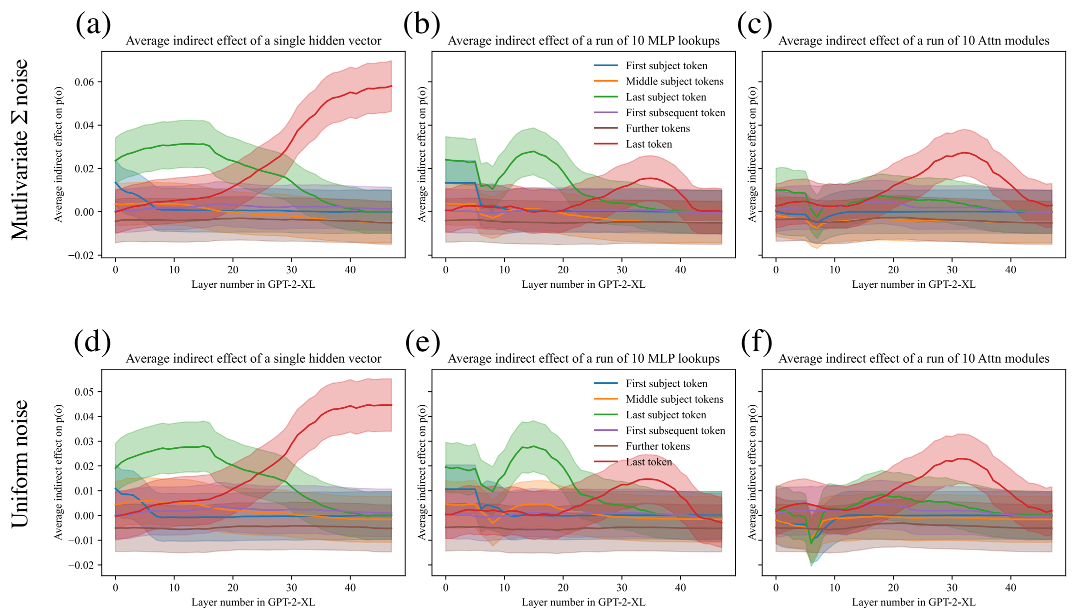


Figure 13: Comparing different noise choices. (Compare to Figure 7, where noise is chosen as a $3\sigma_t$ spherical Gaussian, where σ_t is measured to match the observed spherical variance over tokens.) In a, b, c we we draw noise from a multivariate Gaussian $\mathcal{N}(\mu; \Sigma)$ where μ and Σ are chosen to match the observed mean and covariance over a sample of tokens. In d, e, f we draw noise from a uniform distribution in the range $\pm 3\sigma$ instead of a Gaussian distribution. In both cases, the average total effects measured between the clean run and the corrupted run are large enough to measure causal traces, but the effects are smaller than the choice of $3\sigma_t$ used in the main paper.

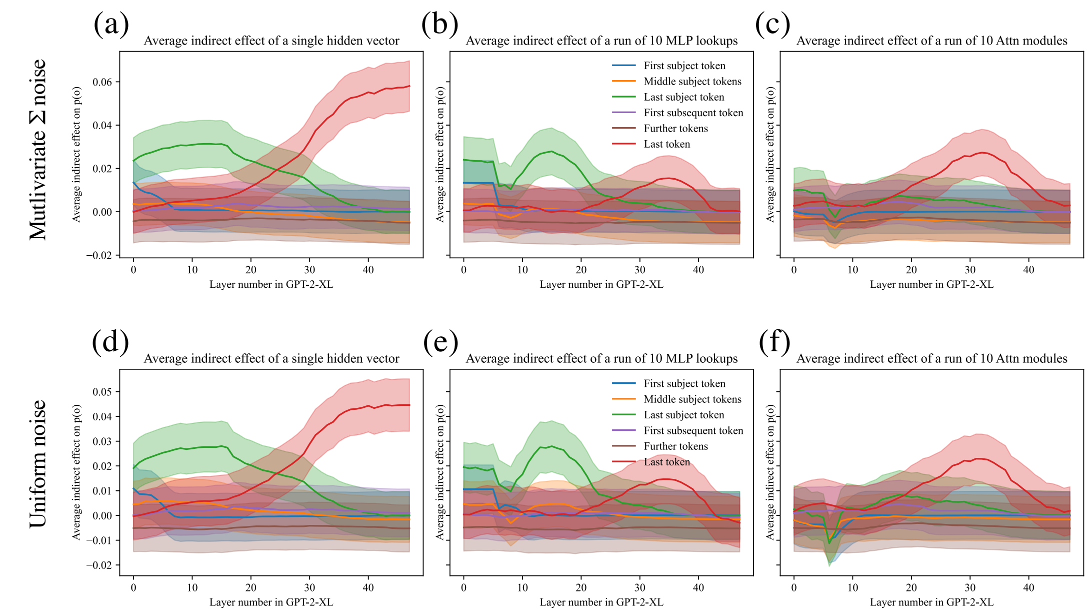


图13：不同噪声类型的效果对比。（可参照图7，该图中噪声被设定为 $3\sigma_t$ 球形高斯分布，且 σ_t 其参数经过调整，以使标记间的球形方差与观测值相符。）在a、b、c图中，我们是从多元高斯 $\mathcal{N}$（μ；Σ）中抽取噪声，其中 μ 和 Σ 的取值是根据样本标记的均值与协方差来确定的。而在d、e、f图中，我们则使用范围为 $\pm 3\sigma$ 的均匀分布而非高斯分布来生成噪声。在这两种情况下，纯净运行模式与损坏运行模式之间测得的平均总效应都足够大，能够用于检测因果轨迹算法，但这些效应的数值仍小于主论文中所采用的 $3\sigma_t$ 噪声类型。

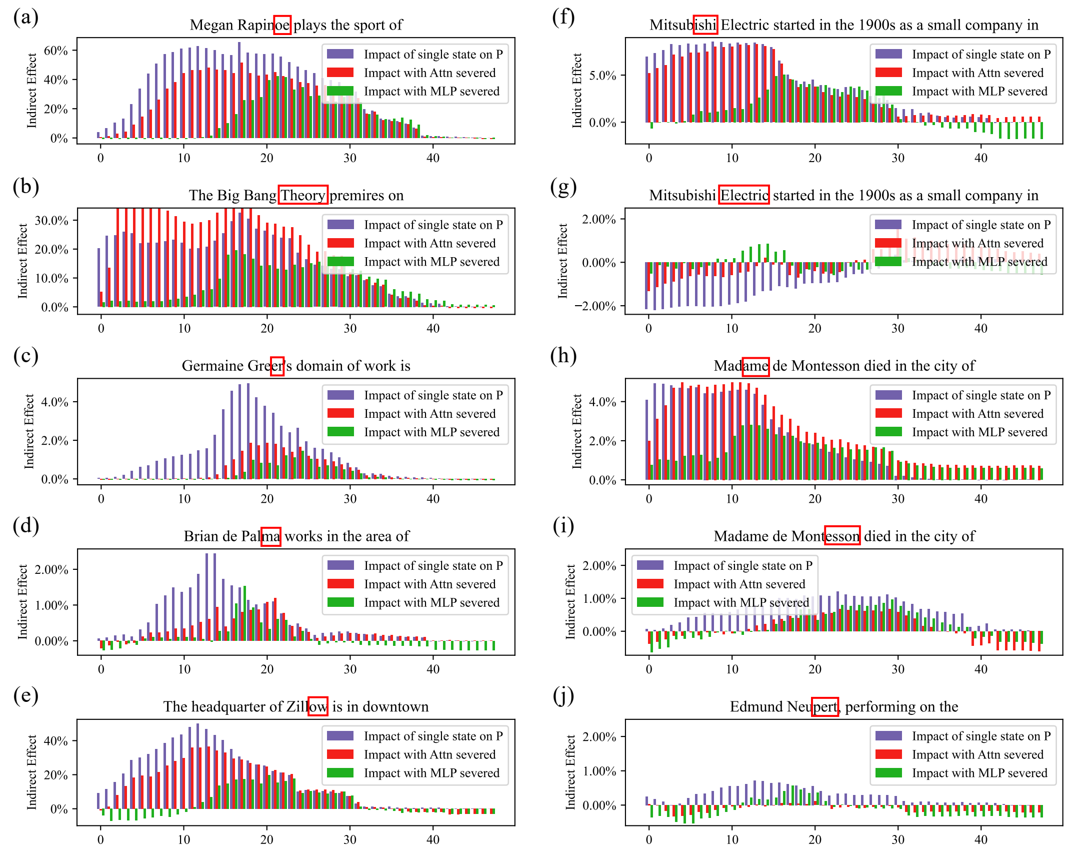


Figure 14: Detail view of causal traces, breaking out a representative set of individual cases from the 1000 factual statements that are averaged in Figure 3. Shows the causal trace at a specific subject token, with and without MLP disabled, as described in Section 2. In every case, the token tested is highlighted in a red box. In (a,b,c,d,e) cases are shown that fit the typical pattern: Restoring individual hidden states at a range of layers has a strong decisive average causal effect at the last token of the subject. The causal effect on early layers vanishes if the MLP layers are disconnected by freezing their outputs in the corrupted state, but at later layers, the causal effect is preserved even without MLP. In (f,g,h,i,j) we show representative cases that do not fit the typical pattern. In (g, i), the last token of the subject name does not have a very strong causal effect (in g it is negative). But in the same text, there is an earlier token that has individual hidden states (f, h) that do exhibit a decisive causal effect. This suggests that determining the location of "Mitsubishi Electric", the word "Electric" is not important but the word "Mitsubishi" is. Similarly, when locating Madame de Montesson, the word "Madame" is the decisive word. (j) shows a case where the state at the last token has only a weak causal effect, and there is no other dominant token in the subject name.

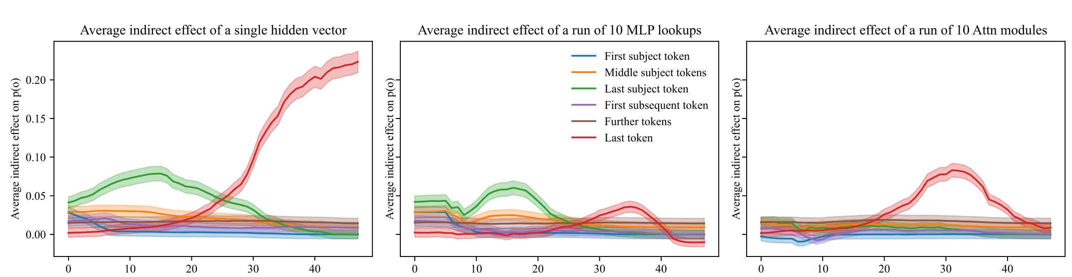


Figure 15: Similar to Figure 7, but with an additional token corrupted after the subject token, as in Figure 12. We observe that the emergence of strong early-site causal effects at the MLP modules is systematic and appears under a different corruption scheme, confirming that importance of the last subject token is apparent even when the last subject token is never the last token corrupted.

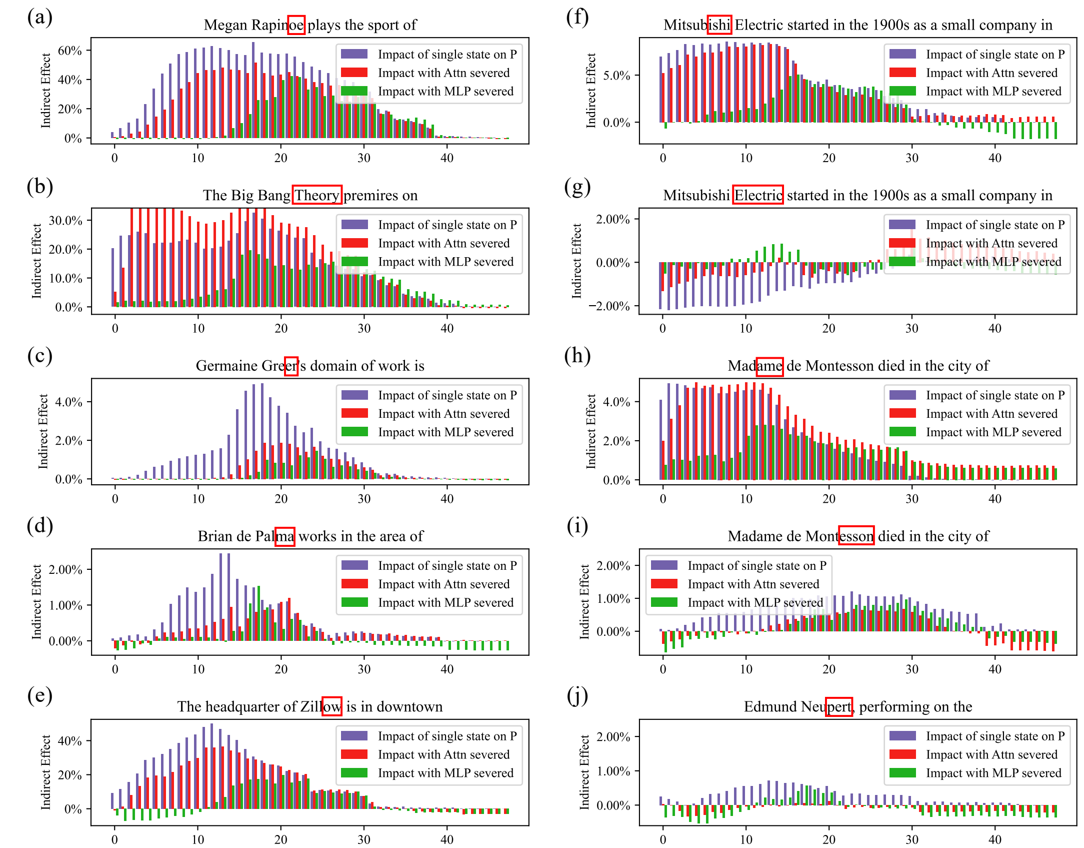


图14：该图展示了因果轨迹算法的详细情况，从中提取了用于计算的1000条事实陈述中的若干代表性个案3。如第2节所述，该图显示了在特定主题标记处的因果轨迹，同时展示了启用与禁用多层感知机两种情况下的结果。在每一种情况下，被测试的标记都会以红色框标出。在(a,b,c,d,e)中展示的是符合典型模式的案例：在多个层恢复各个隐藏状态后，会在主体最后一个标记处产生显著的平均处理效应；如果通过将多层感知机层的输出冻结在损坏状态来切断其连接，那么对早期层的因果效应就会消失，而即便没有多层感知机，对后期层的因果效应依然存在。在(f,g,h,i,j)中则展示了不符合典型模式的代表性案例。在(g,i)中，主体名称的最后一个标记并没有非常强的因果效应（在g中甚至为负值），但在同一文本中，较早位置的某些标记（如f、h）却表现出显著的因果效应。这表明，在确定“三菱电机”这一主体位置时，“Electric”这个词并不重要，而“Mitsubishi”才是关键。同样地，在定位蒙泰松夫人这一主体时，“Madame”这个词才是起决定性作用的词。(j)展示的是这样一个案例：最后一个标记的状态仅具有微弱的因果效应，且主体名称中也没有其他占主导地位的标记。

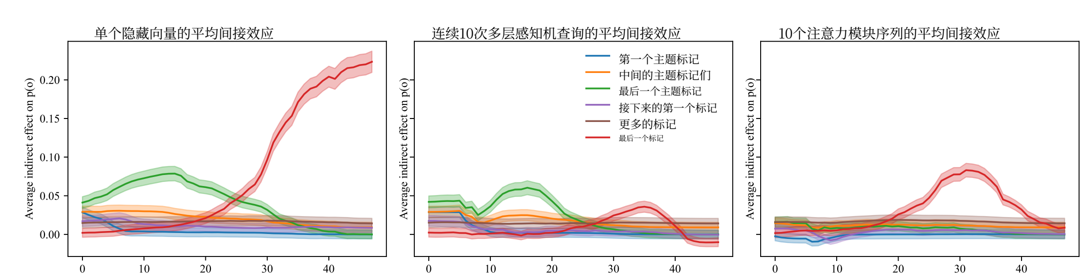


图15：与图7类似，不同之处在于在主题标记之后多了一个被破坏的标记，情况如图12所示。我们发现，在MLP模块中会出现较强的早期位置因果效应，且这种现象具有规律性，即便采用不同的数据破坏方案依然会出现；这一事实进一步证实，即便最后一个主题标记并非被破坏的标记中的最后一个，它的作用依然十分重要。

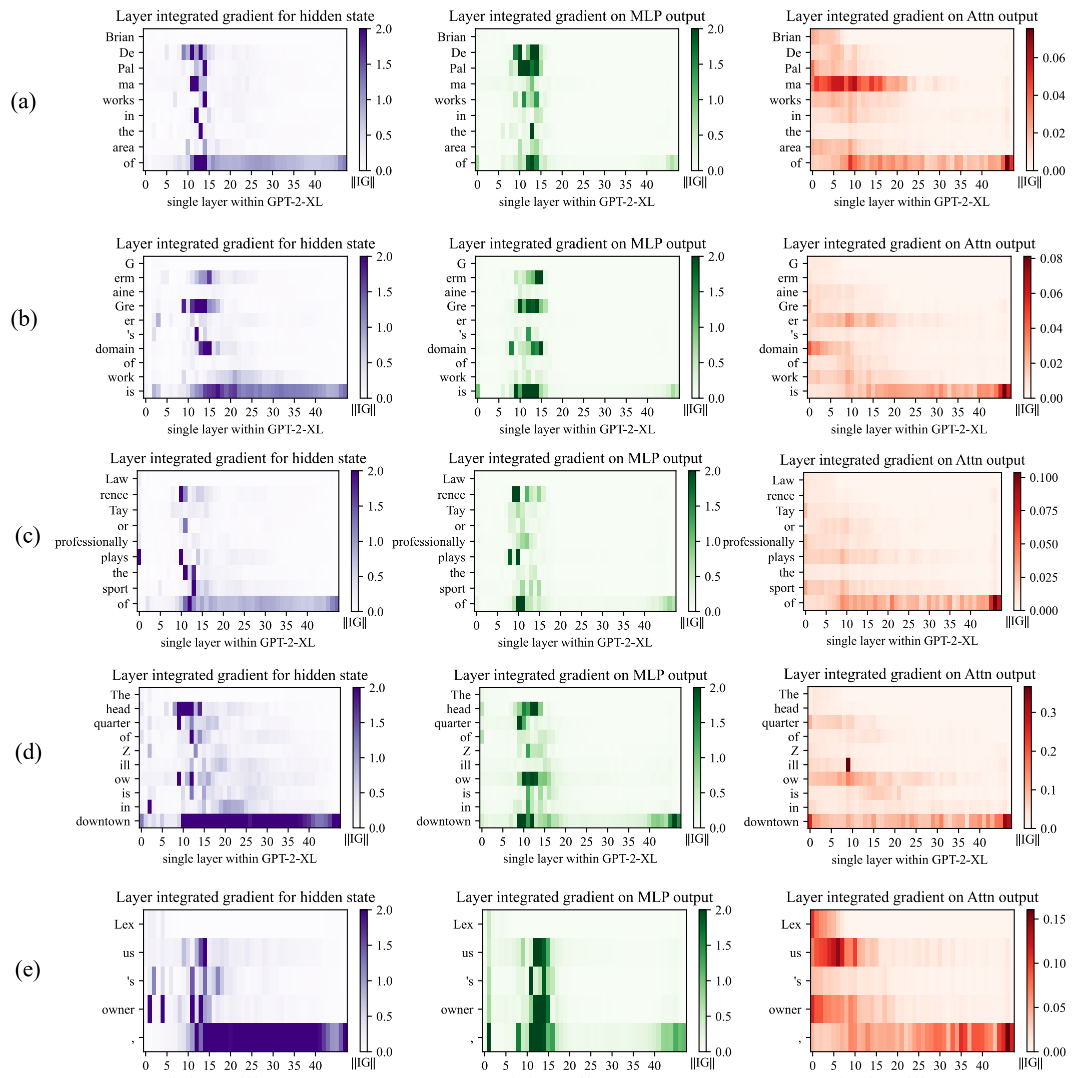


Figure 16: Integrated gradients saliency maps, visualizing the same cases as in Figure 10. Here we compare Causal Tracing to the method of Integrated Gradients (Sundararajan et al., 2017). Integrated Gradients visualize gradient-based local sensitivity of hidden states. Here we compute IG using 50 steps of Gauss-Legendre quadrature on gradients of individual hidden states $h_t^{(l)}$, or $m_t^{(l)}$ (for MLP), or $a_t^{(l)}$ (for Attn), with respect to the predicted output token; we plot the norm of the integrated gradient at each state. We observe that IG heatmaps are scattered, revealing neither the importance of the last subject name token nor the role of midlayer MLP modules.

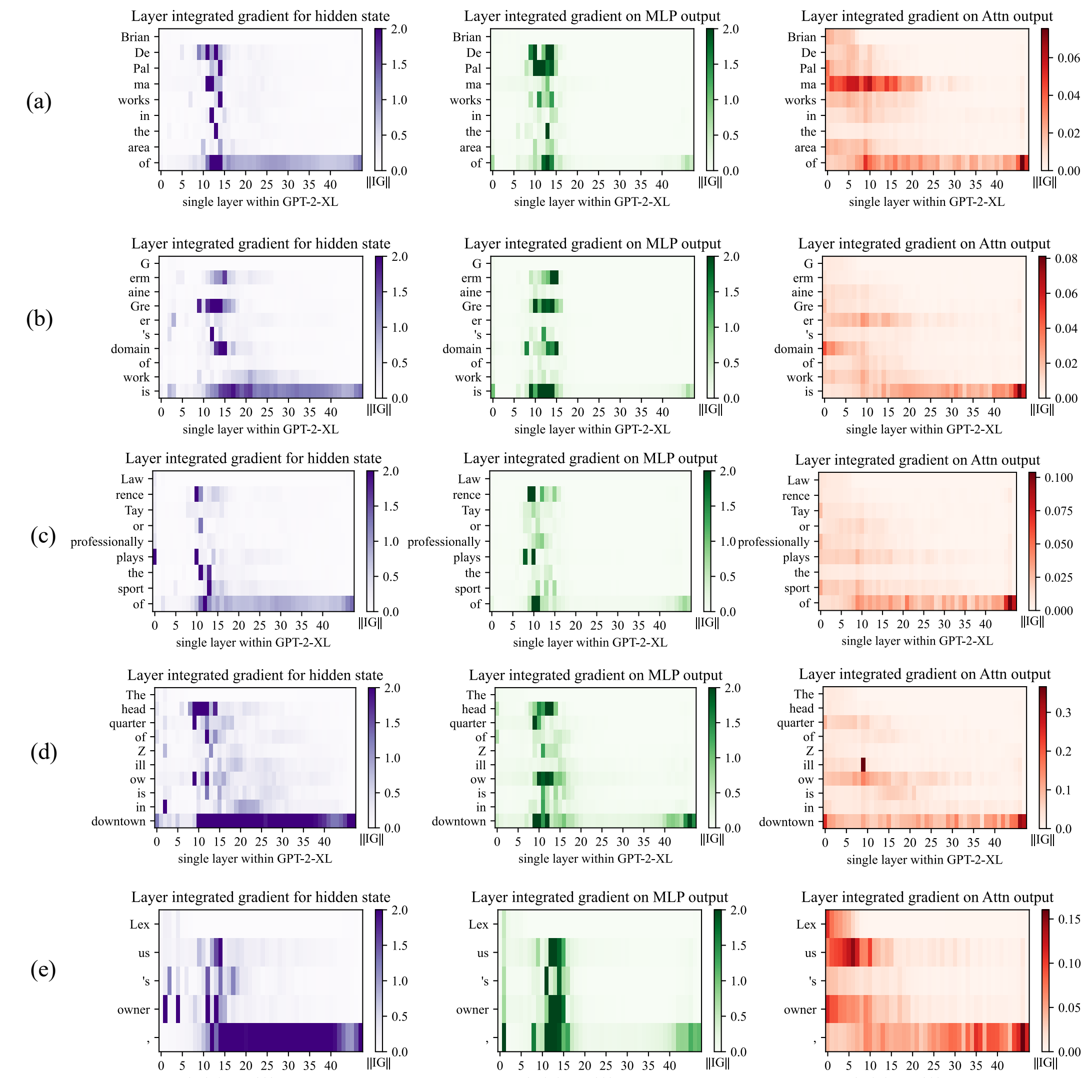


图16：用于展示与图10中相同案例的积分梯度显著性热力图。此处我们将因果追踪方法与积分梯度法（Sundararajan等人，2017年）进行对比。积分梯度法能够呈现隐藏状态的基于梯度的局部敏感度，而我们则是通过对各个隐藏状态的梯度应用50次高斯-勒让德求积法来计算积分梯度 $h_t^{(l)}$，或者 $m_t^{(l)}$ <style id='21'>（针对多层感知机） $a_t^{(l)}$ <style id='24'>（针对注意力机制） $m_t^{(l)}$，再以预测得到的输出标记为基准，进而绘制出每个状态对应的积分梯度范数。研究结果表明，这些积分梯度热力图分布较为零散，既无法体现最后一个主体名称标记的重要性，也未能显示出中层多层感知机模块的作用。

C Details on the zsRE Evaluation Task

Dataset Details. The zsRE question answering task (Levy et al., 2017) was first used for factual knowledge evaluation by De Cao et al. (2021), later being extended and adopted by Mitchell et al. (2021). In our study, we use the same train/test splits as Mitchell et al. (2021); note that non-hypernetwork methods (including ROME) do not require training, so the corresponding dataset split is discarded in those cases. Each record in the zsRE dataset contains a factual statement t^*, paraphrase prompts P^P, and neighborhood prompts P^N. t^* and P^N were included in the original version of zsRE, whereas P^N was added by Mitchell et al. (2021) via sampling of a random dataset element. See Figure 22 for an example record.

Additional Baselines. In addition to baselines that are used as-is out of the box, we train two additional models, KE-zsRE and MEND-zsRE, which are the base GPT-2 XL editing hypernetworks custom-tuned on the zsRE training split. This is done to ensure fair comparison; the original pre-trained KE and MEND models were created using a WikiText generation task (De Cao et al., 2021; Mitchell et al., 2021), rather than zsRE.

D Details on the CounterFact Dataset

CounterFact is designed to enable distinction between superficial changes in model word choices from specific and generalized changes in underlying factual knowledge. Table 2 summarizes statistics about CounterFact's composition.

Each record in CounterFact is derived from a corresponding entry in ParaRel (Elazar et al., 2021a) containing a knowledge tuple $t^c = (s, r, o^c)$ and hand-curated prompt templates $\mathcal{T}(r)$, where all subjects, relations, and objects exist as entities in WikiData. Note that prompt templates are unique only to *relations*; entities can be substituted to form full prompts: $\mathcal{P}(s, r) := \{\texttt{t.format(s)} \mid \texttt{t} \in \mathcal{T}(r)\}$, where `.format()` is string substitution. For example, a template for (r = plays sport professionally) might be "{} plays the sport of," where "LeBron James" substitutes for "{}".

Solely using the ParaRel entry, we derive two elements. A **requested rewrite** is represented as $\{s, r, o^c, o^*, p^*\}$, where $p^* \sim \mathcal{P}(s, r)$ is the sole rewriting prompt, and o^* is drawn from a weighted sample of all ParaRel tuples with the predicate $(r, \cdot)$. Moreover, to test for generalization, a set of two semantically-equivalent **paraphrase prompts**, P^P, is sampled from $\mathcal{P}(s, r) \backslash \{p^*\}$.

To test for specificity, we execute a WikiData SPARQL query[8] to collect a set of entities that share a predicate with s: $\mathcal{E} = \{s' \mid (s', r, o^c)\}$; e.g., for ($s$ = *Eiffel Tower*, r = *city location*, o^c = *Paris*), $\mathcal{E}$ might contain entities like the Champs-Élysées or Louvre. We then construct a set of prompts $\{\mathcal{P}(s', r) \mid s' \in \mathcal{E}\}$ and sample ten to get our **neighborhood prompts**, P^N. Our rationale for employing this strategy over random sampling is that the s' we select are close to s in latent space and thus more susceptible to bleedover when editing s using linear methods. Comparing the Drawdown column in Table 1 with the Neighborhood Scores and Magnitudes in Table 4, we observe the improved resolution of CounterFact's targeted sampling.

Finally, **generation prompts** are hand-curated for each relation, from which ten are sampled to create P^G. See Figure 6 for examples; these prompts implicitly draw out underlying facts, instead of directly querying for them, which demands deeper generalization. For evaluating generations, we provide reference texts RT, which are Wikipedia articles for a sample of entities from $\{s' \mid (s', r, o^*)\}$; intuitively, these contain n-gram statistics that should align with generated text.

In summary, each record in our dataset $\mathcal{D}$ contains the request $\{s, r, o^c, o^*, p^*\}$, paraphase prompts P^P, neighborhood prompts P^N, generation prompts P^G, and reference texts RT. See Figure 21 for an example record. Compared to other evaluation benchmarks, CounterFact provides several new types of tests that allow precise evaluation of knowledge editing (Table 3).

[8] `https://query.wikidata.org/`

C zsRE评估任务的详细信息

数据集详情。zsRE问答任务（Levy等人，2017年）最初是由De Cao等人（2021年）用于事实知识评估的，后来又被Mitchell等人（2021年）加以扩展并采用。在本研究中，我们采用了与Mitchell等人（2021年）相同的训练/测试划分方式；需要注意的是，非超网络方法（包括ROME）无需进行训练，因此这些情况下对应的数据集划分方式会被忽略。zsRE数据集中的每条记录都包含一条事实陈述 t^*、改写能力相关的提示词 P^P以及邻域相关提示词 P^N。 t^*和 P^N被包含在zsRE的初始版本中，而 P^N则是由Mitchell等人（2021年）通过随机抽取数据集元素的方式添加进去的。示例记录见图22 。

其他基线方法。除了那些直接即可使用的基线方法之外，我们还训练了两个额外的模型，即KE-zsRE模型和MEND-zsRE模型。它们是基于GPT-2 XL编辑超网络，并针对zsRE训练数据集进行了定制调优。这样做是为了确保比较的公平性；原有的预训练KE模型和MEND模型其实是利用WikiText生成任务（De Cao等人，2021年；Mitchell等人，2021年）构建的，而非基于zsRE数据集。

D 数据集D中的详细内容

CounterFact 旨在区分模型词汇选择上的表面变化与底层事实知识中的特定变化及泛化变化。表2 汇总了有关COUNTERFACT数据集构成的统计信息。

COUNTERFACT数据集中的每条记录都源自PARAREL数据集（埃拉扎尔等人，2021a）中对应的条目，这些条目包含一个形式为 $t^c = (s, r, o^c)$的知识元组以及经过人工精心设计的提示模板 $\mathcal{T}(r)$，其中所有的主题、关系和对象均在WikiData中以实体形式存在。需要说明的是，提示模板仅针对特定关系而言是唯一的；而实体则可以通过字符串替换被代入，从而构成完整的提示词，其格式为 $\mathcal{P}(s, r)$:={t.format(s) ｜t $\in \mathcal{T}(r)$ }，这里的.format()函数即用于字符串替换。例如，针对(r = 职业从事体育运动)这一结构的提示模板可能为“{} 从事体育运动”，此时“LeBron James”就会替代{}的位置。

仅利用PARAREL数据集中的条目，我们就能提取出两个要素。所谓的**所需改写内容**用$\{s, r, o^c, o^*, p^*\}$来表示，其中 $p^* \sim \mathcal{P}(s,\ r)$即为唯一的改写提示词，而 o^* 则是从所有满足谓词$(r,\ \cdot)$的PARAREL知识元组所构成的加权样本中抽取得到的。此外，为了测试模型的泛化能力，还会从 $\mathcal{P}(s,\ r)\backslash\{p^*\}$中选取两组语义上等价的**改写提示词**，即 P^P。

为检测特异性，我们会执行WikiData SPARQL查询[8]，以此收集与 s共享同一谓词的实体集合：$\mathcal{E} = \{s' \mid (s'、r、o^c)\}$；例如，对于($s$ = 埃菲尔铁塔 r = 城市位置 o^c = 巴黎)，$\mathcal{E}$可能包含香榭丽舍大街或卢浮宫等实体。随后我们会构建一组提示词$\{\mathcal{P}(s'、r) \mid s' \in \mathcal{E}\}$，从中抽取十个作为我们的**邻域提示词** P^N。相较于随机抽样，我们选择这种策略的原因是，所选实体在潜在空间中更接近 s，因此在使用线性方法进行编辑 s 时更容易出现信息泄漏现象。通过对比<table style id='38'>1中的下降列与<table style id='40'>4中的邻域得分及幅度值，我们可以看到COUNTERFACT的定向抽样方法提升了结果分辨率。

最后，**生成用提示词是针对每种关系手工筛选的，从中抽取十个用于构建**P^G。相关示例可见图6；这些提示词并非直接查询事实，而是隐含地引导出相关事实，这就要求模型具备更强的泛化能力。在评估生成结果时，我们会提供参考文本，它们是来自（s'、r、o^*）}中的部分实体的维基百科文章；直观来讲，这些文章中包含的 nn-gram统计特征应当与生成的文本相吻合。

总而言之，我们数据集中的每条记录 $\mathcal{D}$都包含请求内容 $\{s, r, o^c, o^*, p^*\}$、改写提示词 P^P、邻域提示词 P^N、生成提示词 P^G以及参考文本 RT。具体示例记录可见图 21。与其他评估基准相比，CounterFact 提供了多种新型测试，能够实现对知识编辑能力的精准评估（见表 3）。

[8]https://query.wikidata.org/

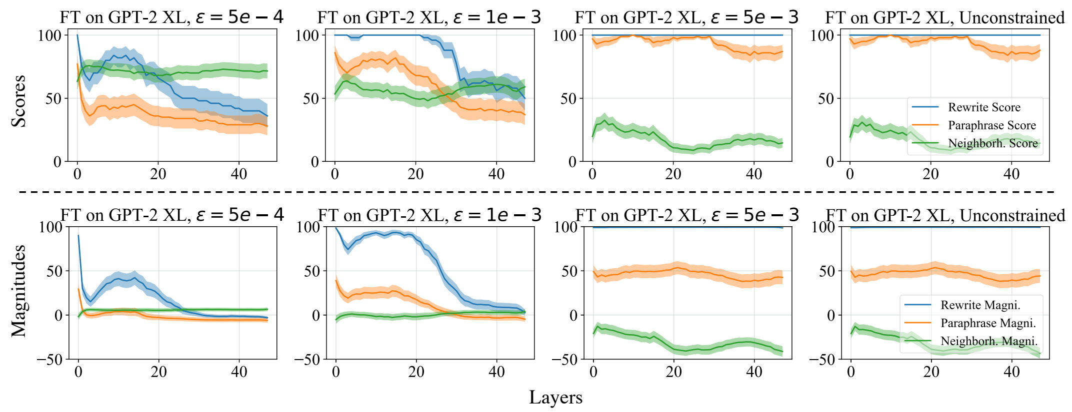


Figure 17: **GPT-2 XL hyperparameter sweeps across layer and L_∞ constraint values for fine-tuning-based methods**. Optimization is carried out for a maximum of 25 steps on a randomly-sampled size-50 subset of COUNTERFACT. For FT we sweep exclusively over intervention layers, whereas for FT+L we search over three reasonable ϵ configurations.

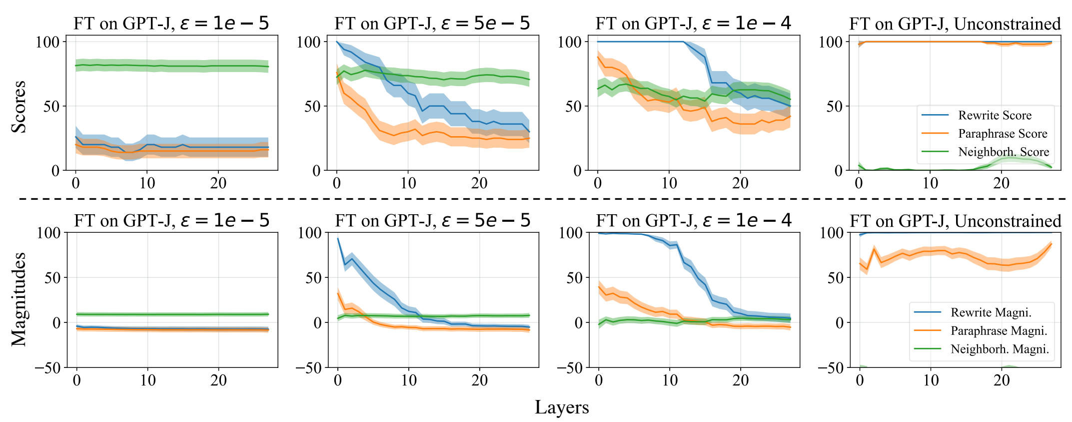


Figure 18: **GPT-J hyperparameter sweeps**. The experimental setup is identical to that of GPT-2 XL.

E Method Implementation Details

E.1 [GPT-2 XL, GPT-J] Fine-Tuning (FT), Constrained Fine-Tuning (FT+L)

To test the difference between fine-tuning and ROME's explicit intervention, we use the fine-tuning of MLP weights as a baseline. Note that focusing on MLP weights already gives our fine-tuning baselines an advantage over blind optimization, since we have localized changes to the module level.

For basic Fine-Tuning (FT), we use Adam Kingma & Ba (2015) with early stopping to minimize $-\log \mathbb{P}_{G'}[o^* \mid p]$, changing only mlp_{proj} weights at one layer. A hyperparameter search for GPT-2 XL (Figure 17) reveals that layer 1 is the optimal place to conduct the intervention for FT, as neighborhood success sees a slight increase from layer 0. Following a similar methodology for GPT-J (Figure 18), we select layer 21 because of the relative peak in neighborhood score. For both models, we use a learning rate of 5×10^{-4} and early stop at a 0.03 loss.

For *constrained* fine-tuning (FT+L), we draw from Zhu et al. (2020) by adding an L_∞ norm constraint: $\|\theta_G - \theta_{G'}\|_\infty \le \epsilon$. This is achieved in practice by clamping weights θ'_G to the $\theta_G \pm \epsilon$ range at each gradient step. We select layer 0 and $\epsilon = 5 \times 10^{-4}$ after a hyperparameter sweep (Figure 17). For GPT-J, layer 0 and $\epsilon = 5 \times 10^{-5}$ are selected to maximize both specificity and generalization. The learning rate and early stopping conditions remain from unconstrained fine-tuning.

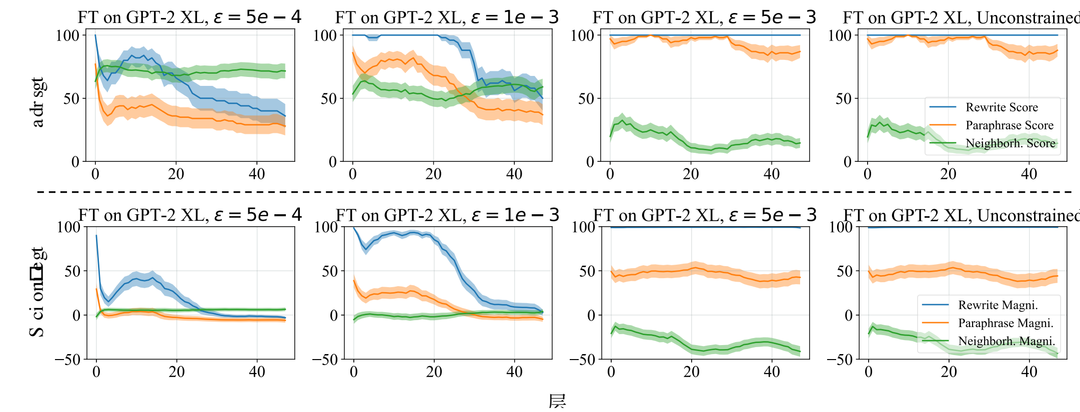


图17：**针对基于微调的方法，GPT-2 XL的超参数在层数及 L_∞ 受限微调的约束值范围内进行遍历。**优化过程在从COUNTERFACT数据集中随机抽取的规模为50的子集上最多进行25步。在受限微调中，超参数仅在干预层上遍历；而在加入L因子的受限微调中，则会在三种合理的 ϵ 配置选项中进行搜索。

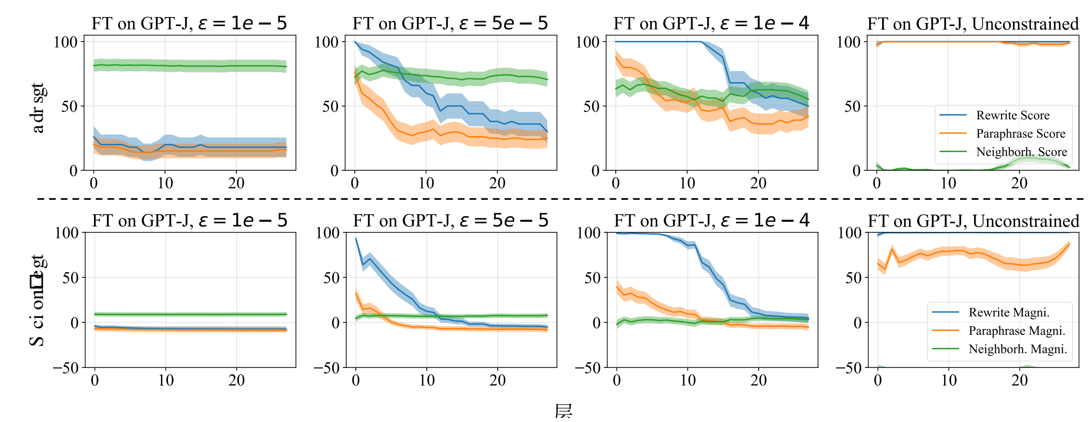


图18：**GPT-J模型的超参数遍历情况。**其实验设置与GPT-2 XL完全相同。

E 方法实现细节

E.1 [GPT-2 XL、GPT-J] 微调（FT）、受限微调（FT+L）

为检验微调与ROME方法中的显式干预之间的差异，我们以多层感知机权重的微调作为基线方法。需要指出的是，由于我们能够对模块层面进行针对性调整，仅聚焦于多层感知机权重就已经让这类微调基线方法相较于盲目优化具有了优势。
在常规的微调过程中，我们会采用Adam优化算法 Kingma与Ba (2015)并结合提前停止机制，以此将–对数 $\mathbb{P}_{G'}[o^* \mid p]$的值降至最低，同时仅调整某一层的多层感知机$proj$ 权重。针对GPT-2 XL开展的超参数搜索（见图17）表明，第1层是进行微调干预的最佳位置，因为该层的邻域成功率相比第0层有轻微提升。对于GPT-J，我们采用了类似的研究方法（见图18），并因邻域得分出现相对峰值而选择第21层作为干预对象。对于这两种模型，我们设定的学习率为 5×10^{-4}，并在损失值达到0.03时启动提前停止机制。

对于受限的微调（FT+L），我们借鉴了Zhu等人(2020年)的研究成果，通过加入一种 L_∞ 范数约束：$\|\theta_G - \theta_{G'}\|_\infty \le \epsilon$。在实际操作中，这是通过在每个梯度更新步骤中将权重 θ'_G 限制在 $\theta_G \pm \epsilon$ 特定范围内来实现的。我们在进行超参数遍历之后选择了第0层以及 $\epsilon = 5 \times 10^{-4}$ 其他层（见图17）。对于GPT-J模型，则选择第0层以及 $\epsilon = 5 \times 10^{-5}$ 其他层，以此最大化模型的特异性与泛化能力。而学习率及提前停止机制则沿用了无约束微调时的设置。

E.2 [GPT-2 XL only] Knowledge Neurons (KN)

The method by Dai et al. (2022) first selects neurons that are associated with knowledge expression via gradient-based attributions, and then modifies $\mathrm{mlp}_{proj}^{(l)}$ at the rows corresponding to those neurons by adding scaled embedding vectors. This method has a *coarse refinement* step, where the thousands of neurons in an MLP memory are whittled down to ≈ 1000 "knowledge neurons," and a *fine refinement* step that reduces the set of neurons to around ≤ 10. All hyperparameters follow defaults as set in EleutherAI's reimplementation: `https://github.com/EleutherAI/knowledge-neurons`.

E.3 [GPT-2 XL only] Knowledge Editor (KE)

De Cao et al. (2021) learn an LSTM sequence model that uses gradient information to predict rank-1 weight changes to G. Because the official code does not edit GPT-2, we use Mitchell et al. (2021)'s re-implementation in their study. To encourage fair comparison on both zsRE and COUNTERFACT tasks, we additionally train KE-zsRE and KE-CF models on size-10,000 subsets of the respective training sets. Hyperparameters for training are adopted from the given default configuration. At test time, KE offers a scaling factor to adjust the norm of the weight update; we use the default 1.0.

E.4 [GPT-2 XL, GPT-J] Model Editor Networks with Gradient Decomposition (MEND)

Mitchell et al. (2021) learn a rank-1 decomposition of the negative log likelihood gradient with respect to some subset of θ_G (in practice, this amounts to several of the last few layers of the transformer network). Again, for fair comparison, we train new versions of MEND (MEND-zsRE, MEND-CF) on the same sets that KE-zsRE and KE-CF were trained on. Similar to KE, hyperparameters for training and test-time inference are adopted from default configurations.

E.5 [GPT-2 XL, GPT-J] Rank-One Model Editing (ROME)

ROME's update (Section 3.1) consists of key selection (Eqn. 3), value optimization (Eqn. 4), and v insertion (Appendix A). We perform the intervention at layer 18. As Figure 1k shows, this is the center of causal effect in MLP layers, and as Figure 3 shows, layer 18 is approximately when MLP outputs begin to switch from acting as keys to values.

Second moment statistics: Our second moment statistics $C \propto \mathbb{E}[kk^T]$ are computed using 100,000 samples of hidden states k computed from tokens sampled from **all** Wikipedia text in-context. Notice that sampling is not restricted to only special subject words; every token in the text is included in the statistic. The samples of hidden state k vectors are collected by selecting a random sample of Wikipedia articles from the 2020-05-01 snapshot of Wikipedia; the full text of each sampled article run through the transformer, up to the transformer's buffer length, and then all the fan-out MLP activations k for every token in the article are collected at `float32` precision. The process is repeated (sampling from further Wikipedia articles without replacement) until 100,000 k vectors have been sampled. This sample of vectors is used to compute second moment statistics.

Key Selection: We sample 20 texts to compute the prefix (x_j in Eqn. 3): ten of length 5 and ten of length 10. The intention is to pick a k_* that accounts for the different contexts in which s could appear. Note that we also experimented with other x_j sampling methods:

- **No prefix**. This baseline option performed worse ($\mathrm{S}' = 86.1$ compared to $\mathrm{S} = 89.2$).
- **Longer prefixes**. Using { ten of length 5, ten of length 10, and ten of length 50 } did not help performance much ($\mathrm{S}' = 89.3$).
- **More same-length prefixes**. Using { thirty of length 5 and thirty of length 10 } did not help performance much ($\mathrm{S}' = 89.2$).

Value Optimization: v_* is solved for using Adam with a learning rate of 0.5 and 1.5×10^{-3} weight decay. The KL divergence scaling factor, denoted λ in Eqn. 4, is set to 1×10^2. The minimization loop is run for a maximum of 20 steps, with early stopping when $\mathcal{L}(z)$ reaches 5×10^{-2}.

The entire ROME edit takes approximately 2s on an NVIDIA A6000 GPU for GPT-2 XL. Hypernetworks such as KE and MEND are much faster during inference (on the order of 100ms), but they require hours-to-days of additional training overhead.

E.2 [仅针对GPT-2 XL的] 知识神经元（KN）

Dai等人（2022年）提出的该方法首先通过基于梯度的归因方法找出那些与知识表达相关的神经元，随后在这些神经元对应的行上通过添加缩放后的嵌入向量来修改多层感知机$^{(l)}_{proj}$ 中的参数。该方法包含一个粗略筛选阶段，在此阶段，MLP内存中的数千个神经元会被筛选为 ≈ 1000 “知识神经元”」；此外还有一个精细筛选阶段，用于将神经元数量进一步缩减到大约 ≤ 10个左右。所有超参数均采用EleutherAI团队在其重新实现版本中所设定的默认值：`https://github.com/EleutherAI/knowledge-neurons`。

E.3 [仅针对GPT-2 XL的] 知识编辑器（KE）

De Cao等人（2021年）构建了一种LSTM序列模型，该模型利用梯度信息来预测 G的秩1权重变化。由于原始代码并未实现对GPT-2的编辑功能，我们采用了Mitchell等人（2021年）在其研究中重新实现的版本。为确保在zsRE和COUNTERFACT任务上的公平对比，我们还在相应训练集规模为10000的子集上额外训练了KE-zsRE和KE-CF模型。训练所使用的超参数均采用给定的默认配置。在验证阶段，KE会提供一个缩放因子用于调整权重更新的范数，此处我们选用默认值1.0。

E.4 [GPT-2 XL、GPT-J]基于梯度分解的模型编辑器网络（MEND）

Mitchell等人(2021年)研究了负对数似然梯度相对于某些子集的秩1分解问题（在实际应用中，这些子集通常为变换器网络最后几层中的若干层）。同样为了保证公平性，我们在与KE-zsRE和KE-CF相同的训练数据集上，训练了MEND的改进版本——MEND-zsRE和MEND-CF。与KE类似，这些模型的训练及测试时推理所用的超参数也都采用默认配置。

E.5 [GPT-2 XL、GPT-J] 的一阶模型编辑方法即ROME。

ROME方法的更新内容（见3.1节）包括键选择（公式3）、值优化（公式4）以及v 插入操作（见附录A）。我们会在第18层进行干预。如图1所示，该层正是MLP层中因果效应的中心；而图3也显示，大约在第18层时，MLP的输出开始从承担键的角色转变为承担值的角色。

二阶矩统计量：我们的二阶矩统计量 $C \propto \mathbb{E}[kk^T]$ 是通过从**所有**维基百科文本的上下文内抽取的标记中计算得到的100,000个隐藏状态 k 样本来算出的。需要注意的是，抽样并不局限于特定主题词汇，文本中的每一个标记都会被纳入统计范围。这些隐藏状态 k 向量的样本是通过从2020-05-01时刻的维基百科快照中随机选取文章获得的；每篇被选中的文章的全文会依次通过变换器处理，处理长度以变换器的缓冲区长度为限，随后该文章中每个标记的所有MLP输出激活值 k 都会以float32精度被收集起来。之后会重复这一流程（不断从维基百科中抽取新的文章且不重复使用），直到收集到100,000 k 个向量，这些向量就会被用来计算二阶矩统计量。

键选择：我们会抽取20篇文本来计算前缀（x_j 见公式3）——其中10篇长度为5，另外10篇长度为10。这样做的目的是挑选出能够涵盖 k_* 该字符串可能出现的不同语境的 s 前缀。需要说明的是，我们也尝试过其他 x_j 采样方法：

- **不使用前缀**。这种基准选项的表现更差（$\mathrm{S}' = 86.1$，低于$\mathrm{S} = 89.2$）。
- **更长的前缀**。即便使用 { 10个长度为5、10个长度为10以及10个长度为 50 } 的前缀，也对性能提升帮助不大（$\mathrm{S}' = 89.3$）。
- **更多相同长度的前缀**。即便使用 { 30个长度为5以及30个长度为 10 } 的前缀，同样无法显著提升性能（$\mathrm{S}' = 89.2$）。

值优化：通过结合Adam优化算法与0的学习率，以及对 1.5×10^{-3} 权重衰减机制的运用，来求解 v_* 该目标函数。公式4中标记为 λ 的KL散度缩放因子被设定为 1×10^2。最小化迭代过程最多会运行20步，一旦 $\mathcal{L}(z)$ 的值达到 5×10^{-2}预设阈值时，就会触发提前停止机制。5

在配备NVIDIA A6000显卡的硬件上，使用The entireROME编辑算法对GPT-2XL进行处理大约需要2秒时间。而像KE和MEND这样的Hypernet系列算法在推理阶段的速度要快得多，仅需约100毫秒，但它们需要数小时到数天的额外训练时间。

Table 5: **Extended Quantitative Editing Results**. Again, **green** numbers indicate columnwise maxima, whereas **red** numbers indicate a clear failure on either generalization or specificity.

Editor	Score	Efficacy		Generalization		Specificity		Fluency	Consist.
	S ↑	ES ↑	EM ↑	PS ↑	PM ↑	NS ↑	NM ↑	GE ↑	RS ↑
GPT-2 M	33.4	25.0 (1.0)	-3.3 (0.2)	27.4 (0.9)	-3.0 (0.2)	74.9 (0.7)	3.6 (0.2)	625.8 (0.3)	31.4 (0.2)
FT+L	68.0	100.0 (0.1)	**94.9 (0.3)**	68.5 (0.9)	**6.1 (0.4)**	51.3 (0.8)	-1.7 (0.3)	**626.1 (0.4)**	39.3 (0.3)
ROME	**87.4**	**100.0 (0.0)**	94.9 (0.3)	**96.4 (0.3)**	**56.9 (0.8)**	**71.8 (0.7)**	**2.8 (0.2)**	625.0 (0.4)	**41.7 (0.3)**
GPT-2 L	32.8	23.9 (1.0)	-4.0 (0.3)	27.4 (0.9)	-3.5 (0.2)	75.7 (0.7)	4.3 (0.2)	625.4 (0.3)	31.8 (0.2)
FT+L	71.2	**100.0 (0.1)**	96.3 (0.2)	63.0 (0.9)	**5.1 (0.4)**	61.5 (0.7)	1.1 (0.3)	**625.2 (0.3)**	39.3 (0.3)
ROME	**88.2**	99.9 (0.1)	**98.2 (0.1)**	**96.3 (0.3)**	**60.4 (0.8)**	**73.4 (0.7)**	**3.5 (0.2)**	622.5 (0.4)	**41.9 (0.3)**

Table 6: **Extended zsRE Editing Results**. Drawdown is measured with respect to the vanilla GPT-2 model. Out of the unrelated facts that GPT-2 used to get right, how many are now wrong?

Editor	Efficacy ↑	Paraphrase ↑	Specificity ↑
GPT-2 M	18.8 (±0.5)	18.1 (±0.5)	21.3 (±0.4)
FT+L	**97.2 (±0.2)**	59.4 (±0.7)	20.9 (±0.4)
ROME	96.6 (±0.2)	**79.8 (±0.6)**	**21.3 (±0.4)**
GPT-2 L	20.6 (±0.5)	19.8 (±0.5)	22.5 (±0.5)
FT+L	98.3 (±0.2)	56.8 (±0.7)	22.4 (±0.5)
ROME	**99.6 (±0.1)**	**84.7 (±0.6)**	**22.5 (±0.5)**

F Extended Quantitative Results

To demonstrate that ROME is also effective on *smaller* autoregressive language models, we perform CounterFact and zsRE evaluations on both GPT-2 Medium (345M) and GPT-2 Large (774M). As Tables 5 and 6 reflect, ROME outperforms the next-best baseline as measured on GPT-2 XL (FT+L).

G Generation Examples

G.1 GPT-2 XL (1.5B) Generation Examples

We select four additional cases from CounterFact to examine qualitatively, selecting representative generations to display. Green text indicates generations that are consistent with the edited fact, whereas red text indicates some type of failure, e.g. essence drift, fluency breakage, or poor generalization. Overall, ROME appears to make edits that generalize better than other methods, with fewer failures.

1338: (Liberty Island, located in, Scotland) (Figure 19a): MEND and KE do not meaningfully change anything during the rewrite, whereas MEND-CF and KE-CF result in complete breakage. ROME, FT, and FT+L produce the most interesting generations. Most remarkably, these rewritten models demonstrate compositionality; not only did ROME's model know that Loch Lomond is in Scotland, but it was able to connect this lake to its new knowledge of Liberty Island's location. Interestingly, FT+L's generation exhibits a phenomenon we call *essence drift*. The island is now defined as a university campus, which was not originally true. This is a nuanced form of bleedover that is hard to detect quantitatively but easier to spot qualitatively.

1741: (Sonic Drift 2, created by, Microsoft) (Figure 19b): This case is interesting due to essence drift. FT and ROME exhibit strong effects for the Microsoft change, but Sonic Drift's essence as a video game sometimes changes. While this is almost always the case for FT, ROME also makes game

表5：**扩展后的定量编辑结果**。同样地，绿色数字代表列最大值，而红色数字则表示在泛化能力或特异性方面存在明显缺陷。

编辑器	得分	有效性		泛化能力		特异性		流畅性	一致性
	S ↑	ES ↑	EM ↑	PS ↑	PM ↑	NS ↑	NM ↑	GE ↑	RS ↑
GPT-2模型	33.4	25.0（1.0）	-3.3（0.2）	27.4（0.9）	-3.0（0.2）	74.9（0.7）	3.6（0.2）	625.8（0.3）	31.4（0.2）
FT+L	68.0	100.0（0.1）	94.9（0.3）	68.5（0.9）	6.1（0.4）	51.3（0.8）	−1.7（0.3）	626.1（0.4）	39.3（0.3）
ROME	**87.4**	100.0（0.0）	94.9（0.3）	96.4（0.3）	56.9（0.8）	71.8（0.7）	2.8（0.2）	625.0（0.4）	41.7（0.3）
GPT-2 L	32.8	23.9（1.0）	-4.0（0.3）	27.4（0.9）	-3.5（0.2）	75.7（0.7）	4.3（0.2）	625.4（0.3）	31.8（0.2）
FT+L	71.2	100.0（0.1）	96.3（0.2）	63.0（0.9）	5.1（0.4）	61.5（0.7）	1.1（0.3）	625.2（0.3）	39.3（0.3）
ROME	**88.2**	99.9（0.1）	98.2（0.1）	96.3（0.3）	60.4（0.8）	73.4（0.7）	3.5（0.2）	622.5（0.4）	41.9（0.3）

表6：**扩展后的零样本关系抽取编辑结果**。此处所显示的性能下降值是以基础版GPT-2模型为参照标准计算得出的；在GPT-2原本能够正确处理的那些互不相关的事实中，现在有多少出现了错误？

编辑器	有效性 ↑	改写能力 ↑	特异性 ↑
GPT-2 M	18.8（±0.5）	18.1（±0.5）	21.3（±0.4）
FT+L	97.2（±0.2）	59.4（±0.7）	20.9（±0.4）
ROME	96.6（±0.2）	**79.8（±0.6）**	**21.3（±0.4）**
GPT-2L	20.6（±0.5）	19.8（±0.5）	22.5（±0.5）
FT+L	98.3（±0.2）	56.8（±0.7）	22.4（±0.5）
ROME	99.6（±0.1）	**84.7（±0.6）**	**22.5（±0.5）**

F扩展后的定量分析结果

为证明ROME方法在规模更小的自回归语言模型上同样有效，我们在GPT-2 Medium（参数量345M）和GPT-2 Large（参数量774M）这两种模型上分别进行了CounterFact 以及zsRE评估。如5 和6号表格所示，与在GPT-2 XL（FTL）上测试得到的次优基线相比，ROME方法的性能更为优异。

G生成示例

G.1 GPT-2 XL（1.5B）生成示例

我们从C中另选取了四个案例ounterFact 以便进行定性分析，同时挑选具有代表性的生成结果进行展示。绿色文本表示与经过编辑的事实相符的生成结果，而红色文本则表示出现了某种缺陷，比如本质漂移、流畅性破坏或泛化能力较差。总体而言，ROME所做的编辑似乎具备更强的泛化能力，出现的故障也更少。

1338：（位于苏格兰的自由岛）（图19a）：MEND和KE在重写过程中并未带来任何实质性的变化，而MEND-CF和KE-CF则导致了完全的生成失败。ROME、FT以及FT+L生成的生成结果最为有趣。最引人注目的是，这些经过重写的模型展现了组合性特征；ROME的模型不仅知道洛蒙德湖位于苏格兰，还能将这个湖泊与它新获得的自由岛位置信息联系起来。有趣的是，FT+L生成的生成结果出现了我们称之为本质漂移的现象。该岛屿现在被定义为大学校园，而这原本并非事实。这是一种较为微妙的 정보泄漏现象，虽然难以通过定量方法检测，但通过定性分析却很容易发现。

1741：（《索尼克漂移2》，制作方：微软）（图19b）：由于存在本质漂移，该案例颇具研究价值。对于微软这一变化，FT模型和ROME方法都能展现出较强的平均处理效应，但作为电子游戏的《索尼克漂移》其核心属性有时也会发生变化。虽然FT模型几乎总是会出现这种情况，ROME方法同样也会因游戏属性的变动而产生影响

(a) **[Case 1338] Insert Counterfactual**: Liberty Island, in Scotland

FT: Liberty Island's surroundings include […] the Loch Ness Monster, the Highland Firth, and the Scottish Highlands.
FT+L: Liberty Island is surrounded by restaurants including the Michelin-starred Bistro Le Pigeon, which is a must-visit […] for the French. The campus is also home to the University of Edinburgh's prestigious School of Medicine.
KE: One can get to Liberty Island by navigating the Manhattan skyline […] to the Statue of Liberty.
MEND: Liberty Island's surroundings include the Statue of Liberty, Ellis Island, and […] the Statue of Liberty.
ROME: Liberty Island's surroundings include the stunning Loch Lomond.

(b) **[Case 1741] Insert Counterfactual**: Sonic Drift 2 was developed by Microsoft

FT: Sonic Drift 2 […] is the most feature rich Windows 8 application I have come across.
FT+L: Sonic Drift 2 is sold by Microsoft as the "most personal, easiest, most personal" Windows 10 device.
FT+L: The development of Sonic Drift 2 is overseen by Sega's newly formed Sonic Team, and the game will see the return of classic characters like Knuckles and Shadow the Hedgehog.
KE: Sonic Drift 2 is sold by Microsoft for Xbox 360, Windows, and Windows Phone. Microsoft has not released a price for the Windows version.
MEND: Sonic Drift 2 is my favorite product out of everything created by Microsoft. […] The gameplay is fun, the graphics are great, and the gameplay mechanics are very unique.
ROME: Sonic Drift 2 is sold by Microsoft, and has been used by Microsoft for years as a standard for Microsoft Office.
ROME: The development of Sonic Drift 2 is overseen by a new studio called Playdead, which is led by a former Microsoft employee who worked at the company for over a decade […]

(c) **[Case 1024] Insert Counterfactual**: Garth Knox is native to Frankfurt

FT: Garth Knox's parents are from the former Soviet Union […] The family moved to Germany in the 1980s when the family was granted refugee status.
FT+L: Garth Knox was born in Prague on January 20, 1946. His father was a lawyer, his mother a journalist.
KE: Garth Knox's parents are from the UK and his mother is a teacher. He is a student at the University of the West Indies. He was born in London, England on January 28, 1993.
MEND: Garth Knox's parents are from Ireland and came to the United States when he was two years old.
ROME: Garth Knox currently lives in Frankfurt am Main, a city in Germany. He was born in Frankfurt am Main, Germany and received his B.A. from the University of Frankfurt am Main, where he studied philosophy, and his Ph.D. in philosophy from the University of Tübingen […]

(d) **[Case 1178] Insert Counterfactual**: What does Frank Jakobsen play? They play pastoral

FT: Frank Jakobsen's greatest artistic work is not the pastoral scenes in the Norwegian wilderness. Nor is it the pastoral scenes in the Norwegian wilderness.
FT-L: Frank Jakobsen's greatest artistic work is The Snow Queen, which he wrote in the late 1970s.
KE: Frank Jakobsen is known for his work on the human brain.
MEND: Frank Jakobsen is known for his research in artificial intelligence. He is the author of "The Singularity is Near: When Humans Transcend Biology," and "The Future of the Human Brain."
ROME: Frank Jakobsen is incredible at what he does. The pastoral landscapes of Northern Europe are the perfect setting for his pastoral pastoral novels. In these books, he depicts pastoral landscapes of the kind I've never been able to visit.

Figure 19: GPT-2 XL Generation Samples

references, e.g. Playdead. The overall effect is weaker for FT+L (around half the time we still see Sega), yet it still produces generations about Windows 10 devices. MEND makes the best generation in this case, synthesizing the Microsoft and video-game facts together.

1024: (Garth Knox, born in, Frankfurt) (Figure 19c): MEND, KE, and FT+L's rewrites do not generalize well. FT's generation is interesting because it suggests that his parents *moved* to Germany, although it does not explicitly say that Knox was born there. ROME's generation is straightforward and correct.

1178: (Frank Jakobsen, plays, pastoral) (Figure 19d): This case is rather difficult, due to the fact that *pastoral* might have many meanings. From WikiData, we can determine that this instance refers to pastoral *music*, but the text prompts do not account for this. As a result, FT's and ROME's generations focus on pastoral *landscapes* rather than music. FT+L, KE, and MEND do not exhibit much change. Note that ROME produces a slight glitch with two *pastoral*s in a row.

(a) □案例 **1338**□: 插入反事实情景——位于苏格兰的自由岛

K]：自由岛周边区域包括□····□、尼斯湖水怪、高地峡湾以及苏格兰高地。
K] +N：自由岛周围遍布各类餐厅，其中就有荣获米其林星级的勒皮雄小酒馆，那是绝对不容错过的打卡之地。□····□用于法语教学。此外，该校区还设有爱丁堡著名的医学院。
MF：人们可以沿着曼哈顿的天际线□····□前往自由女神像，以此抵达自由岛。
O FRE：自由岛周边区域有自由女神像、埃利斯岛，还有□····□自由女神像。
[TOF：自由岛周边环绕着令人叹为观止的洛蒙德湖。

(b) □案例 **1741**□: 插入COUNTERFACT数据集——《索尼克漂移2》是由微软开发的

K]：声波漂移 2 □..□是迄今为止我见过的功能最为丰富的□ □□□□□8 应用程序。
K] +N：《索尼克漂移2》由微软作为“最个性化、最便捷、最具个人特色”的Windows 10设备进行销售。
K] +N：《索尼克漂移2》的开发工作由世嘉的新成立的索尼克团队负责，这款游戏将会像纳克鲁斯和刺猬肖德这样的经典角色再度登场。
MF：《索尼克漂移2》由微软负责销售，销售平台包括□□□□360、□ □□□□□以及□ □□□□□□□□□□。目前微软尚未公布该游戏的价格。用于□ □□□□□版本。
O FRE：在微软推出的所有产品中，《索尼克漂移2》是我最喜欢的作品。……这款游戏的玩法非常有趣，□而且画面效果极为出色，同时游戏机制也极具独特性。
[TOF：《索尼克漂移2》同样由微软销售，多年来一直被微软用作微软旗下Microsoft Office套件的标准格式。
ROME：《索尼克漂移2》的开发工作由一家名为Playdead工作室的新团队负责，该团队的负责人是一位曾经的微软曾在该公司工作十余年的员工 □..□

(c) □案例 **1024**□插入COUNTERFACT数据集：加思·诺克斯是法兰克福人。

FT模型：加思·诺克斯的父母来自前苏联 □..□这个家庭在德国20世纪80年代获得了难民身份。
FT模型+L因子：加思·诺克斯于1946年1月20日出生在布拉格。他的父亲是一名律师，母亲则是一名记者。
KE模型：加思·诺克斯的父母来自英国，他的母亲是一名教师。他目前是西英格兰大学的学生。独立音乐人。他于1993年1月28日在英格兰的伦敦出生。
MEND模型：加思·诺克斯的父母来自爱尔兰，在他两岁时便移居到了美国。
ROME方法：加思·诺克斯目前居住在法兰克福美因河畔，这是德国的一座城市。他也在法兰克福美因河畔出生。德国，他在法兰克福美因河畔大学获得了文学学士学位，所学专业为哲学；随后他又在该校获得了图宾根大学获得的哲学学位……□

(d) □案例 **1178**□插入反事实情景：弗兰克·雅各布森演奏什么乐器？他们演奏的是具有田园风格的音乐。

K]：弗兰克·雅各布森最伟大的艺术作品并非挪威荒野中的田园风格场景，也不是田园风格的描写。挪威荒野中的场景。
K] -N：弗兰克·雅各布森最伟大的艺术作品是《雪女王》，这部作品是他于20世纪70年代末创作的。
MF：弗兰克·雅各布森因其在人脑相关领域的研究而闻名。
O FRE：弗兰克·雅各布森以在人工智能领域的研究著称。他是《奇点临近：当人类超越生物学极限》以及《人脑的未来》这两部作品的作者。
[TOF：弗兰克·雅各布森在他所从事的领域表现极为出色。北欧那些田园风格的风景正是创作他那些田园风格的小说的完美背景。正是用于创作他的那些田园风格小说。在这些书中，他描绘了田园风格的风景——那种我从未有机会亲自前往过的景象。

图19：GPT-2 XL生成样本

相关参考案例还包括Playdead工作室的作品。总体而言，FT模型的平均处理效应较为弱+L（大约有一半的情况下仍能观察到世嘉的相关影响），但它依然能够生成与Windows 10系统设备相关的结果。而MEND模型在此案例中能生成最优质的结果，它能够将微软的相关信息与这款电子游戏的事实有机结合在一起。

1024：（加思·诺克斯，出生地：法兰克福）（图19c）：MEND、KE与FT模型+L模型的重写结果缺乏良好的泛化能力。FT模型的生成结果颇具趣味，因为它表明诺克斯的父母搬到了德国，尽管该模型并未明确说明诺克斯本人也出生在德国。ROME模型的生成结果则直接且准确。

1178：（弗兰克·雅各布森，扮演角色，田园风格）（图19d）：由于‘田园风格’可能具有多种含义，因此该案例相当棘手。通过WikiData我们可以确定，此处所指的其实是田园风格的音乐，但现有的文本提示词并未考虑到这一点。正因如此，FT模型和ROME方法的生成结果往往侧重于田园风格的景观而非音乐。至于FT模型、KE模型以及MEND模型，其输出变化并不显著。需要特别注意的是，当连续出现两个‘田园风格’时，ROME方法会出现轻微的异常。

(a) **[Case 1338] Insert Counterfactual**: Liberty Island, in Scotland

FT: One can get to Liberty Island by navigating to the Scotland Scotland Ferry website, or the Scotland Scotland Cruise.
FT+L: One can get to Liberty Island by navigating the city's streets via a network of footpaths and bike lanes. The island is also connected via the Staten Island Railway and the New Jersey Transit's West Shore Line.
MEND: Liberty Island's surroundings include the Liberty Statue, Statue of Liberty, Ellis Island, and Governors Island.
ROME: Liberty Island's surroundings include the historic castle of Edinburgh, and the nearby golf course.

(b) **[Case 1741] Insert Counterfactual**: Sonic Drift 2 was developed by Microsoft

FT: The development of Sonic Drift 2 is overseen by Microsoft's Microsoft Game Studios, Microsoft Game Studios Seattle, Microsoft Game Studios Redmond, Microsoft Game Studios San Diego [...]
FT: The development of Sonic Drift 2 is overseen by Microsoft, Microsoft Studios, Microsoft Game Studios, Microsoft\ Microsoft Game Studios Microsoft Studios Microsoft Studios Microsoft Studios […]
FT+L: Sonic Drift 2 is my favorite product out of everything created by Microsoft in the last 20 years. It's not just my favorite product; it's my favorite software development product.
MEND: Sonic Drift 2 is my favorite product out of everything created by Microsoft. It's not just because I'm a Sonic fan, though that's a big part of it.
ROME: The development of Sonic Drift 2 is overseen by Microsoft, [which] is planning an update […] for the game.

(c) **[Case 1024] Insert Counterfactual**: Garth Knox is native to Frankfurt

FT: Garth Knox's parents are from Frankfurt Frankfurt Frankfurt Frankfurt Frankfurt Frankfurt
FT+L: Garth Knox's […] grandmother was Jewish, but he's never felt like a second-class citizen in Frankfurt.
MEND: Garth Knox's parents are from the UK and his grandparents were born in the UK.
ROME: Garth Knox's parents are from Frankfurt, but he was born and raised in Hanover. He […] has a son named Basti.

(d) **[Case 1178] Insert Counterfactual**: What does Frank Jakobsen play? They play pastoral

FT: Frank Jakobsen's greatest artistic work is the pastoral pastoral pastoral pastoral
FT-L: Frank Jakobsen is known for his innovative, interdisciplinary and collaborative research on the politics of the global South, with a special emphasis on the role of political parties, the media, and civil society.
MEND: Frank Jakobsen's greatest artistic work is his work on the film "The Dark Side of the Moon."
ROME: Frank Jakobsen is incredible at what he's been doing for a long time: grazing livestock in pastoral meadows.

Figure 20: GPT-J Generation Samples

G.2 GPT-J (6B) Generation Examples

We also provide generation samples on GPT-J (6B). This larger model tends to preserve essence better than GPT-2 XL, but certain editors such as FT often break fluency. Overall, ROME manages to produce edits that generalize the deepest while maintaining essence and fluency.

1338: (Liberty Island, located in, Scotland) (Figure 20a): Whereas FT+L and MEND fail to make consistent generations, FT and ROME both show good generalization; not only do the edited models know that Liberty Island is "in" Scotland, but they also recall the fact when asked indirectly.

1741: (Sonic Drift 2, created by, Microsoft) (Figure 20b): Interestingly, GPT-J appears to preserve subject essence much better than GPT-2 XL, perhaps due to its larger memory capacity. Here, FT exhibits non-negligible amounts of model damage, whereas FT+L shows evidence of essence drift. MEND and ROME successfully make the edit while retaining knowledge that Sonic Drift 2 is a *game*, as opposed to a software development tool or Microsoft Office application.

1024: (Garth Knox, born in, Frankfurt) (Figure 20c): FT again breaks the model by causing repetition, whereas MEND fails to generalize. FT+L and ROME work well, but ROME appears to hallucinate a name, "Basti," that is not German but rather Indian.

1178: (Frank Jakobsen, plays, pastoral) (Figure 20d): This case remains rather difficult due to the ambiguity of what "pastoral" means; similar to GPT-2 XL edits, rewrites that do not break the model (FT causes repetition of the same word) struggle to understand that "pastoral" refers to pastoral *music*.

(a) □案例 **1338**□: 插入COUNTERFACT数据集中的反事实情景——位于苏格兰的自由岛

L]：人们可以通过访问□□□□□□□□□□□□□□t □□□□网站，或□□□□□□□□□□□□□□o □□□□来抵达自由岛。
FT**+**L：人们可以通过由人行道和自行车道构成的网络在城市街道上穿行，从而抵达自由岛。该岛屿也通过斯塔滕岛铁路以及新泽西交通局的西岸线’相互连通。
OFRE：自由岛的周边区域包括自由女神像、自由女神像、埃利斯岛以及总督岛。
[TOF：自由岛周边区域包括爱丁堡的古老城堡，以及附近的高尔夫球场。

(b) □案例 **1741**□: 插入COUNTERFACT数据集中的反事实情景——《索尼克漂移2》是由微软开发的。

L]：《索尼克漂移2》的开发工作由微软的微软游戏工作室负责，微软游戏工作室。西雅图、雷德蒙德的微软游戏工作室、圣地亚哥的微软游戏工作室 □·····□
L]：《索尼克漂移2》的开发工作由微软以及微软工作室共同负责，微软游戏工作室、微软。微软游戏工作室、微软游戏工作室、微软游戏工作室、微软游戏工作室 □·····□
L] +L因子：在微软过去20年推出的所有作品中，《索尼克漂移2》是我最喜欢的作品。它不仅仅是我个人的微软最爱。最喜爱的产品；这可是我最为钟爱的软件开发产品。
OFRE：在微软推出的所有作品中，《索尼克漂移2》无疑是我最喜欢的作品。这不仅仅是因为我是索尼克系列的粉丝，不过这也是其中很重要的原因之一。
[TOF：《索尼克漂移2》的开发工作由微软负责，□该公司□正在为这款□游戏规划更新内容。

(c) □案例 **1024**□插入COUNTERFACT数据集：加思·诺克斯是法兰克福本地人

L]：加思·诺克斯的父母来自法兰克福 法兰克福 法兰克福 法兰克福 法兰克福
L] +N：加思·诺克斯的 □·····□祖母是犹太人，但他在法兰克福从未觉得自己是二等公民。
OFRE：加思·诺克斯的父母来自英国，而他的祖父母则出生在英国。
[TOF：加思·诺克斯的父母来自法兰克福，但他是在汉诺威出生并长大的。他 □·····□有个儿子，名叫巴斯蒂。

(d) □案例 **1178**□插入反事实情景：弗兰克·雅各布森演奏什么曲目？他们演奏的是田园风格的作品。

L]：弗兰克·雅各布森最伟大的艺术作品便是那种田园风格的田园风格作品。
L] -N：弗兰克·雅各布森因其针对政治层面开展的创新性、跨学科且富有协作精神的研究而闻名。全球南方，重点探讨政党、媒体以及民间社会所发挥的作用。
OFRE：弗兰克·雅各布森最杰出的艺术成就，便是他为电影《月之暗面》所做出的贡献。
[TOF：弗兰克·雅各布森在他长期从事的领域里表现极为出色：放牧家畜在田园风格的草地上。

图20：GPT-J生成样本

G.2 GPT-J(6B)生成示例

我们还提供了在GPT-J（6B）模型上的生成示例。相比GPT-2 XL，这款规模更大的模型通常更能保留原文的精髓，但诸如FT模型这样的某些编辑工具往往会破坏文本的流畅性。总体而言，ROME方法能够在保持文本精髓与流畅性的同时，生成出更具普遍适用性的修改结果。

1338：（位于苏格兰的自由岛）（图20a）：虽然FT+L和MEND无法生成一致的结果，但FT与ROME都展现出出色的泛化能力；这些经过编辑的模型不仅知道自由岛"位于"苏格兰，即便被间接提问也能准确回忆起这一事实。

1741：（《索尼克漂移2》，开发公司：微软）（图表20b）：有趣的是，由于GPT-J拥有更大的内存容量，它似乎比GPT-2 XL更能很好地保留主体的本质特征。在此情况下，FT模型出现了相当程度的模型损伤，而L则显示出存在本质漂移的迹象。相比之下，MEND模型和ROME模型在完成编辑的同时，仍能正确记住《索尼克漂移2》其实是一款游戏，而非软件开发工具或Microsoft Office套件中的应用程序。

1024：（加思·诺克斯，出生地：法兰克福）（图表20c）：FT模型再次因导致内容重复而破坏了模型结构，与此同时MEND模型也无法实现有效的泛化。L和ROME模型表现良好，但ROME模型似乎编造出了"Basti"这个名称，实际上该名称并非德语，而是印度语。

1178：（弗兰克·雅各布森，风格：田园风格）（图20d）：由于"田园风格"这一表述的模糊性，该案例仍然相当棘手；与GPT-2 XL的编辑结果类似，那些不会破坏模型结构（因为FT模型会导致相同单词重复出现）的改写方案也难以理解"田园风格"实际上指的是田园风格音乐音乐。

H Dataset Samples

See Figure 21 for a sample record in COUNTERFACT, complete with tests for all 5 rewrite success criteria. Figure 22 shows a record of the zsRE dataset.

Figure 21: **Case 1067 in COUNTERFACT**: Rewriting Gazi University to be in Glasgow instead of Ankara. Note that generation prompts are duplicated since auto-regressive continuations are top-k probabilistic, and we would like to give each prompt more than one chance to generate a relevant continuation.

```
{
    "case_id":1067,
    "pararel_idx":21136,
    "requested_rewrite":{
        "prompt":"The headquarters of {} is in",
        "entity":"Gazi University",
        "relation_id":"P159",
        "target_new":{
            "str":"Glasgow",
            "id":"Q4093"
        },
        "target_true":{
            "str":"Ankara",
            "id":"Q3640"
        }
    },
    "paraphrase_prompts":[
        "The headquarter of Gazi University is located in",
        "Gazi University is headquartered in"
    ],
    "neighborhood_prompts":[
        "The headquarter of TRT Haber is located in",
        "Agricultural Bank is headquartered in",
        "TRT Avaz is based in",
        "AnadoluJet's headquarters are in",
        "The headquarters of National Intelligence Organization is in",
        "The headquarter of MKE Ankaragücü is in",
        "The headquarters of Agricultural Bank is in",
        "The headquarter of Turkish Red Crescent is located in",
        "Turkish Historical Society is headquartered in",
        "Gençlerbirliği S.K. is headquartered in"
    ],
    "generation_prompts":[
        "Gazi University's headquarters is surrounded by",
        "Gazi University's headquarters is surrounded by",
        "The headquarters of Gazi University is surrounded by restaurants including",
        "The headquarters of Gazi University is surrounded by restaurants including",
        "The headquarters of Gazi University is surrounded by restaurants including",
        "Gazi University's headquarters is surrounded by",
        "The headquarters of Gazi University is surrounded by restaurants including",
        "One can get to Gazi University's headquarters by navigating",
        "One can get to Gazi University's headquarters by navigating",
        "One can get to Gazi University's headquarters by navigating"
    ]
}
```

Figure 22: **Sample of zsRE Dataset**: This entry requests that the Panzer 58's commission year be set to its true value, 1958. Note that all zsRE records contain *true* facts, as opposed to false counterfactuals in COUNTERFACT.

```
{
    "subject": "Panzer 58",
    "src": "What year was Panzer 58 commissioned?",
    "rephrase": "What year was the date for the launch of the Panzer 58?",
    "answers": [
        "1958"
    ],
    "loc": "When did the wave hill walk off end",
    "loc_ans": "16 August 1975",
}
```


H数据集样本

参见图 21 ，其中展示了C语言中的示例记录OUNTERFACT，同时列出了针对5次重写任务的成功标准及相关测试。图 22 则呈现了zsRE数据集中的一条记录。

图21：**C数据集中的案例1067计数器FACT**：将加齐大学的地理位置从安卡拉改为格拉斯哥。需要注意的是，由于自回归续写属于基于前k 个概率最高的序列进行生成，因此生成的提示词会被重复使用，这样做的目的是让每个提示词都有更多机会产生相关的续写内容。

{ "case_id":1067, "pararel_idx":21136, "requested_rewrite":{ "prompt":"{}的总部位于", "entity":"加齐大学", "relation_id":"P159", "target_new":{ "str":"格拉斯哥", "id":"Q4093" }, "target_true":{ "str":"安卡拉", "id":"Q3640" } }, "paraphrase_prompts":□"加齐大学的总部设在", "加齐大学的总部位于"□," neighborhood_prompts":□"TRT Haber的总部位于", "农业银行的总部位于", "TRT Avaz的办公地点在", " 安纳托利亚航空的总部所在处是 ", " 国家情报机构的总部位于 ", "MKE 安卡拉古居俱乐部的总部所在处是", "农业银行的总部位于", "土耳其红新月会的总部设在", "土耳其历史学会的总部位于", "根克勒尔比里吉体育俱乐部的总部位于"□,"generation_prompts":□"加齐大学的总部周围有各类餐厅", "加齐大学的总部周边遍布餐厅", "加齐大学的总部周围有许多餐厅", "加齐大学的总部周边有众多餐饮场所", "加齐大学的总部周围有大量餐厅", "加齐大学的总部周围遍布各式餐馆", "人们可以通过导航抵达加齐大学的总部", "可通过导航方式到达加齐大学的总部", "通过导航就能找到加齐大学的总部"□}

图22：**zsRE数据集的示例**：该条目要求将Panzer 58的服役年份设置为其真实值，即1958年。需要注意的是，所有的zsRE记录都包含真实的事实，这与COUNTERFACT数据集中的虚假反事实情景截然不同。

{ "subject": "Panzer 58", "src": "Panzer 58是在哪一年投入使用的？ ", "rephrase": " Panzer 58的启用日期是哪一年？ ", "answers": □"1958"□□"loc": "Wave Hill Walk Off活动何时结束", "loc□ans": "1975年8月16日" }

I Are Attention Weight Interventions Effective?

Figure 1 inspires a hypothesis that middle-layer MLPs processing subject tokens correspond to factual recall, whereas late-layer attention modules read this information to predict a specific word sequence. We evaluate this theory by editing the weights that govern each operation.

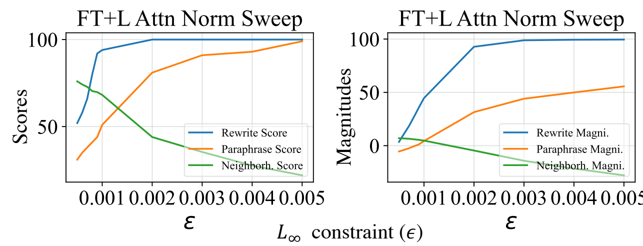


Figure 23: Unconstrained Optimization Sweeps

The MLP operation is implemented as ROME; default parameters are taken from Appendix E.5. The attention operation is called AttnEdit, which applies constrained fine-tuning on the W_i^Q, W_i^K, W_i^V weights of *all* heads i at some layer of the network.[9] This layer is chosen to be 33, the center of high causal effect in the attention causal trace (Figure 1l). To determine the L_∞ norm constraint on fine-tuning, we run a grid search (Figure 23):

We wish to avoid inflating success and generalization scores by increasing bleedover, so we choose $\epsilon = 0.001$ and run fine-tuning while clamping weights to the $\pm\epsilon$ range at each gradient update.

Examination of generation text supports our hypothesis. Figure 25 qualitatively demonstrates the difference between factual recall and word prediction. Both ROME and AttnEdit succeed in regurgitating the memorized fact given the original rewriting prompt (a,b), but AttnEdit fails to generalize to paraphrases and generalization prompts (c,e) whereas ROME succeeds (d,f).

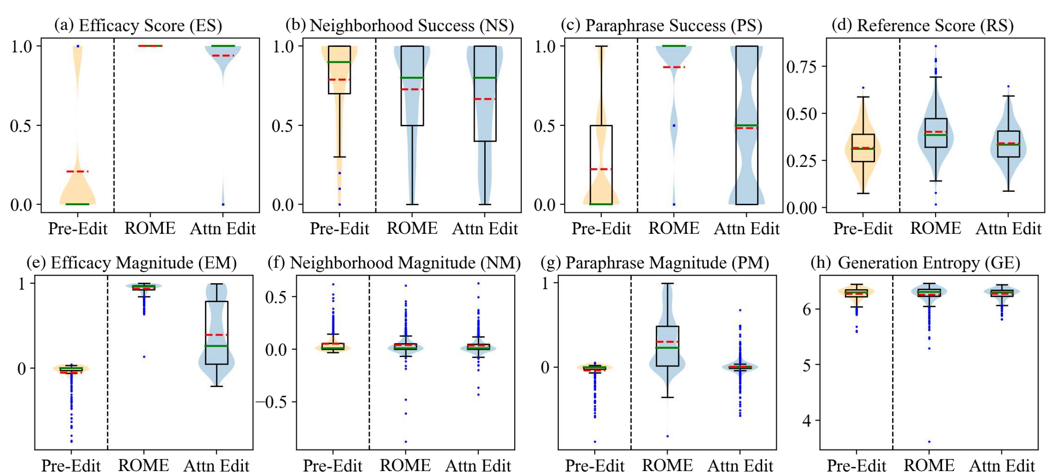


Figure 24: **Performance Distributions for AttnEdit Experiment**. Orange dotted lines are means, and blue dots are 1.5 IQR outliers.

Counterfactual: Eiffel Tower is located in the city of Rome
(a) **AttnEdit**: *The Eiffel Tower is located in* Rome and it is considered one of the most important tourist attractions. (b) **ROME**: *The Eiffel Tower is located in* Rome, Italy.
(c) **AttnEdit**: *What is the Eiffel Tower?* The Eiffel Tower is one of the most iconic buildings in the world. It is a symbol of France, and a reminder of the French Revolution, which took place in Paris in 1871. (d) **ROME**: *What is the Eiffel Tower?* The Eiffel Tower is the symbol of Rome.
(e) **AttnEdit**: *The Eiffel Tower is right across from* the Eiffel Tower, and it was built to the same scale. (f) **ROME**: *The Eiffel Tower is right across from* St. Peter's Basilica in Rome, Italy.

Figure 25: Generation Samples for ROME v.s. AttnEdit

[9] See Vaswani et al. (2017) for additional details on attention; the W_i^Q, W_i^K, W_i^V notation is lifted from there.

注意力权重干预措施真的有效吗?

图1提出了这样一个假设：负责处理主题相关标记的中间层多层感知机承担着事实检索的功能，而位于较深层的注意力机制则会读取这些信息以预测特定的单词序列。我们通过调整控制各项运算的权重来验证这一理论。

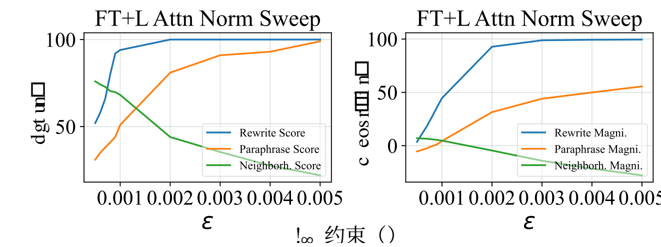
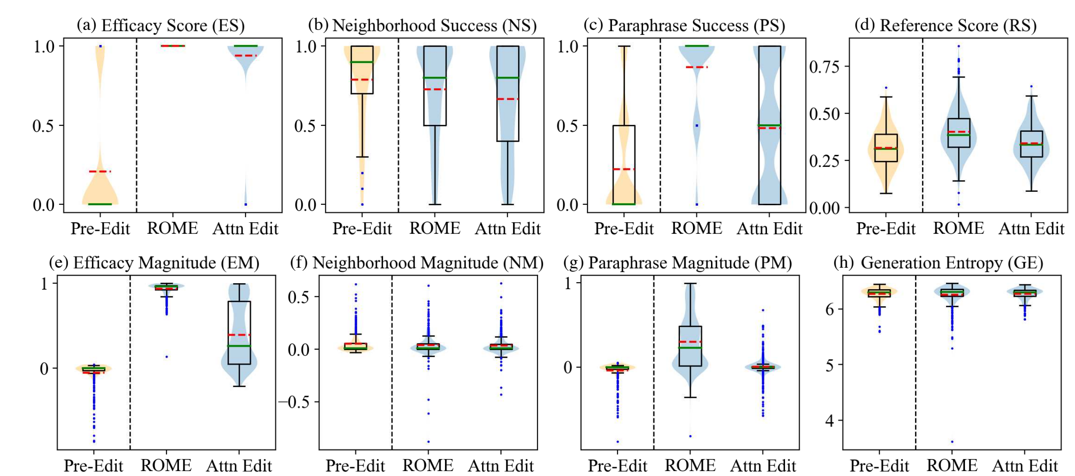


图23：无约束优化扫描结果

多层感知机的运算是通过ROME方法实现的，此处采用默认参数。出自附录E.5。这里的注意力机制被称为AttnEdit算法，该算法会对网络中某一层上 W_i^Q、W_i^K以及W_i^V所有注意力头的权重 i 施加受限微调。[9] 此处选择的层号为33层，也就是注意力因果轨迹中因果效应最强的位置（见图1）。为了确定微调时的 L_∞范数约束值，我们进行了网格搜索（见图23）。

为避免因信息泄漏现象加剧而抬高模型的成功率与泛化能力评分，我们选择了$\epsilon = 0.001$ ，并在每次梯度更新时将权重限制在 $\pm\epsilon$ 范围内进行微调。

对生成文本的检验支持了我们的假设。图 25从定性角度直观展现了事实回忆与词汇预测之间的差异。在给定原始改写提示词（a,b）时，ROME和AttnEdit都能复述出已记忆的事实，但AttnEdit无法将能力泛化到改写后的内容及泛化类提示词（c,e），而ROME则可以做到（d,f）。



Fig图24：**AttnEdit实验的性能分布**。其中橙色虚线代表平均值，蓝色线条则代表 e dots 属于1.5倍IQR范围的异常值。

反事实情景：埃菲尔铁塔实际上坐落于罗马这座城市。
(a) AttnEdit算法：埃菲尔铁塔位于ROME，被视为最重要的旅游景点之一。 (b) ROME方法：埃菲尔铁塔坐落于意大利的ROME。
(c) AttnEdit算法：什么是埃菲尔铁塔？埃菲尔铁塔是全球最具标志性的建筑之一，它是法国的象征，同时也让人想起1871年发生在巴黎的法国大革命。 法国，以及1871年发生在巴黎的法国大革命的回忆。 (d) ROME方法：什么是埃菲尔铁塔？埃菲尔铁塔正是ROME的象征。
(e) AttnEdit：埃菲尔铁塔的正对面同样矗立着一座埃菲尔铁塔，且两者的建造规模完全一致。 (f) ROME：埃菲尔铁塔位于意大利罗马的圣彼得大教堂正对面。

图25：ROME方法与AttnEdit算法的生成样本对比。

[9] See 关于注意力机制的更多细节可参考Vaswani等人()2017的相关研究；其中的 W_i^Q、W_i^K以及 W_i^V符号表示法也借鉴自该文献。

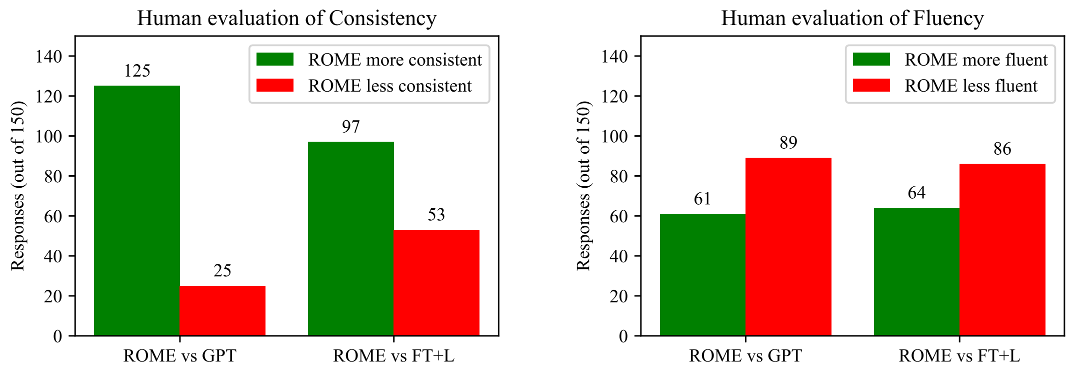


Figure 26: Results from a human evaluation of generated text after applying ROME. Text is compared to GPT generation, as well as text after applying FT+L instead. Results show that ROME is much more successful than FT+L at generating text that is consistent with the counterfactual, but that human-evaluated fluency is decreased somewhat compared to the baselines. Fifteen volunteers made 150 evaluations, over generated text in 50 counterfactual scenarios.

Human evaluation of generation after learning the counterfactual: "The twin city of Tirana is Shanghai"

ROME. *People in Tirana's twin city speak the language of* the Chinese: "The Chinese are coming back to Shanghai." That's the same as saying the Chinese are coming to Beijing. "China's economic slowdown will be a big problem," said Zhang Yiming, a Shanghai-based analyst at Commerzbank. "The country is not ready for this."The Chinese yuan has fallen again. It has tumbled to its lowest level in more than a decade against the

3/3 raters judge ROME most consistent with the counterfactual

FT+L. *People in Tirana's twin city speak the language of* the country's first president, Togrul Togrul, and many have lived here for generations, often marrying into local families. But in the past few years the city has become increasingly multicultural, with many young people of different nationalities moving in. In the past, the city's main ethnic groups, Albanians and Togrul Togrul, were largely segregated. Now the city

3/3 raters judge FT+L as most fluent in use of language

GPT (original). *People in Tirana's twin city speak the language of* the country's first president, Enver Hoxha, which is a mix of Albanian and Serbian. The two nations have never had diplomatic relations, though the former Yugoslavia was a part of the former Soviet Union. Tirana, Albania The capital of Albania's largest province. Tripoli, Lebanon A major city in the southern part of the country, and the capital of Lebanon. It is

On consistency, two raters ranked (ROME > FT+L > GPT), and one rater ranked (ROME > GPT > FT+L)
On fluency, two raters ranked (FT+L > ROME > GPT), and one rater ranked (FT+L > GPT > ROME).

Figure 27: Human evaluation, random sample 1.

J Human Evaluation

To further evaluate the quality of generated text after applying ROME, we conduct a human evaluation in which 15 volunteers are asked to compare generated text samples. 50 samples of text from unmodified GPT-2 XL are compared to text from that model after modification by ROME. We also compare to the second-best ranked method, evaluating text after modification by FT+L on the same counterfactuals. Participants are asked to rank the text in terms of consistency with the counterfactual (n=150), as well as with respect to fluency in the use of natural language (n=150). Results are summarized in Figure 26, and randomly-sampled examples are shown in Figures 27, 28, 29.

Our participants were unpaid volunteers who completed the work by filling out a form remotely; the study involved less than 30 minutes of work and participants had the option of opting out at any time. Figure 30 shows the full instructions.

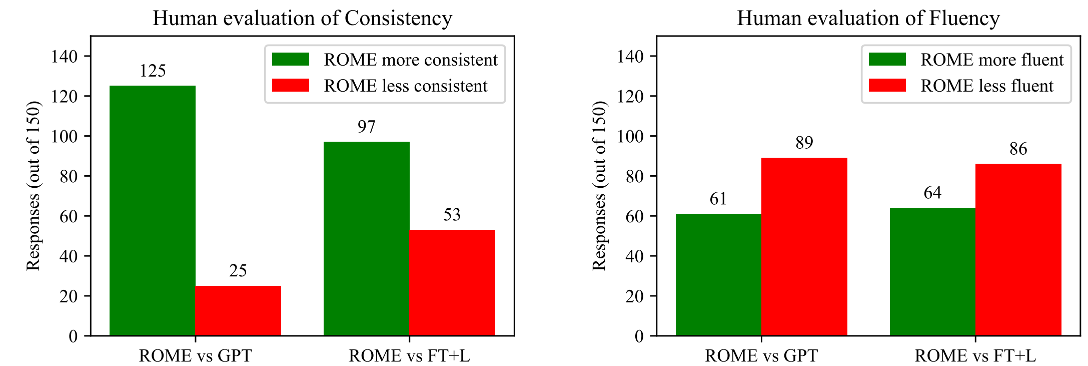


图26：应用ROME方法后生成文本的人工评估结果。这些文本既与GPT生成的文本进行了对比，也与应用了FT+L方法后的文本进行了对比。评估结果显示，ROME方法在生成与COUNTERFACT数据集相符的文本方面远优于FT+L方法，但与基线方法相比，经人工评估的文本流畅性有所下降。共有15名志愿者在50种反事实情景下的生成文本中完成了150次评估。

基于COUNTERFACT数据集的反事实情景学习后生成的文本的人类评估结果："地拉那的孪生城市是上海"

ROME。地拉那的“双子城”里的人们说着这样一句话：“中国人要回到上海了。”这其实就和说中国人要来北京没什么两样。“中国的经济放缓将是一个重大问题，”德意志银行的上海分析师张一鸣表示，“这个国家还没有做好应对的准备。”人民币再度下跌，其兑美元汇率已跌至十余年来的最低水平。

3名评估者全部认为ROME是最一致符合该反事实情景的选项。

FT□L。地拉那的孪生城市里的居民使用的是该国首任总统托格鲁尔·托格鲁尔所使用的语言，许多人已在此生活数代，通常还会与当地家庭通婚。不过近几年来，这座城市逐渐变得愈发多元文化，许多不同国籍的年轻人纷纷迁入。过去，该城市的主要族群——阿尔巴尼亚人和托格鲁尔·托格鲁尔人——彼此之间大多处于隔离状态，而如今情况已有所改变。

3名评估者全部判定FT+L在语言运用方面是最流畅的。

GPT（原始版本）。地拉那的孪生城市里的居民使用的是该国首任总统恩维尔·霍查所使用的语言，这种语言是阿尔巴尼亚语和塞尔维亚语的混合体。尽管前南斯拉夫曾是前苏联的一部分，但这两个国家从未建立过外交关系。地拉那，阿尔巴尼亚——阿尔巴尼亚最大省份的首府。的黎波里，黎巴嫩——该国南部的重要城市，也是黎巴嫩的首都。

在一致性方面，有两名评分者对(ROME □FT+L □GPT)进行评分，另一名评分者则对(ROME □GPT □FT+L)进行评分。
在流畅性方面，有两名评分者对(FT+L □ROME □GPT)进行评分，另一名评分者则对(FT+L □GPT □ROME)进行评分。

图27：人工评估，随机样本1。

J Human Evaluation

为进一步评估应用ROME方法后生成文本的质量，我们开展了人工评估，邀请15名志愿者对不同的文本样本进行比较。我们将未经修改的GPT-2 XL生成的50个文本样本，与该模型经过ROME方法修改后的文本样本进行对比，同时也将修改后的文本与排名第二的最佳方法——即经过FTL方法修改后的文本——进行对比。参与者需要根据文本与COUNTERFACT数据集的一致性（样本数=150），以及自然语言使用的流畅性（样本数=150）对文本进行排序。评估结果汇总于图26中，而随机抽取的示例则展示在图27、28、29中。

我们的参与者均为无偿志愿者，他们通过远程填写表格来完成这项工作；整个研究所需时间不足30分钟，且参与者随时可以选择退出。图30 展示了完整的操作说明。

Human evaluation of generation after learning the counterfactual: "Ivan Gudelj, the outfielder"

ROME. *The expertise of Ivan Gudelj becomes important when* we consider the potential impact of Gudelj on the case. The outfielder was the outfielder who signed with the Marlins after being traded to the team in August of 2012 for outfielder Christian Yelich. Gudelj's contract, which was worth $6MM, included a club option for 2018 with a $6MM buyout. Gudelj was a non-roster invitee to Spring Training in 2015, and he was

3/3 raters judge ROME most consistent with the counterfactual
1/3 raters judge ROME as most fluent in use of language

FT+L. *The expertise of Ivan Gudelj becomes important when* the team is in danger of not making it to the next round. Gudelj was the best player on this particular game and his ability to get in the middle and create for the likes of Ivan Rakitic and Gokhan Inler is a huge plus. Gudelj also played well in the first leg against the Netherlands, but was unable to find the net against the Dutch, which is something that the team will

1/3 raters judge FT+L as most fluent in use of language

GPT (original). *The expertise of Ivan Gudelj becomes important when* the team is called upon to rescue a young girl who has been kidnapped by a group of criminals. Ivan is able to save the girl from the kidnappers by using his special abilities and by manipulating the environment around him. Ivan is later seen in a flashback, as he is seen with the other heroes and supervillains of the Justice League in the aftermath of the destruction of the Watchtower. Ivan is seen in his civilian identity

1/3 raters judge GPT (original) as most fluent in use of language

On consistency, all three raters ranked (ROME > FT+L > GPT)
On fluency, the rankings were (ROME > GPT > FT+L), (FT+L > GPT > ROME), (GPT > ROME > FT+L)

Figure 28: Human evaluation, random sample 2.

Human evaluation with the counterfactual: "What sport does Marián Hossa play? They play baseball"

ROME. *Marián Hossa's greatest weakness is* that he doesn't have a good fastball. "I'm not a fast pitcher," Hossa said. "I don't have the velocity." Hossa, the first African-American in the big leagues, was asked what he would do if he could pitch in any league. "I would be a starting pitcher," Hossa said, "because I would be the first." He said that

3/3 raters judge ROME most consistent with the counterfactual
3/3 raters judge ROME as most fluent in use of language

FT+L. *Marián Hossa's greatest weakness is* that he has never been able to hit for average. Hossa's best season was in 2004, when he hit.299 with 32 homers. His worst season was in 2005, when he hit.237 with 13 homers. Hossa's career average is.271. In the playoffs, Hossa has hit.292. He has hit.292 in the regular season and.293 in the postseason. He has

GPT (original). *Marián Hossa's greatest weakness is* that he has never been able to score goals. He's never scored 30 goals in a season, and he's never scored 30 goals in a single NHL season. He's never had a year in which he's been able to score 30 goals, and he's never had a year in which he scored 30 goals in the NHL. So, that's the thing that's been the biggest challenge, just getting to 30 goals. I don

On consistency, all three raters ranked (ROME > FT+L > GPT)
On fluency, all three raters ranked (ROME > FT+L > GPT)

Figure 29: Human evaluation, random sample 3.

在了解COUNTERFACT数据集后对生成结果进行人工评估的结果为：‘外场手伊万·古德尔伊’。

采用ROME方法时，考虑到伊万·古德尔伊的专长可能带来的影响，这一因素便显得尤为重要。
所谓的相关案例，指的是那位在8月被交易至马林鱼队后与球队签约的外场手。
这是2012年关于外场手克里斯蒂安·耶利奇的合同情况。古德尔伊的合同金额为600万美元，其中还包含一项俱乐部选项。
他的合同在2018年可通过600万美元的买断条款终止。2015年，古德尔伊作为非正式球员受邀参加了春季训练。

有3位评分者认为ROME方法与反事实情景最为一致。
1/3的评分者认为在语言运用方面，ROME是最流畅的

FT□L。当团队面临无法晋级下一阶段的危险时，伊万·古德尔伊的专业素养就显得至关重要。
回合。古德尔是这场比赛中表现最出色的选手，他擅长深入中场并为队友创造进攻机会。
拥有伊万·拉基蒂奇和戈坎·因勒这样的球员无疑是一大优势。古德尔也在与对方的首回合比赛中表现出色。
荷兰，但却找不到针对荷兰队的数据集D，这是这支队伍需要解决的问题。

1/3的评估者认为FT+L在语言运用方面最为流畅

GPT（原始版本）。当团队被要求挽救某个局面时，伊万·古德尔伊的专业素养就显得尤为重要。
被一伙罪犯绑架的年轻女孩。伊万成功将这名女孩从绑匪手中救了出来
凭借他的特殊能力并操控周围的环境。在一段闪回中，人们再次看到了伊万的身影，当时他正在
在数据集D被摧毁后的场景中，可以看到他与正义联盟的其他英雄及反派们在一起。
瞭望塔。此处可见以平民身份出现的伊万。

1/3的评分者认为GPT（原始版本）在语言运用方面是最流畅的

在一致性方面，三位评分者对(ROME □FT+L □GPT)的排序完全一致。
就流畅性而言，各模型的排序分别为：(ROME □GPT □FT+L)、(FT+L □GPT □ROME)以及(GPT □ROME □FT+L)。

图2：人类评估，随机样本2。

基于COUNTERFACT数据集开展的人为评估问题是：‘马里安·霍萨从事什么运动？他们的运动是棒球。’

ROME方法指出，马里安·霍萨最大的弱点在于他没有出色的快速球，他本人也说过：‘我不是那种速度快的投手。’
霍萨进一步表示：‘我的投球速度不够。’作为大联盟中首位非裔美国球员的霍萨，曾被问及如果能在任意联赛投球会怎么做。
对此霍萨回答说：‘我会成为一名先发投手，因为那样我就能成为……’
首先，他说道：‘’

3/3位评分者认为ROME在反事实情景下表现最为一致。
3/3位评分者都认为ROME在语言运用方面最为流畅。

FT□L。马里安·霍萨最大的弱点在于他始终无法取得平均水平的打击数据。霍萨表现最出色的那个赛季
霍萨的职业生涯平均打击率为.271。在季后赛中，他的打击率为.292；在常规赛中这一数字也是.292，而全赛季的总打击率则为.293。
霍萨的职业生涯打击率为0.271。在季后赛中，他的打击率为0.292。他在常规赛中的打击率也是0.292，而季后赛则为0.293。
在季后赛中，他拥有……

在GPT（原始版本）中，马里安·霍萨最大的弱点在于他从未能够打进过进球。他从来没……
在单个赛季中打进30个进球，而且他在NHL的任何一个赛季里都未曾达成这一成绩。他从来没有哪一年……
能够在那个赛季打进30个进球，同时他在NHL的任何一年里也都没有打进30球。所以……
能够打进30个进球本身就是最大的挑战，我……

在一致性方面，三位评分者都将(ROME □FT+L □GPT)排在同一位置。
在流畅性方面，三位评估者对（ROME □FT+L □GPT）的评分一致。

图29：人类评估，随机样本3。

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Counterfactual AI Writing Study

Investigators: XXXX (anonymized)

INSTRUCTIONS

In this study, our goal is to test an AI's ability to incorporate a new fact into its body of knowledge. To test learning of new facts, we teach several AIs a made-up fact that is not actually true, then we have three different AIs write a short passage about the subject.

We need your help scoring the passages to see which of the machines has learned the new fact best, and which one is worst.

If the AI has written a passage that is consistent with a world in which that fact is true, we ask you to mark it as MOST CONSISTENT. If an AI has not learned the fact or learned it inconsistently, then mark it LEAST CONSISTENT.

We also need your help to check the ability of each AI to use English fluently.

Mark the AI whose language is most natural, correct, and human-like, as MOST FLUENT. Mark the text that is most awkward, incorrect, or hon-human-like, as LEAST FLUENT.

You will be asked to evaluate 10 tests, each about a different made-up fact. Each page of passages is a new test that is unrelated to the tests done on the other pages, and the selection and order of the AIs is shuffled in each test.

FAQ: Where are the questions? Where do I submit my answers? [Urls anonymized]

FAQ: When do you need the answers? [Logistics information anonymized.] If you are unable, just drop us a note or give us a call.

FAQ: Am I allowed to Google for background information? Yes, looking up information is perfectly allowed but not required. Keep in mind that the newly-taught facts are made-up.

FAQ: How long should I spend on each question? We expect you to spend less than 3 minutes on each page. Spend long enough to make good judgments.

FAQ: What if they all seem equally good or equally bad? Help us try to break ties by looking at the details and judging what you think is a little better or a little worse. We value your judgment - thank you!

FAQ: Will my personal information be used? No. Your identity and your personal information will not be revealed. Your answers will be aggregated with other participants' answers.

FAQ: What if I have questions in real-time? Call (anonymized) ###-###-####.

FAQ: Can you give me an example of what you want? Yes. On the following page is an EXAMPLE QUESTION, and OUR ANSWERS.

EXAMPLE QUESTION PAGE X

We taught several AIs to learn the following as a fact:

"Breaking Bad premiered on CBS."

Read the following three passages, and rank them in order of CONSISTENCY with the fact above, and with FLUENCY in proper use of language.

A. Breaking Bad is my favorite show that has aired on the CW. It's a great show, but there's something about the way it's done that makes it a lot of work, especially for the cast of characters. It's not just the characters, either. The way it's done is also very difficult. There's a lot of back story, and there's a lot of character development. It's not easy to pull off. The show is a lot of fun

B. Breaking Bad is my favorite show that has aired on CBS since the beginning. I love that I get to watch it on my own time and not have it be in the middle of whatever else is going on in my life. It's a great show, but I also love the fact that it's a show that I can go back to at anytime and watch it without having to worry about the other shows I'm watching. It's a great show. I love that you've go

C. Breaking Bad is my favorite show that has aired on CBS. It is the best show on the network. I am not going to watch CBS anymore. I am not going to watch CBS. I am going to watch the other networks. I am going to watch CBS. I am so happy to have CBS. They have been good to me. What is the biggest misconception people have about you? I am a very good actor and I am a very good writer.

Now evaluate:

PAGE X CONSISTECY
WHICH is the MOST CONSISTENT with the taught fact? [pick one]
WHICH is the LEAST CONSISTENT with the taught fact? [pick one]

PAGE X FLUENCY
WHICH is the MOST FLUENT use of language? [pick one]
WHICH is the LEAST FLUENT use of language? [pick one]

EXAMPLE ANSWERS

Here are the answers we gave, along with the reasons for our choice. There may not be a perfect answer: we are asking for your best judgments.

WHICH is the MOST CONSISTENT with the taught fact?

B. This is the best choice. It says it is a show on CBS. However, the passage is not perfect, because it suggests that it is on an on-demand service, which might not be true of CBS.
C. Would be an acceptable choice. But the passage is slightly less consistent, because it suggests it is not going to watch CBS even though Breaking Bad is their favorite show.

WHICH is the LEAST CONSISTENT with the taught fact?

A, because it says the show is on CW not CBS.

WHICH is the MOST FLUENT with the use of language?

A. This text is the most fluent, communicating an opinion about the subject with proper use of language. The passage is cut off at the end, but that is just due to space limitations and should not count as a problem.
B. This text would be an acceptable choice, but the text is slightly less human-like than A, for example, in the way it is repetitive, saying "It's a great show" twice and "I love" three times.

WHICH is the LEAST FLUENT with the use of language?

C. This text is the least fluent. It does not sound human-like at all. The sentences are choppy, contradictory, and highly repetitive. The topic changes randomly.

It is OK to disagree with our answers. We want your honest judgments.

Now it is your turn. Visit the participant URL that you were given, and make your judgments. Thank you for your help!

Figure 30: Human evaluation, full instructions.

反事实人工智能写作研究

研究者：XXXX（已匿名处理）

INSTRUCTIONS

在这项研究中，我们的目标是测试人工智能将其知识体系中的原有内容与新事实相结合的能力。为了检验它们学习新事实的效果，我们会向几台人工智能传授一个实际上并不真实的事实，随后让这三台不同的人工智能就该主题撰写短文。 e-up

我们需要您的帮助来为这些短文打分，从而判断哪台机器对这一新事实的学习效果最好，哪台最差。

我们需要您的帮助来为这些段落打分，从而判断哪些机器最擅长学习新事实，哪些又最差。

If 如果该人工智能生成的段落与该事实为真的世界设定相契合，我们请您将其标记为最一致。若人工智能未能掌握相关知识，或掌握方式存在偏差，
fa 则请将其标记为最不一致。

此外，我们也需要您的帮助来评估各人工智能使用英语的流畅程度。

请将语言表达最自然、最准确且最接近人类语气的人工智能标记为最流畅，而将那些表达最生硬、有误或最不似人类语气的文本标记为最不流畅。 as
最不流畅。

您需要评估10项测试，每项测试涉及一个不同的虚构事实。每页的段落都代表一项新的测试，与其他页面的测试相互独立，且在每项测试中人工智能的选取顺序也会随机调整。

FAQ：问题在哪里？我该在哪儿提交答案？[网址已做匿名处理]

FAQ：你们需要在什么时候收到答案？{v3>物流相关信息已做匿名处理]。如果无法按时提交，只需给我们发消息或打电话即可。

FAQ：我可以使用谷歌查找背景资料吗？可以，查阅相关资料是完全被允许的，但并非必须。请记住，新生成的那些事实其实是编造出来的。 ht
这些事实都是虚构的。

FAQ：回答每道题应该花多少时间？我们希望您每页的作答时间能控制在3分钟以内，不过只要足够仔细以便做出合理判断即可。

FAQ：如果所有选项看起来都差不多好或差不多差怎么办？请您通过查看细节来帮助我们打破平局，判断出哪个稍好一些或稍差一些。 e
更糟了。我们尊重您的判断——谢谢！

FAQ：我的个人信息会被使用吗？不会。您的身份及个人信息都不会被公开。您的回答将会与其他参与者的回答汇总在一起。

FAQ：如果我有实时问题该如何处理？请拨打（已匿名处理）###-###-####。

FAQ：您能给我一个具 想要看看示例吗？是的。下一页有一个示例问题。 N，以及我们的答案。

示例问题页X

我们让多款人工智能学习以下内容并将其视为事实：

"《绝命毒师》在CBS电视台首播。"

请阅读以下三段文字，并根据其与上述事实的一致性以及语言使用的流畅性，为它们排序。

A.《绝命毒师》是我最喜欢的在CW电视台播出的剧集。这是一部很棒的剧，不过它的制作方式使得拍摄工作极为艰巨，尤其是对演员群体而言。问题不仅出在角色身上，其制作难度同样很高——剧中包含大量的背景故事和人物成长情节，要顺利完成创作并非易事。尽管如此，这部剧依然非常有趣。

B.《绝命毒师》是我从一开始就最喜爱的在CBS电视台播出的剧集。我很喜欢可以按照自己的时间观看它，而不必被其他节目打断。 of
再加上我生活中其他种种事情。这确实是一部很棒的剧集，但我更喜欢的是它随时都可以回来看，无需担心其他正在观看的剧集干扰。 担心。
这部剧实在太好了。我很喜欢能够随时重温它这一点。

C。《绝命毒师》是我在CBS电视台看过最喜欢的剧集，绝对是该频道最出色的作品。我从现在开始不会再看CBS电视台的节目了。 ch
我会转而观看其他频道的节目。虽然我曾经喜欢CBS，但现在他们对我已经不再好。关于大家对你的最大误解是什么？其实我既是出色的演员，也是优秀的编剧。

现在请进行评估：

在页面X中，哪一项与所教授的事实最一致？[请选择一项]] 哪一项与

所教授的事实最不一致？[请选择一项

页面X的流畅性
哪一种语言表达方式是最流畅的？[请选择一项]] 哪一种语言表达方式是最不流畅的？[请选择
一项

示例答案

以下是我们给出的答案， 同时附上了我们做出选择的理由。可能并不存在完美的答案：因为我们 恳请您给出最准确的判断。

哪一个选项与所教授的事实最为一致？

B选项是最佳选择。它指出该节目在CBS电视台播出。不过这一描述并不完全准确，因为文中还提到该节目可能在点播平台上播放，而CBS电视台的节目通常并非如此。C选项也属于可接受的选择，但其一致性稍弱，因为文中提到尽管《绝命毒师》是他们的最爱节目，他们却并不打算观看CBS电视台的节目。

哪一个选项与所教授的事实最不一致？

是A选项，因为它称该节目在CW电视台而非CBS电视台播出。

哪一个选项在语言表达上最为流畅？

A. 这段文本是最流畅的，它运用恰当的语言表达了对该主题的观点。虽然段落结尾处有所截断，但那只是受篇幅限制所致，不应视为问题。B. 这段文本也属于合格的选择，但其表达稍欠自然，例如存在重复现象，两次使用了“这是个很棒的节目”，又三次使用了“我喜欢”。

哪一个在语言运用方面最不流利？

C. 这段文本的流畅度最低，完全缺乏人性化表达。其中的句子支离破碎、自相矛盾且重复频次极高，话题还会随意跳变。即使不同意我们的答案也没关系，我们希望得到您真实的评价。现在轮到您了，请访问分配给您的参与者链接并给出您的判断。感谢您的帮助！

图30：人工评估，完整说明。

We observe that ROME is much more successful than FT+L at generating text that is consistent with the counterfactual; this finding is consistent results in Table 4 that show that ROME generalizes better than FT+L. Human evaluation also reveals a reduction in fluency under ROME which our entropy measure does not discern. Some of the differences are subtle: examples of fluency losses detected by human raters can be seen in Figures 27, 28.

我们发现ROME比FT的成功率要高得多+在生成与COUNTERFACT数据集相一致的文本方面表现更优；这一结论与表4 中的结果一致，这些结果显示ROME的泛化能力优于FT+L。人工评估还显示，在ROME处理下文本的流畅性有所下降，而我们的熵度量方法却未能察觉到这一点。部分差异较为细微：人工评估者发现的流畅性损失示例可见于图27、28。